

Exercises

Level - 1

(Problems Based on Fundamentals)

Evaluation of Definite Integrals

1. Evaluate : $\int_0^{\pi/2} \sin^2 x \, dx$

2. Evaluate : $\int_0^{\pi/4} \frac{dx}{1 + \sin x}$

3. Evaluate : $\int_0^2 \frac{dx}{\sqrt{x+2} + \sqrt{x}}$

4. Evaluate : $\int_0^{\pi/2} \frac{\sin x}{\sin x - \cos x} dx$

5. Evaluate : $\int_0^2 \frac{dx}{x(1+x^4)}$

6. Evaluate : $\int_0^1 \frac{dx}{x^2 + x + 1}$

7. Evaluate : $\int_0^1 x e^x dx$

8. Evaluate : $\int_0^{\pi/2} \frac{dx}{\sin x + \cos x}$

9. Evaluate : $\int_0^{\pi/2} \frac{dx}{(3\sin^2 x + 4\cos^2 x)}$

10. Evaluate : $\int_0^1 (1-x)x^n dx$

11. If $I_n = \int_0^{\pi/2} \tan^n x \, dx$, prove that $I_n + I_{n-2} = \frac{1}{n-1}$

12. Prove that

$$\int_0^x \left(\frac{\sin nx}{\sin x} \right) dx = \begin{cases} 0, & \text{when } n \text{ is even} \\ \pi, & \text{when } n \text{ is odd} \end{cases}$$

13. Prove that $\int_0^{\pi/2} \left(\frac{\sin \theta}{\sin \theta} \right)^2 d\theta = n\pi$, where $n \in \mathbb{W}$

14. $\int_0^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)}$

15. $\int_0^1 x(1-x)^5 dx$.

16.
$$\int_0^{\pi/2} \left(\frac{\sin^2 x}{\sin^4 x + \cos^4 x} \right) dx$$

17.
$$\int_1^2 \frac{dx}{x(x^3 + 1)}$$

Evaluation of Definite Integrals by Substitution

18. Evaluate:
$$\int_2^4 \frac{x}{\sqrt{x-2}} dx$$

19. Evaluate:
$$\int_0^1 x(1-x)^{2012} dx$$

20. Evaluate:
$$\int_1^2 \frac{dx}{x(x^3 + 1)}$$

21. Evaluate:
$$\int_0^{\pi/2} \frac{dx}{(2 + \cos x)}$$

22. Evaluate:
$$\int_0^{\pi/2} \frac{dx}{(2\sin^2 x + \cos^2 x)}$$

23. Evaluate:
$$\int_0^{\pi/2} \left(\frac{\cos x}{6 - 5\sin x + \sin^2 x} \right) dx$$

24. Evaluate:
$$\int_0^{\pi/2} \left(\frac{\sin^2 x}{\sin^4 x + \cos^4 x} \right) dx$$

25. Evaluate:
$$\int_0^{\pi/4} \left(\frac{\sin x + \cos x}{16 + \sin^2 x} \right) dx$$

26. Evaluate:
$$\int_0^{\pi/2} \left(\frac{1}{\sqrt{\cos x} + \sqrt{\sin x}} \right)^4 dx$$

27. Evaluate:
$$\int_0^2 \frac{dx}{\sqrt{x+1} + \sqrt{(x+1)^3}}$$

28. Evaluate:
$$\int_0^{\pi/2} \left(\frac{\cos^2 x}{4\sin^2 x + \cos^2 x} \right) dx$$

29. Evaluate:
$$\int_1^2 \frac{dx}{(2x+1)\sqrt{x^2+x}}$$

30. Evaluate:
$$\int_0^4 \frac{dx}{x + \sqrt{x}}$$

31. Evaluate:
$$\int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} dx$$

32. Evaluate:
$$\int_0^{\pi/2} \frac{\cos \theta}{(1 + \sin \theta)(2 + \sin \theta)} d\theta$$

33. Evaluate:
$$\int_0^{\pi/2} (\sqrt{\tan x} + \sqrt{\cot x}) dx$$

34. Evaluate:
$$\int_0^{\pi/2} \frac{dx}{4\sin^2 x + 5\cos^2 x}$$

35. Evaluate:
$$\int_0^{\pi/4} \left(\frac{\sin x + \cos x}{9 + 16 \sin^2 x} \right) dx$$

36. Evaluate:
$$\int_0^{\pi/2} \left(\frac{\sin^2 x}{\sin^4 x + \cos^4 x} \right) dx$$

37. Evaluate:
$$\int_0^1 \left(\sqrt{\frac{1-x}{1+x}} \right) dx.$$

Geometrical Interpretation of the Definite Integral

38. Evaluate:
$$\int_{-1}^4 (2x - 3) dx$$

39. Evaluate:
$$\int_{-2}^2 f(x) dx,$$

where $f(x) = \min\{x - [x], -x - [-x]\}$

40. Evaluate:
$$\int_0^{2\pi} f(x) dx,$$

where $f(x) = \sin^{-1}(\sin x).$

41. Evaluate:
$$\int_0^{2\pi} [\sin x] dx$$

42. Evaluate:
$$\int_{-2}^2 f(x) dx,$$

where $f(x) = \min\{|x-2|, |x|, |x+2|\}$

43. Evaluate:
$$\int_{-1}^1 f(x) dx,$$

where $f(x) = \{|x| - 1, 1 - |x|\}$

44. Evaluate:
$$\int_{\pi/4}^{\pi/2} \left[\sin x + \left\lfloor \frac{2x}{\pi} \right\rfloor \right] dx$$

45. Evaluate:
$$\int_0^1 [xd(x - [x])].$$

46. Evaluate:
$$\int_0^2 [x^2] dx.$$

47. Evaluate:
$$\int_0^{1.5} [x^2] dx$$

48. Evaluate:
$$\int_0^{1.8} [x^2] dx$$

49. Evaluate:
$$\int_0^2 [x^2 - x + 1] dx.$$

50. Evaluate: $\int_1^2 [2x^2 - 3] dx$

51. Evaluate: $\int_1^2 [x^3 - 1] dx$

52. Evaluate: $\int_0^n [x] dx$

53. Evaluate: $\int_0^{10} [x] dx$

54. Evaluate: $\int_0^{10} [x + 2] dx$

55. Evaluate: $\int_0^2 [2x + 3] dx$

56. Evaluate: $\int_0^2 [3x - 2] dx$

57. Evaluate: $\int_0^n \{x\} dx$

58. Evaluate: $\int_2^0 (x[x]) dx$

59. Evaluate: $\int_1^6 \left[\frac{\log x}{3} \right] dx$

60. Evaluate: $\int_0^\pi [2\sin x] dx$

61. Evaluate: $\int_0^\pi [2\sin x] dx$

62. Evaluate: $\int_0^{\pi/3} [\sqrt{3} \tan x] dx$

63. Evaluate: $\int_1^{1.5} (\sin[x]) dx$

64. Evaluate: $\int_0^{102} [\tan^{-1} x] dx$

65. Evaluate: $\int_0^{2n\pi} [\sin x + \cos x] dx$

66. Evaluate: $\int_0^{5\pi/12} [\tan x] dx$

67. Evaluate: $\int_{-2}^1 \left[x \left\{ 1 + \cos \left(\frac{\pi x}{2} \right) \right\} \right] dx$

68. Evaluate: $\int_0^3 \left([x] + \left[x + \frac{1}{2} \right] + \left[x + \frac{2}{3} \right] \right) dx$

69. Evaluate: $\int_0^{10\pi} ([\sec^{-1} x] + [\tan^{-1} x]) dx$

70. Evaluate: $\int_0^{\pi/4} [\sin x + \{\cos x + (\tan x + \sec x)\}] dx$

71. Evaluate: $\int_{\pi/4}^{\pi/2} \left[\sin x + \frac{2x}{\pi} \right] dx$

Definite Integral as the Limit of Sum

Q. Evaluate each of the following integrals from the first principle.

72. $\int_0^3 x dx$

73. $\int_0^2 x^2 dx$

74. $\int_1^3 (x^2 + x + 2) dx$

75. $\int_0^1 e^{mx} dx$

76. $\int_0^{\pi/2} \sin x dx$

77. $\int_1^2 (2x + 3) dx$

78. $\int_0^2 (2x^2 + 1) dx$

79. $\int_0^2 (x^2 + 3x) dx$

80. $\int_0^3 (x^2 - 5x + 4) dx$

81. $\int_0^4 (x^3 - 4) dx$

82. $\int_0^4 (3x^3 + 2) dx$

83. $\int_0^4 (2x^3 - 3) dx$

84. $\int_0^1 e^x dx$

85. $\int_0^1 (e^3 x + e^x) dx$

86. $\int_a^b \cos x dx$

(Q.No. 77 to 86 for Board Exams Only)

Evaluation of the Limit of Sum using Newton-Leibnitz Formula

Q. Evaluate each of the following limit by N-L Formula.

87. $\lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n} + 3 + \dots + \frac{1}{2n} \right]$
88. $\lim_{n \rightarrow \infty} \left[\frac{n}{n^2} + \frac{n}{n^2+1^2} + \frac{n}{n^2+n^2} + \frac{n}{n^2+3^2} + \dots + \frac{n}{n^2+(n-1)^2} \right]$
89. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \dots + \frac{n}{n^2+(2n-1)^2} \right]$
90. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{1+\frac{1}{n}} + \sqrt{1+\frac{2}{n}} + \dots + \sqrt{1+\frac{n}{n}} \right)$
91. $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \frac{1}{\sqrt{4n^2-9}} + \dots + \frac{1}{\sqrt{3n^2}} \right)$
92. $\lim_{n \rightarrow \infty} \left(\frac{1m+2m+3m+\dots+nm}{n^{m+1}} \right), m > -1$
93. $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \dots + \frac{1}{n} \right)$
94. $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right)$
95. $\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{1/n}$
96. $\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2} \right) \left(1 + \frac{2^2}{n^2} \right) \left(1 + \frac{3^2}{n^2} \right) \dots \left(1 + \frac{n^2}{n^2} \right) \right]^{1/n}$
97. $\lim_{n \rightarrow \infty} \left(\frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{4n} \right)$
98. $\lim_{n \rightarrow \infty} \left(\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8n} \right)$
99. $\lim_{n \rightarrow \infty} \left(\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \frac{n+3}{n^2+3^2} + \dots + \frac{3}{5n} \right)$
100. $\lim_{n \rightarrow \infty} \left(\frac{n^2}{(n^2+1)^{3/2}} + \frac{n^2}{(n^2+2^2)^{3/2}} + \dots + \frac{n^2}{(n^2+(n-1)^2)^{3/2}} \right)$
101. $\lim_{n \rightarrow \infty} \left(\frac{1^2}{n^3+1^3} + \frac{2^2}{n^3+2^3} + \frac{3^2}{n^3+3^3} + \dots + \frac{1}{2n} \right)$
102. $\lim_{n \rightarrow \infty} \left(\frac{2^k+4^k+6^k+\dots+(2n)^k}{n^{k+1}} \right), k \neq -1$
103. $\lim_{n \rightarrow \infty} \left(\frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+\dots+2\sqrt{n}}{n\sqrt{n}} \right)$
104. $\lim_{n \rightarrow \infty} \left(\frac{1}{na} + \frac{1}{na+1} + \frac{1}{na+2} + \dots + \frac{1}{na+n} \right)$
105. $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{2n}} \right)$

106. $\lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \left(\frac{1}{n+r} \right)$
107. $\lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \left(\frac{1}{\sqrt{n^2-r^2}} \right)$
108. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{\sqrt{n^2+3rn}} \right)$
109. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2} \right)$
110. $\lim_{n \rightarrow \infty} \sum_{r=1}^{10n} \left(\frac{1}{\sqrt{r}\sqrt{n}} \right)$
111. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{n}{(r+n)\sqrt{r^2+2nr}} \right)$

Properties of Definite Integrals

Property III

Q. 112 to 128, Evaluate each of the following integrals.

112. $\int_0^4 |x-2| dx$
113. $\int_0^4 (|x|+|x-2|) dx$
114. $\int_{-3}^3 (|x+2|+|x|+|x-2|) dx$
115. $\int_1^4 (|x-1|+|x-2|+|x-3|) dx$
116. $\int_0^3 |x^2-3x+2| dx$
117. $\int_{-4}^3 |x^2-4| dx$
118. $\int_0^\pi |\sin x| dx$
119. $\int_0^\pi ||\sin x| - |\cos x|| dx$
120. $\int_{-1}^{3/2} |x \sin \pi x| dx$
121. $\int_{-\pi/2}^{2\pi} \sin^{-1}(\sin x) dx$
122. $\int_{1/e}^e ||\log x| dx$
123. $\int_0^{100} [\tan^{-1} x] dx$, where $[.] = \text{GIF}$

$$124. \int_0^{1/2} [2\sin^{-1}x] dx$$

$$125. \int_{-2}^1 \left[x \left[1 + \cos\left(\frac{\pi x}{2}\right) \right] + 1 \right] dx$$

$$126. \int_0^{\sqrt{3}} \tan^{-1}\left(\frac{2x}{1-x^2}\right) dx$$

$$127. \int_{-\pi/2}^{\pi/2} (\cos|x| + \sin|x|) dx$$

$$128. \int_0^2 \left| \cos\left(\frac{\pi x}{2}\right) \right| dx$$

129. Let $f(x) = \max\{x + |x|, x - [x]\}$ where $[.] = \text{GIF}$, find

the value of $\int_{-2}^2 f(x) dx$.

$$130. \int_{-1}^1 [x[1 + \sin\pi x] + 1] dx, \text{ where } [.] = \text{GIF}.$$

$$131. \text{ For } x > 0, \text{ let } f(x) = \int_1^x \frac{\ln t}{1+t} dt.$$

Find the function $f(x) + f\left(\frac{1}{x}\right)$ and show that

$$f(e) + f\left(\frac{1}{e}\right) = \frac{1}{2}.$$

Here, $\ln t = \log_e t$.

$$132. \text{ Prove that } \int_0^{\sin^2 x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\cos^2 x} \cos^{-1}(\sqrt{t}) dt = \frac{\pi}{4}$$

$$133. \text{ Evaluate: } \int_0^2 |x^2 + 2x - 3| dx$$

$$134. \text{ Evaluate: } \int_{1/e}^e |\log x| dx$$

$$135. \text{ Evaluate: } \int_1^{10\pi} ([\sec^{-1}x] + [\cot^{-1}x]) dx$$

$$136. \text{ Evaluate: } \int_{-\pi/2}^{2\pi} ([\cot^{-1}x]) dx$$

$$137. \text{ Evaluate: } \int_1^2 [2x^2 - 3] dx$$

$$138. \text{ Prove that } \int_0^{\pi} ||\sin x| - |\cos x|| dx = 4(\sqrt{2} - 1)$$

$$139. \text{ Prove that } \int_0^2 [x^2 - x + 1] dx = \left(\frac{5 - \sqrt{5}}{2}\right), \text{ where } [.] = \text{GIF}$$

$$140. \text{ Prove that } \int_0^{\frac{5\pi}{12}} [\tan x] dx = \frac{\pi}{4}$$

$$141. \text{ Prove that } \int_0^{102} [\tan^{-1}x] dx = 102 - \tan(1)$$

$$142. \text{ Prove that } \frac{\int_0^n [x] dx}{\int_0^n \{x\} dx} = n - 1$$

$$143. \text{ Prove that } \sum_{n=1}^{1000} \int_{n-1}^n e^{x-[x]} dx = 1000(e - 1)$$

$$144. \text{ Prove that } \int_{1/e}^{\tan x} \frac{1}{(1+t^2)} dt + \int_{\frac{1}{e}}^{\cot x} \frac{dt}{t(1+t^2)} = 1.$$

Property IV

$$145. \text{ Evaluate: } \int_0^{\pi/2} \left(\frac{\sin x}{\sin^n x + \cos^n x} \right) dx.$$

$$146. \text{ Evaluate: } \int_0^{14} \left(\frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} \right) dx$$

$$147. \text{ Evaluate: } \int_0^{\infty} \frac{dx}{(1+x^a)(1+x^2)}$$

$$148. \text{ Evaluate: } \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$$

$$149. \text{ Evaluate: } \int_0^1 \tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right) dx$$

$$150. \text{ Evaluate: } \int_0^1 \cot^{-1}(1-x+x^2) dx$$

$$151. \text{ Evaluate: } \int_0^{\pi} \left(\frac{dx}{1+2^{\tan x}} \right)$$

$$152. \text{ Evaluate: } \int_0^1 \log\left(\frac{1}{x} - 1\right) dx$$

$$153. \text{ Evaluate: } \int_0^1 \frac{\log(1+x)}{(1+x^2)} dx$$

$$154. \text{ Evaluate: } \int_0^1 x(1-x)^{2013} dx$$

$$155. \text{ Evaluate: } \int_0^{\pi} \left(\frac{x}{1 + \sin x} \right) dx$$

$$156. \text{ Evaluate: } \int_0^{\pi/2} \cos(\pi \sin^2 x) dx$$

157. Let $M = \int_0^1 \frac{e^t}{1+t} dt$ and $N = \int_0^1 e^t \log(1+t) dx$. Prove that $M + N = e \times \log 2$.

158. Let $a_n = \int_0^{\pi/2} \frac{\cos^2 nx}{\sin x} dx$. Prove that $a_2 - a_1, a_3 - a_2, a_4 - a_3$ are in AP.

Q.No. 159 to 181 Evaluate each of the following integrals.

159. $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

160. $\int_0^{\pi/2} \frac{\sin^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x} dx$

161. $\int_0^{\pi/2} \log(\tan x) dx$

162. $\int_0^{\theta} \log(1 + \tan \theta \tan x) dx, \theta \in (0, \frac{\pi}{2})$

163. If $\int_0^{\pi} \log \sin x dx = k$, find the value of

$\int_0^{\pi/4} \log(1 + \tan x) dx.$

164. $\int_0^{\pi/2} \log\left(\frac{4 + 3\sin x}{4 + 3\cos x}\right) dx$

165. $\int_0^{\pi/2} \left(\frac{\sin x - \cos x}{1 + \sin x \cos x}\right) dx$

166. $\int_0^1 x(1-x)^n dx$

167. $\int_0^1 \left[\log\left(\frac{1}{x} - 1\right)\right] dx$

168. $\int_0^{\pi} \left(\frac{x \sin x}{1 + \cos^2 x}\right) dx$

169. $\int_0^{\pi} \left(\frac{x}{1 + \sin x}\right) dx.$

170. $\int_0^{\pi/2} \left(\frac{x \sin x \cos x}{\sin^4 x + \cos^4 x}\right) dx$

171. $\int_0^1 \cot^{-1}(1-x+x^2) dx$

172. $\int_0^1 \tan^{-1}(1-x+x^2) dx$

173. $\int_0^{\infty} \frac{dx}{(1+x^a)(1+x^2)}, (a > 0)$

174. $\int_0^{\pi/2} \frac{dx}{\sqrt{\tan x} - \sqrt{\cot x}}$

175. $\int_0^{\pi} \left(\frac{x}{a^2 \cos^2 x + b^2 \sin^2 x}\right) dx$

176. $\int_{1/2}^2 \frac{1}{x} \operatorname{cosec}^{101}\left(x - \frac{1}{x}\right) dx$

177. $\int_0^{\pi} \frac{2 \sin x \times \cos x}{2^{[\sin x]}} dx$

178. $\int_0^{\pi} x \log(\sin x) dx$

179. For $n > 0$, $\int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$

180. $\int_0^{\pi/2} \left(\frac{\cos x}{1 + \sin x + \cos x}\right) dx$

181. If $I = \int_0^{\pi/2} \frac{dx}{1 + \sin x}$, prove that

$\int_0^{\pi} \frac{x^2 \cos x}{(1 + \sin x)^2} dx = \pi I - \pi^2.$

182. If $\lim_{t \rightarrow a} \frac{\int_a^t f(t) dt - \left(\frac{t-a}{2}\right)(f(t) + f(a))}{(t-a)^3} = 0$, prove that

the maximum degree of $f(x)$ is 1.

Property V

183. Evaluate: $\int_a^b \left(\frac{f(x)}{f(x) + f(a+b-x)}\right) dx$

184. Evaluate: $\int_3^5 \left(\frac{[x^2]}{[x^2 - 16x + 64] + [x^2]}\right) dx$

185. Evaluate: $\int_{-2}^2 \left(\frac{dx}{1 + 3^x}\right)$

186. Evaluate: $\int_{5\pi/6}^{\pi/6} \left(\frac{dx}{1 + e^{\tan x}}\right)$

187. Evaluate: $\int_{-\pi}^{\pi} \left(\frac{\cos^2 x}{1 + a^x}\right) dx$

188. If $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \cos^{-1}\left(\frac{2x}{1+x^2}\right)\right) dx$

$$= k \int_0^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \right) dx, \text{ find } k$$

189. Let f be a positive function and

$$I_1 = \int_{1-k}^k f(x(1-x)) dx,$$

$$I_2 = \int_{1-k}^k f(x(1-x)) dx, \text{ where } 2k-1 > 0,$$

$$\text{find } \frac{I_1}{I_2}.$$

190. Evaluate: $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \left(\frac{x \sin(x^2)}{\sin x^2 + \sin(\ln(6-x^2))} \right) dx$

191. Evaluate: If $I_n = \int_{-\pi}^{\pi} \left(\frac{\sin nx}{(1+\pi^x) \sin x} \right) dx, n \in W,$

find the value of

$$(i) \sum_{m=1}^{10} I_{2m+1} \quad (ii) \sum_{m=1}^{10} I_{2m}$$

192. Evaluate: $\int_0^1 \left(\frac{x^4(1-x)^4}{1+x^2} \right) dx$

Q. Evaluate each of the following definite integrals.

193. If $f(a+b-x) = f(x)$, prove that

$$\int_a^b x f(x) = \frac{a+b}{2} \int_a^b f(x) dx$$

194. Prove that $\int_4^{10} \frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} dx = 3$

195. Prove that $\int_{\pi/4}^{3\pi/4} \left(\frac{dx}{1+\cos x} \right) = 2.$

196. Prove that $\int_{-\pi/2}^{\pi/2} \left(\frac{\cos x}{1+e^x} \right) dx = 1$

197. Prove that $\int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x} + 1} = \frac{\pi}{2}$

198. Prove that $\int_{50}^{100} \frac{\ln x}{\ln x + \ln(150-x)} dx = 25$

199. For any t in R and f be a continuous function.

$$\text{Let } I_1 = \int_{\sin^2 t}^{1+\cos^2 t} x f(x(2-x)) dx$$

$$\text{and } I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f(x(2-x)) dx,$$

$$\text{prove that } \frac{I_1}{I_2} = 1.$$

200. If $f(x) = \frac{e^x}{1+e^x}$ and let

$$I_1 = \int_{f(-a)}^{f(a)} x g\{x(1-x)\} dx \text{ and}$$

$$I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx,$$

$$\text{prove that } \frac{I_1}{I_2} = 2$$

201. Prove that $\int_0^{4\pi} \sin^{2013} x dx = 0$

202. Prove that $\int_0^{2\pi} \cos^{2013} x dx = 0$

Property VI

203. Evaluate: $\int_{-\pi/2}^{\pi/2} \sin^5 x dx$

204. Evaluate: $\int_{-\pi/4}^{\pi/4} x^3 \tan^4 x dx$

205. Evaluate: $\int_{-2}^2 \log \left(\frac{2-\sin x}{2+\sin x} \right) dx$

206. Evaluate: $\int_0^2 x(x-1)(x-2) dx$

207. Evaluate:

$$\int_{2010}^{2014} [(x-2010)(x-2011)(x-2012)(x-2013)(x-2014)] dx$$

208. Find the value of

$$\int_{-\pi/4}^{\pi/4} \left(\frac{x^{2013} - 3x^{2011} + 5x^{2009} - 7x^{2007} + 1007}{\cos^2 x} \right) dx$$

209. Let $f(x) = \frac{\sin^{2013} x}{x^{2014} - x^{2012} + 1}.$

$$\text{Find } \int_{-2010}^{2010} f(x) dx$$

210. Find the value of

$$\int_{-10}^{10} \left(\tan^{2013} x + \left(\frac{2013\sqrt{x}}{1+x^{2014}} \right) + 1007 \right) dx$$

211. Let $f(x) = x^3 \left(\frac{x^2}{e^{x^2} - 1} + \frac{x^2}{2} + 1 \right)$

$$\text{Find the value of } \int_{\sqrt{\ln(1/2)}}^{\sqrt{\ln(2)}} f(x) dx$$

212. Find the value of the integral

$$\int_{-2}^0 [x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)] dx$$

213. For any real number x , let $[x]$ denotes the largest integer less than or equal to x . Let f be a real function defined on $[-10, 10]$ by

$$f(x) = \begin{cases} x - [x] & : \text{if } [x] \text{ is odd} \\ 1 + [x] - x & : \text{if } [x] \text{ is even} \end{cases}$$

Find the value of

$$\frac{\pi^2}{10} \times \int_{10}^{10} f(x) \cos(\pi x) dx$$

214. Find the value of

$$\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5) dx.$$

215. Find the value of $\int_{-\pi}^{\pi} \left(\frac{2x(1 + \sin x)}{1 + \cos^2 x} \right) dx$

216. Find the value of

$$\int_{-\pi/3}^{\pi/3} \left(\frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} \right) dx$$

217. Find the value of

$$\int_{-1}^1 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right) + \tan^{-1}\left(\frac{1+x^2}{x}\right) \right) dx$$

Q. Evaluate each of the following integrals.

$$218. \int_{-\pi/2}^{\pi/2} [f(x) + f(-x)] \times [g(x) - g(-x)] dx$$

$$219. \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4} \cos^{-1}\left(\frac{2x}{1+x^2}\right) dx$$

$$220. \int_{-1}^1 \log\left(\frac{2-x}{2+x}\right) \sin^2 x dx$$

$$221. \int_{\log(1/3)}^{\log 3} \tan\left(\frac{e^x - 1}{e^x + 1}\right) dx$$

$$222. \int_{-\pi}^{\pi} \left(\frac{\cos^2 x}{1 + a^x} \right) dx$$

$$223. \int_{-1/2}^{1/2} \sqrt{\left[\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2 - 2\right]} dx.$$

$$224. \int_{-\pi/2}^{\pi/2} \sin\left(\log(x + \sqrt{x^2 + 1})\right) dx$$

$$225. \int_{-1/2}^{1/2} \left[[x] + \log\left(\frac{1-x}{1+x}\right) \right] dx$$

$$226. \int_{-1/2}^{1/2} \left(\cos x \times \log\left(\frac{1-x}{1+x}\right) \right) dx$$

$$227. \int_{-1}^1 \sin^{10} x \cos^{11} x dx$$

$$228. \int_{-1}^1 \sin^{2013} x dx$$

$$229. \int_2^4 (x-1)(x-2)(x-3)(x-4)(x-5) dx$$

$$230. \int_{2009}^{2011} [(x-2007)(x-2008)(x-2009)(x-2010)(x-2011)(x-2012)(x-2013)] dx$$

$$231. \text{ If } n \in N, \text{ prove that } \int_{-n}^n (-1)^{[x]} dx = 0$$

232. Prove that

$$\int_{-1}^1 \frac{\sin x - x^2}{3 - |x|} dx = 2 \int_0^1 \frac{-x^2}{3 - [x]} dx$$

$$233. \int_{-2}^2 [\sqrt{2012 + 2013x + 2014x^2} - \sqrt{2012 - 2013x + 2014x^2}] dx$$

$$234. \text{ If } f(x) = \begin{cases} e^{\cos x} \sin x & : -2 \leq x \leq 2 \\ 2 & : \text{otherwise} \end{cases},$$

$$\text{prove that } \int_{-2}^2 f(x) dx = 2.$$

$$235. \text{ Prove that } \int_{-1/2}^{1/2} \left| x \cos\left(\frac{\pi x}{2}\right) \right| dx = \frac{1}{\pi^2} (\pi\sqrt{2} + 4\sqrt{2} - 8)$$

236. Prove that

$$\int_0^{\pi} \left(\frac{x \sin 2x \cdot \sin\left(\frac{\pi}{2} \times \cos x\right)}{2x - \pi} \right) dx = \frac{8}{\pi^2}$$

Property VII

$$237. \text{ Evaluate: } \int_0^{\pi} \sin x dx$$

$$238. \text{ Evaluate: } \int_0^{\pi} \cos x dx$$

$$239. \int_0^{\pi} \cos^{2013} x dx$$

$$240. \text{ Evaluate: } \int_0^{\pi} (\sin^3 x + \cos^5 x + \tan^7 x) dx$$

241. Evaluate: $\int_0^{\pi/2} \log \sin x \, dx$

242. Evaluate: $\int_0^{\pi} x \log(\sin x) \, dx$

243. Evaluate: $\int_0^{\pi/2} \sin 2x \log(\tan x) \, dx$

244. Evaluate: $\int_0^{\pi} \left(\frac{x \, dx}{1 + \cos \alpha \cdot \sin x} \right)$

245. Evaluate: $\int_{-\pi/4}^{\pi/4} \log(\sin x + \cos x) \, dx$

246. Evaluate: $\int_0^{\pi} \frac{dx}{3\sin^2 x + 4\cos^2 x}$

247. Evaluate: $\int_0^{\pi} \left(\frac{x \sin x}{1 + \cos^2 x} \right) dx$

248. Evaluate: $\int_0^{\infty} \log \left(x + \frac{1}{x} \right) \frac{dx}{1 + x^2}$

249. Evaluate: If $I_1 = \int_0^1 x^{1006} (1 - x)^{1006} \, dx$

and $I_2 = \int_0^1 x^{1006} (1 - x^{2014})^{1006} \, dx$

Find the value of $2^{2014} \times \left(\frac{I_2}{I_1} \right)$

Q. Evaluate each of the following integrals.

250. $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) \, dx$

251. $\int_0^1 \frac{\log x}{\sqrt{1 - x^2}} \, dx$

252. $\int_0^{\pi} \log(1 - \cos x) \, dx$

253. $\int_0^{\infty} \log \left(x + \frac{1}{x} \right) \frac{dx}{1 + x^2}$

254. $\int_0^{\pi} \frac{x}{1 + \cos^2 x} \, dx$

Property VIII

255. Evaluate: $\int_0^{400\pi} \sqrt{1 - \cos^2 x} \, dx$

256. Evaluate: $\int_0^{1000} e^{x-[x]} \, dx$

257. Evaluate: $\int_0^{2000\pi} \left(\frac{dx}{1 + 5^{\sin x}} \right)$

258. Evaluate: $\int_0^{1007} \{2x\} \, dx$

259. Evaluate: $\int_0^{\pi} |\sin x + \cos x| \, dx$

260. Evaluate: $\int_0^{100\pi} (|\sin x| + |\cos x|) \, dx$

261. Evaluate: $\int_0^{100\pi} (|\sin x| - |\cos x|) \, dx$

262. Evaluate: $\int_{-503\pi}^{504\pi} |\cos x| \, dx$

263. Evaluate: $\int_0^{50\pi/3} \sqrt{\left(\frac{1 - \cos 2x}{2} \right)} \, dx$

264. Given $\int_0^1 \frac{\sin t}{1+t} \, dt = \alpha$,

prove that

$$\int_{4\pi-2}^{4\pi} \left(\frac{\sin\left(\frac{t}{2}\right)}{4\pi+2-t} \right) dt = -\alpha$$

265. If $\int_0^1 \frac{e^t}{(t+1)} \, dt = a$,

prove that

$$\int_{b-1}^b \left(\frac{e^{-t}}{(t-b-1)} \right) dt = -ae^{-b}.$$

266. If $F(x)$ be a periodic function such that

$$F(x) + F\left(\frac{1}{x} + 2\right) = 10,$$

find the value of $\int_0^{201} F(x) \, dx$

267. If $\int_0^{50\pi} (\sin^4 x + \cos^4 x) \, dx = k \int_0^{\pi/2} \left(\frac{3}{4} + \frac{1}{4} \cos 4x \right) \, dx$,

find k .

268. If $g(x) = \int_0^x \cos^4 t \, dt$, find the value of $g(x + \pi)$.

Q. Evaluate each of the following integrals

269. $\int_0^{2014\pi} \sqrt{1 - \cos 2x} \, dx$

270. $\int_0^{2012} e^{x-[x]} \, dx$

271. $\int_0^{4\pi} |\cos x| \, dx$

$$272. \int_0^{10\pi} |\sin x| dx$$

$$273. \int_0^{[x]} dx, \text{ where } [.] = \text{GIF}$$

$$274. \int_0^{4\pi} [\sin x + \cos x] dx$$

$$275. \int_1^{e^{37}} \frac{\pi \sin(\pi \log x)}{x} dx$$

$$276. \int_0^{\frac{32\pi}{3}} \sqrt{1 + \cos 2x} dx$$

$$278. \int_0^{10\pi/3} |\sin x| dx$$

$$280. \int_{-\pi}^{199\pi} \sqrt{\frac{1 - \cos 2x}{2}} dx$$

$$281. \int_{-20\pi}^{20\pi} |\cos x| dx$$

$$282. \int_0^{100} \sin(x - [x]) dx$$

$$283. \int_0^{2014\pi} (|\sin x| + |\cos x|) dx$$

$$284. \int_0^{2014\pi} (|\sin x| - |\cos x|) dx$$

$$285. \text{ Prove that } \int_0^{n\pi+V} |\sin x| dx = (2n+1) - \cos V$$

$$286. \text{ Prove that } \int_0^{2n\pi} [\sin x + \cos x] dx = -n\pi,$$

$$287. \text{ Prove that } \int_0^{2\pi} \sin^2(100x) dx = \pi, \text{ where } [.] = \text{GIF}$$

$$288. \text{ Prove that } \int_{10\pi+\pi/6}^{10\pi+\pi/3} (\sin x + \cos x) dx = \sqrt{3} - 1$$

$$289. \text{ Prove that } \int_{0.1 + (x - [x])^2}^5 \frac{\tan^{-1}(x - [x])}{dx} = \frac{5\pi^2}{32},$$

where $[.] = \text{GIF}$

$$290. \text{ Let } F(x) \text{ be a non-negative continuous function defined on } R \text{ such that } F(x) + F\left(x + \frac{1}{2}\right) = 3, \text{ prove}$$

$$\text{that } \int_0^{1500} F(x) dx = 2250.$$

$$291. \text{ If for every integer } n,$$

$$\int_0^{n+1} f(x) dx = n^2,$$

$$\text{prove that } \int_{-2}^4 f(x) dx = 19$$

$$292. \text{ If } \int_0^{\pi/2} \sin^m x \cos^m x dx = k \int_0^{\pi/2} \sin^m x dx,$$

$$\text{prove that } k = 2^{-m}.$$

$$293. \text{ If } M = \int_0^{\pi} f(\cos^2 x) dx \text{ and } N = \int_0^{3\pi} f(\cos^2 x) dx, \text{ then prove that } N = 3M.$$

Property IX

$$294. \text{ Let } \frac{d}{dx}(F(x)) = \frac{e^{\sin x}}{x}, x > 0.$$

$$\text{If } \int_1^4 \frac{2e^{\sin^2 x}}{x} dx = F(k) - F(1), \text{ find } k.$$

$$295. \text{ Find the equation of the tangent to the curve}$$

$$y = \int_{x^2}^{x^3} \frac{dt}{\sqrt{1+t^2}} \text{ at } x = 1$$

$$296. \text{ Find the interval where the function}$$

$$f(x) = \int_0^x e^t(t-1)(t-2) dt \text{ is strictly increasing.}$$

$$297. \text{ Find the point of maxima and minima for the}$$

$$\text{function } f(x) = \int_0^x t(t+1)(t-2) dt.$$

$$298. \text{ Find the point of inflection for the curve}$$

$$f(x) = e^t(t-1)^3(t-2)^2 dt$$

$$299. \text{ Prove that the equation}$$

$$y = \int_{1/10}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{1/10}^{\sin^2 x} \cos^{-1} \sqrt{t} dt$$

is a straight line parallel to x -axis, where

$$0 \leq x \leq \frac{\pi}{2}$$

$$300. \text{ If } F(x) = \int_x^{2x} \sqrt{5-3\sin^2 t} dt + \int_0^y \sin t dt, \text{ find } F'(x).$$

$$301. \text{ Let } f(x) = \int_1^x \sqrt{2-t^2} dt. \text{ Find the real roots of the}$$

$$\text{equation } x^2 - f'(x) = 0.$$

$$302. \text{ If } f(x) = \int_{1/x^2}^{x^2} \cos \sqrt{t} dt, \text{ the value of } f'(1) \text{ is ...}$$

303. If $\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x$, $x \in (0, \frac{\pi}{2})$, find the

value of $f\left(\frac{1}{\sqrt{3}}\right)$

304. Let $f: (0, \infty) \rightarrow R$ and $F(x) = \int_0^x f(t) dt$

If $F(x^2) = x^2(1+x)$, find the value of $f(4)$.

305. If $\int_0^{t^2} x f(x) dx = \frac{2}{5} t^5$, find the value of $f\left(\frac{4}{25}\right)$.

306. Prove that the value of

$$\int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt \text{ is } \frac{\pi}{4}.$$

307. If $\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$, prove that the value of $f(1)$ is $1/2$.

308. Prove that the value of

$$\lim_{x \rightarrow 0} \frac{1}{x} \left[\int_y^a e^{\sin^2 t} dt - \int_{x+y}^a e^{\sin^2 t} dt \right] = e^{\sin^2 y}.$$

Advance Properties of Definite Integral

309. (i) Prove that $F(x) = \int_0^x \log\left(\frac{1-t}{1+t}\right) dt$ is an even function

(ii) Prove that $F(x) = \int_0^x \left(\frac{t}{e^t - 1} + \frac{t}{2 + 1} \right) dt$ is an odd function.

(iii) Prove that the function

$f(t) = \int_0^t \sin(\log(\sqrt{x^2 + 1} + x)) dx$ is an even function.

(iv) Prove that the function

$f(x) = \int_0^x \left(\cos(\sin(\log(\sqrt{t^2 + 1} + t))) \right) dt$ is an odd function.

(v) Prove that the function $g(x) = \int_0^x f(t) dt$ is an even function, where $f(x)$ satisfies the relation $f(x+y) = f(x) + f(y)$.

310. Evaluate: $\int_0^{\pi} \sin^{100} x \cos^{99} x dx$

311. If $f(x) = \int_0^x \frac{e^t}{t} dt$, $x > 0$, prove that

$$\int_1^x \frac{e^t}{(t+\alpha)} dt = e^{-\alpha} [f(x+\alpha) - f(1+\alpha)].$$

312. Prove that $\int_{-4}^{-5} e^{(x+5)^2} dx + \int_{1/3}^{2/3} e^{9(x-2/3)^2} dx = 0$.

313. Prove that $\int_0^{\pi} \sin^{2014} x \cos^{2013} x dx = 0$.

314. Evaluate: $\int_0^1 |\cos \pi x| dx$

315. Prove that $\int_0^1 |\sin 2\pi x| dx = \frac{2}{\pi}$

316. Prove that

$$\int_{-3/4}^{3/2} |x \cos \pi x| dx = \frac{1}{\pi^2} \left\{ \frac{7\pi}{2} - \frac{3\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} - 2 \right\}$$

317. Evaluate: $\int_{-\pi/12}^{-5\pi/12} \left(\frac{\sin^{2014} x}{\sin^{2014} x + \cos^{2014} x} \right) dx$

318. Prove that $\int_{-\pi/3}^{-\pi/6} \frac{\sin^{2n} x}{\cos^{2n} x + \sin^{2n} x} dx = \frac{\pi}{12}$.

319. Prove that $\int_{-\pi/6}^{-\pi/3} \left(\frac{\sin^3 x + \cos^3 x}{\sin^{10} x + \cos^{10} x} \right) dx = 0$

320. Given $\int_0^1 \frac{\sin t}{1+t} dt = \alpha$,

prove that

$$\int_{4\pi-2}^{4\pi} \frac{\sin(t/2)}{4\pi + 2 - t} dt = -\alpha$$

321. Find the value of $\int_0^{\pi} \left(\frac{|\sin(2x)|}{|\sin x| + |\cos x|} \right) dx$

322. If $\int_a^b \left(\frac{\sin(x-a) - \cos(x-a)}{\sin(b-x) - \cos(b-x)} \right) dx = m \int_a^b \left(\frac{\sin(b-x) - \cos(b-x)}{\sin(x-a) - \cos(x-a)} \right) dx$,

find the value of m .

323. If $\int_0^1 \frac{e^t}{(t+1)} dt = a$,

prove that

$$\int_{b-1}^b \frac{e^{-t}}{(t-b-1)} dt = -ae^{-b}.$$

324. Find the value of

$$\int_0^{\pi} \frac{x \sin 2x \sin\left(\frac{\pi}{2} \cos x\right)}{2x - \pi} dx = \frac{8}{\pi^2}$$

325. Let a, b and c be three non-zero real numbers such that $\int_0^1 (1 + \cos^8 x)(ax^2 + bx + c) dx$. Prove that the equation has at least one root in $(1, 2)$.

326. Estimate the absolute value of the integral $\int_{10}^{19} \frac{\sin x}{1+x^8} dx$.

327. Estimate the absolute value of the integral $\int_5^{10} \frac{\cos x}{1+x^4} dx$.

328. Prove that $4 \leq \int_1^3 \sqrt{x^2 + 3} dx \leq 4\sqrt{3}$.

329. Prove that the value of the integral $\int_0^1 e^{x^2} dx$ lies in between $[1, e]$

330. Prove that the value of the integral $\int_0^1 \frac{x}{x^3 + 16} dx$ lies in between $[0, 1/17]$

331. Prove that $4 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30}$

332. Prove that $\frac{\pi}{6} < \int_0^1 \left(\frac{dx}{\sqrt{4-x^2-x^3}} \right) < \frac{\pi}{4\sqrt{2}}$

333. Prove that $1 \leq \int_0^1 \sqrt{1+x^3} dx \leq \frac{5}{4}$

334. Prove that $\frac{1}{17} \leq \int_1^2 \frac{dx}{1+x^4} \leq \frac{7}{24}$

335. Evaluate $\int_0^\infty \left(\frac{\tan^{-1} ax - \tan^{-1} x}{x} \right) dx$

336. Evaluate $\int_0^{\pi/2} \ln \left(\frac{1+a \sin x}{1-a \sin x} \right) \frac{dx}{\sin x}$, ($|a| < 1$)

337. Prove that $I(b) = \int_0^1 \frac{x^b - 1}{\log x} dx = \log(b+1)$

338. Let $f(x)$ be a continuous function for all x such that

$$(f(x))^2 = \int_0^x f(t) \cdot \frac{2 \sec^2 t}{4 + \tan t} dt \text{ and } f(0) = 0,$$

$$\text{prove that } f\left(\frac{\pi}{4}\right) = \log\left(\frac{5}{4}\right).$$

339. Prove that

$$f(x) = \int_0^{\pi/2} \frac{\log(1+x \sin^2 \theta)}{\sin^2 \theta} dx, x \geq 0 = \pi(\sqrt{1+x} - 1)$$

340. Prove that

$$\int_0^1 \left(\frac{x^{\cos \alpha} - 1}{\log_e x} \right) dx = \log |1 + \cos \alpha|,$$

$[\alpha \neq (2n+1)\pi]$, where α is a parameter

341. Prove that

$$\int_0^\pi \ln(1 + b \cos x) dx = \pi \ln \left(\frac{1 + \sqrt{1-b^2}}{2} \right).$$

342. Find the maximum value of

$$\int_0^2 \sqrt{(1+x)(1+x^4)} dx.$$

343. Prove that the maximum value of

$$\int_0^1 \sqrt{(1+x)(1+x^3)} dx \text{ is } \sqrt{\frac{15}{8}}.$$

344. Prove that the value of

$$\int_a^{a+\frac{\pi}{2}} (\sin^4 x + \cos^4 x) dx \text{ is independent of } a.$$

345. Find the mean value of $f(x) = \sin^2 x$ over $[0, \pi]$.

Q. Find the mean value.

346. $f(x) = \frac{1}{e^x + 1}$ on $[0, 2]$

347. $f(x) = \sqrt[3]{x}$ on $[0, 1]$

348. $f(x) \sin^3 x$ on $[0, 2\pi]$

349. $f(x) = \frac{1}{x+x^2}$ on $[0, 1]$

350. $f(x) = \frac{e^x}{1+e^x}$ on $[0, 1]$

Gamma and Beta Functions

351. Evaluate: $\int_0^1 \log x dx$

352. Evaluate: $\int_0^\infty e^{-x} x^6 dx$

353. Evaluate: $\int_0^1 \left(\log \left(\frac{1}{x} \right) \right)^{10} dx$

Q. Evaluate each of the following integrals.

354. $\int_0^\infty e^{-x} x^3 dx$

355. $\int_0^1 \left(\log \left(\frac{1}{x} \right) \right)^{n-1} dx$

356. Evaluate: $\int_0^1 x^4 \sqrt{1-x^2} dx$

357. Evaluate: $\int_0^{\pi/2} \sin^8 x \cdot \cos^4 x dx$

Q.No. 358 to 362, Evaluate each of the following integrals.

$$358. \int_0^1 x^6 \sqrt{1-x^2} dx$$

$$359. \int_0^{\pi/2} \sin^4 x \cos^6 x dx$$

$$360. \int_0^1 x^{10} \sqrt{1-x^2} dx$$

$$361. \int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}} \times \int_0^{\pi/2} \sqrt{\sin x} dx$$

$$362. \int_0^{\infty} e^{-a^2 x^2} dx$$

$$363. \text{ Evaluate: } \int_0^{\pi/2} \sin^7 x \cdot \cos^5 x dx$$

$$364. \text{ Evaluate: } \int_0^{\pi/2} \sin^{10} x dx$$

$$365. \text{ Evaluate: } \int_0^{\pi/2} \cos^7 x dx$$

$$366. \text{ Evaluate: } \int_0^{\pi/4} 8 \cos^4 x \cdot \sin^4 x dx$$

$$367. \text{ Evaluate: } \int_0^{\pi/2} \sin^6 x \cdot \cos^5 x dx$$

$$368. \text{ Evaluate: } \int_0^{\pi/2} \sin^5 x \cdot \cos^3 x dx$$

Q.No. 369 to 373, Evaluate each of the following integrals.

$$369. \int_0^{\pi/2} \sin^6 x \cdot \cos^4 x dx$$

$$370. \int_0^{\pi/2} \sin^7 x \cdot \cos x dx$$

$$371. \int_0^{\pi/2} \sin^7 x \cdot \cos^5 x dx$$

$$372. \int_0^{\pi/2} \sin^{11} x dx$$

$$373. \int_0^{\pi/2} \cos^8 x dx$$

HINTS AND SOLUTIONS

Level I

1. We have,

$$\begin{aligned} \int_0^{\pi/2} \sin^2 x \, dx &= \frac{1}{2} \int_0^{\pi/2} \sin^2 x \, dx \\ &= \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2x) \, dx \\ &= \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) \Big|_0^{\pi/2} \\ &= \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) \\ &= \frac{\pi}{4} \end{aligned}$$

2. We have,

$$\begin{aligned} \int_0^{\pi/4} \frac{dx}{1 + \sin x} &= \int_0^{\pi/4} \frac{dx}{1 + \cos\left(\frac{\pi}{2} - x\right)} \\ &= \int_0^{\pi/4} \frac{dx}{2 \cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)} \\ &= \frac{1}{2} \int_0^{\pi/4} \sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) dx \\ &= \frac{1}{2} \times \left[-2 \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \right]_0^{\pi/4} \\ &= -\left[\tan\left(\frac{\pi}{4} - \frac{\pi}{8}\right) - 1 \right] \\ &= -\left[\tan\left(\frac{\pi}{8}\right) - 1 \right] \\ &= -[(\sqrt{2} - 1) - 1] \\ &= 2 - \sqrt{2} \end{aligned}$$

3. We have,

$$\begin{aligned} \int_0^2 \frac{dx}{\sqrt{x+2} + \sqrt{x}} &= \int_0^2 \frac{\sqrt{x+2} - \sqrt{x}}{(\sqrt{x+2} + \sqrt{x})(\sqrt{x+2} - \sqrt{x})} dx \\ &= \int_0^2 \frac{(\sqrt{x+2} - \sqrt{x})}{x+2-x} dx \\ &= \frac{1}{2} \int_0^2 (\sqrt{x+2} - \sqrt{x}) dx \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left[\frac{2}{3} (x+2)^{3/2} - \frac{2}{3} x^{3/2} \right]_0^2 \\ &= \frac{1}{3} \left(8 - \frac{2}{3} 2^{3/2} \right) \end{aligned}$$

4. We have,

$$\begin{aligned} \int_0^{\pi/2} \frac{\sin x}{\sin x - \cos x} dx &= \frac{1}{2} \int_0^{\pi/2} \frac{2 \sin x}{\sin x - \cos x} dx \\ &= \frac{1}{2} \int_0^{\pi/2} \frac{(\sin x - \cos x) + (\sin x + \cos x)}{\sin x - \cos x} dx \\ &= \frac{1}{2} \int_0^{\pi/2} \left(1 + \frac{(\sin x + \cos x)}{(\sin x - \cos x)} \right) dx \\ &= \frac{1}{2} (x + \log |\sin x - \cos x|) \Big|_0^{\pi/2} \\ &= \frac{\pi}{4} \end{aligned}$$

5. We have,

$$\begin{aligned} \int_1^2 \frac{dx}{x(1+x^4)} &= \int_1^2 \frac{x^3 dx}{x^4(1+x^4)} \\ &= \frac{1}{4} \log \left| \frac{x^4}{x^4+1} \right|_1^2 \\ &= \frac{1}{4} \left[\log\left(\frac{16}{17}\right) - \log\left(\frac{1}{2}\right) \right] \\ &= \frac{1}{4} \log\left(\frac{32}{17}\right) \end{aligned}$$

6. We have,

$$\begin{aligned} \int_0^1 \frac{dx}{x^2 + x + 1} &= \int_0^1 \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) \Big|_0^1 \\ &= \frac{2}{\sqrt{3}} \left[\tan^{-1}(\sqrt{3}) - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \right] \\ &= \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \\ &= \frac{\pi}{3\sqrt{3}} \end{aligned}$$

7. We have,

$$\begin{aligned} \int_0^1 x e^x dx &= x \int e^x dx \Big|_0^1 - \int_0^1 e^x dx \\ &= (x e^x - e^x) \Big|_0^1 \\ &= (e - e) - (0 - 1) = 1 \end{aligned}$$

8. We have,

$$\begin{aligned} \int_0^{\pi/2} \frac{dx}{\sin x + \cos x} &= \sqrt{2} \int_0^{\pi/2} \frac{dx}{\left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x\right)} \\ &= \sqrt{2} \int_0^{\pi/2} \frac{dx}{\sin x \left(x + \frac{\pi}{4}\right)} \\ &= \sqrt{2} \int_0^{\pi/2} \operatorname{cosec} \left(x + \frac{\pi}{4}\right) dx \\ &= \sqrt{2} \log \left| \left(\frac{x}{2} + \frac{\pi}{8}\right) \right|_0^{\pi/2} \\ &= \sqrt{2} \log 3 \end{aligned}$$

9. We have

$$\begin{aligned} \int_0^{\pi/2} \frac{dx}{(3 \sin^2 x + 4 \cos^2 x)} &= \int_0^{\pi/2} \frac{\sec^2 x dx}{3 \tan^2 x + 4} \\ &= \frac{1}{3} \int_0^{\pi/2} \frac{\sec^2 x dx}{\tan^2 x + \left(\frac{2}{\sqrt{3}}\right)^2} \\ &= \frac{1}{3} \times \frac{\sqrt{3}}{2} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{2}\right) \Big|_0^{\pi/2} \\ &= \frac{1}{2\sqrt{3}} \left(\frac{\pi}{2} - 0\right) \\ &= \frac{\pi}{4\sqrt{3}} \end{aligned}$$

10. We have,

$$\begin{aligned} \int_0^1 (1-x)x^n dx &= \int_0^1 (x^n - x^{n+1}) dx \\ &= \left(\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2}\right) \Big|_0^1 \\ &= \frac{1}{n+1} - \frac{1}{n+2} \\ &= \frac{1}{(n+1)(n+2)} \end{aligned}$$

11. We have,

$$\begin{aligned} I_n &= \int_0^1 \tan^n x dx \\ &= \int_0^1 \tan^{n-2} x \cdot \tan^2 x dx \\ &= \int_0^1 \tan^{n-2} x (\sec^2 x - 1) dx \\ &= \int_0^1 \tan^{n-2} x \cdot \sec^2 x dx - \int_0^1 \tan^{n-2} x dx \\ &= \left(\frac{\tan^{n-1} x}{n-1}\right) \Big|_0^1 - I_{n-2} \\ \Rightarrow I_n + I_{n-2} &= \frac{1}{n-1} \end{aligned}$$

Hence, the result.

12. We have,

$$\begin{aligned} \sin nx - \sin(n-2)x &= 2 \cos(n-1)x \sin x \\ \Rightarrow \frac{\sin nx}{\sin x} - \frac{\sin(n-2)x}{\sin x} &= 2 \cos(n-1)x \\ \Rightarrow \frac{\sin nx}{\sin x} &= 2 \cos(n-1)x + \frac{\sin(n-2)x}{\sin x} \\ \Rightarrow \int_0^{\pi} \frac{\sin nx}{\sin x} dx &= \int_0^{\pi} 2 \cos(n-1)x dx \\ &\quad + \int_0^{\pi} \frac{\sin(n-2)x}{\sin x} dx \\ \Rightarrow I_n &= 0 + I_{n-2} = I_{n-4} = \dots = I_2 \text{ or } I_1 \end{aligned}$$

according as n is even or odd.

Case I: When n is even

$$I_n = I_2 = \int_0^{\pi} \left(\frac{\sin 2x}{\sin x}\right) dx = 0$$

Case II: When n is odd

$$I_n = I_1 = \int_0^{\pi} \left(\frac{\sin 2x}{\sin x}\right) dx = x \Big|_0^{\pi} = \pi$$

13. Let $I_n = \int_0^{\pi/2} \left(\frac{\sin n\theta}{\sin \theta}\right)^2 d\theta$

Now,

$$\begin{aligned} I_n - I_{n-1} &= \int_0^{\pi/2} \left(\frac{\sin^2 n\theta - \sin^2(n-1)\theta}{\sin^2 \theta}\right) d\theta \\ &= \int_0^{\pi/2} \left(\frac{\sin(2n-1)\sin \theta}{\sin^2 \theta}\right) d\theta \\ &= \int_0^{\pi/2} \left(\frac{\sin(2n-1)\theta}{\sin \theta}\right) d\theta \\ &= \pi \end{aligned}$$

$$\begin{aligned}
 14. \int_0^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)} \\
 &= \frac{1}{(b^2 - a^2)} \int_0^{\infty} \left(\frac{1}{x^2 + a^2} - \frac{1}{x^2 + b^2} \right) dx \\
 &= \frac{1}{(b^2 - a^2)} \left(\frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) - \frac{1}{b} \tan^{-1} \left(\frac{x}{b} \right) \right) \Big|_0^{\infty} \\
 &= \frac{1}{(b^2 - a^2)} \left(\frac{\pi}{2a} - \frac{\pi}{2b} \right) \\
 &= \frac{\pi}{2(b^2 - a^2)} \left(\frac{1}{a} - \frac{1}{b} \right) \\
 &= \frac{\pi}{2(b^2 - a^2)} \left(\frac{b - a}{ab} \right) \\
 &= \frac{\pi}{2(a + b)ab}
 \end{aligned}$$

$$\begin{aligned}
 15. \int_0^1 x(1 - x)^5 dx \\
 &= \int_0^1 x(1 - 5x + 10x^2 - 10x^3 + 5x^4 - x^5) dx \\
 &= \int_0^1 (x - 5x^2 + 10x^3 - 10x^4 + 5x^5 - x^6) dx \\
 &= \left(\frac{x^2}{2} - \frac{5x^3}{3} + \frac{10x^4}{4} - \frac{10x^5}{5} + \frac{5x^6}{6} - \frac{x^7}{7} \right) \Big|_0^1 \\
 &= \frac{1}{2} - \frac{5}{3} + \frac{10}{4} - \frac{10}{5} + \frac{5}{6} - \frac{1}{7} \\
 &= \frac{1}{50}
 \end{aligned}$$

$$\begin{aligned}
 16. \int_0^{\pi/2} \left(\frac{\sin 2x}{\sin^4 x + \cos^4 x} \right) dx \\
 &= \int_0^{\pi/2} \left(\frac{2 \sin x \cos x}{\sin^4 x + \cos^4 x} \right) dx \\
 &= \int_0^{\pi/2} \left(\frac{2 \tan x \sec^2 x}{\tan^4 x + 1} \right) dx \\
 &= \left[\tan^{-1}(\tan x) \right]_0^{\pi/2} \\
 &= \left(\frac{\pi}{2} - 0 \right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 17. \int_1^2 \frac{dx}{x(x^3 + 1)} \\
 &= \int_1^2 \frac{x^2 dx}{x^3(x^3 + 1)}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \left| \log \left| \frac{x^3}{x^3 + 1} \right| \right|_1^2 \\
 &= \frac{1}{2} \left| \log \left| \frac{16}{9} \right| \right| \\
 &= \log \left(\frac{4}{3} \right)
 \end{aligned}$$

18. We have,

$$\int_2^4 \frac{x}{\sqrt{x-2}} dx$$

$$\text{Let } x - 2 = t^2 \Rightarrow dx = 2t dt$$

$$\text{When } x = 0, \text{ then } t = 0$$

$$\text{and } x = 4, \text{ then } t = \sqrt{2}$$

Thus, the given integral reduces to

$$\begin{aligned}
 &\int_0^{\sqrt{2}} \left(\frac{t^2 + 2}{t} \right) 2t dt \\
 &= 2 \left(\frac{t^3}{3} + 2t \right) \Big|_0^{\sqrt{2}} \\
 &= 2 \left(\frac{2\sqrt{2}}{3} + 2\sqrt{2} \right) \\
 &= \frac{16\sqrt{2}}{3}
 \end{aligned}$$

19. We have,

$$\int_0^1 x(1 - x)^{2012} dx$$

$$\text{Let } 1 - x = t \Rightarrow dx = -dt$$

$$\text{When } x = 0, \text{ then } t = 1$$

$$\text{and } x = 1, \text{ then } t = 0$$

Thus, the given integral reduces to

$$\begin{aligned}
 &-\int_1^0 (1 - t)t^{2012} dt \\
 &= \int_0^1 (t^{2013} - t^{2012}) dt \\
 &= \left(\frac{t^{2014}}{2014} - \frac{t^{2013}}{2013} \right) \Big|_0^1 \\
 &= \frac{1}{2014} - \frac{1}{2013} \\
 &= -\frac{1}{2013 \times 2014}
 \end{aligned}$$

20. We have,

$$\begin{aligned}
 &\int_1^2 \frac{dx}{x(x^3 + 1)} \\
 &= \int_1^2 \frac{dx}{x(x^3 + 1)} \\
 &= \int_1^2 \frac{x^2 dx}{x^3(x^3 + 1)}
 \end{aligned}$$

$$\text{Let } x^3 = t \Rightarrow 3x^2 dx = dt$$

$$\text{When } x = 0, \text{ then } t = 0$$

$$\text{and } x = 1, \text{ then } t = 1$$

$$= \frac{1}{3} \int_1^2 \frac{dt}{t(t+1)}$$

$$= \frac{1}{3} \log \left| \frac{t}{t+1} \right|_1^2$$

$$= \frac{1}{3} \log \left| \frac{4}{3} \right|$$

21. We have,

$$\int_0^{\pi/2} \frac{dx}{(2 + \cos x)}$$

$$= \int_0^{\pi/2} \frac{dx}{2 + \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}}$$

$$= \int_0^{\pi/2} \left(\frac{\sec^2(x/2)}{3 + \tan^2(x/2)} \right) dx$$

$$\text{Let } \tan\left(\frac{x}{2}\right) = t \Rightarrow \sec^2\left(\frac{x}{2}\right) dx = 2dt$$

$$\text{When } x = \pi/2, \text{ then } t = 1$$

$$\text{and } x = 0, \text{ then } t = 0.$$

$$= 2 \int_0^1 \left(\frac{dt}{t^2 + (\sqrt{3})^2} \right)$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t}{\sqrt{3}} \right) \Big|_0^1$$

$$= \frac{\pi}{3\sqrt{3}}$$

22. We have,

$$\int_0^{\pi/2} \frac{dx}{(2\sin^2 x + \cos^2 x)}$$

$$= \int_0^{\pi/2} \frac{dx}{(2\sin^2 x + \cos^2 x)}$$

$$= \int_0^{\pi/2} \frac{\sec^2 x dx}{(2\tan^2 x + 1)}$$

$$\text{Let } \tan x = t \Rightarrow \sec^2 x dx = dt$$

$$\text{When } x = 0, \text{ then } t = 0$$

$$\text{and } x = \frac{\pi}{2}, \text{ then } t = \infty$$

$$= \frac{1}{2} \int_0^{\infty} \frac{dt}{t^2 + (1/\sqrt{2})^2}$$

$$= \frac{1}{2} \times \frac{\sqrt{2}}{1} \tan^{-1}(t\sqrt{2}) \Big|_0^{\infty}$$

$$= \frac{\pi}{2\sqrt{2}}$$

23. We have,

$$\int_0^{\pi/2} \left(\frac{\cos x}{6 - 5\sin x + \sin^2 x} \right) dx$$

$$\text{Let } \sin x = t \Rightarrow \cos x dx = dt$$

$$\text{When } x = 0, \text{ then } t = 0$$

$$\text{and } x = \pi/2, \text{ then } t = 1$$

$$= \int_0^1 \frac{dt}{t^2 - 5t + 6}$$

$$= \int_0^1 \frac{dt}{(t-2)(t-3)}$$

$$= \int_0^1 \left(\frac{1}{t-3} - \frac{1}{t-2} \right) dt$$

$$= \log \left| \frac{t-3}{t-2} \right|_0^1 = \log \left| \frac{4}{3} \right|$$

24. We have,

$$\int_0^{\pi/2} \left(\frac{\sin 2x}{\sin^4 x + \cos^4 x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{2\sin x \cos x}{\sin^4 x + \cos^4 x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{2\tan x \cdot \sec^2 x}{\tan^4 x + 1} \right) dx$$

$$\text{Let } \tan^2 x = t \Rightarrow 2\tan x \sec^2 x dx = dt$$

$$\text{When } x = 0, \text{ then } t = 0$$

$$\text{and } x = \pi/2, \text{ then } t = \infty$$

$$= \int_0^{\infty} \frac{dt}{t^2 + 1}$$

$$= \tan^{-1}(t) \Big|_0^{\infty}$$

$$= \pi/2$$

$$25. \text{ We have, } \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{16 + \sin 2x} \right) dx$$

$$\text{Let } \sin x - \cos x = t$$

$$\Rightarrow (\cos x + \sin x) dx = dt$$

$$\text{When } x = 0, \text{ then } t = -1$$

$$\text{and } x = \pi/4, \text{ then } t = 0$$

$$\text{Also, } t^2 = 1 - \sin 2x \Rightarrow \sin 2x = 1 - t^2$$

$$\begin{aligned}
 &= \int_{-1}^0 \frac{dt}{16 + 1 - t^2} \\
 &= - \int_{-1}^0 \frac{dt}{t^2 - (\sqrt{17})^2} \\
 &= -\frac{1}{\sqrt{17}} \log \left| \frac{t - \sqrt{17}}{t + \sqrt{17}} \right|_{-1}^0 \\
 &= \frac{1}{2\sqrt{17}} \log \left| \frac{9 - 2\sqrt{7}}{4} \right|
 \end{aligned}$$

26. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \left(\frac{1}{(\sqrt{\cos x} + \sqrt{\sin x})^4} \right) dx \\
 &= \int_0^{\pi/2} \frac{dx}{(\sqrt{\cos x} + \sqrt{\sin x})^4} \\
 &= \int_0^{\pi/2} \frac{\sec^2 x dx}{(1 + \sqrt{\tan x})^4} \\
 \text{Let } \tan x &= t^2 \Rightarrow \sec^2 x dx = 2t dt \\
 \text{When } x &= 0, \text{ then } t = 0 \\
 \text{and } x &= \frac{\pi}{2}, \text{ then } t = \infty \\
 &= \int_0^{\infty} \frac{dt}{(t + 1)^4} \\
 &= -\frac{1}{3(t + 1)^3} \Big|_0^{\infty} \\
 &= \frac{1}{3}
 \end{aligned}$$

27. We have,

$$\begin{aligned}
 &\int_0^2 \frac{dx}{\sqrt{x+1} + \sqrt{(x+1)^3}} \\
 \text{Let } (x+1) &= t^2 \Rightarrow dx = 2t dt \\
 \text{When } x &= 0, \text{ then } t = 0 \\
 \text{and } x &= 2, \text{ then } t = \sqrt{3} \\
 &= \int_0^{\sqrt{3}} \frac{2t dt}{t + t^3} \\
 &= \int_0^{\sqrt{3}} \frac{2 dt}{t^2 + 1} \\
 &= 2 \tan^{-1}(t) \Big|_0^{\sqrt{3}} = 2 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = \frac{\pi}{6}
 \end{aligned}$$

28. We have,

$$\int_0^{\pi/2} \left(\frac{\cos^2 x}{4 \sin^2 x + \cos^2 x} \right) dx$$

$$\begin{aligned}
 &= \int_0^{\pi/2} \frac{dx}{(4 \tan^2 x + 1)} \\
 &= \int_0^{\pi/2} \frac{\sec^2 x dx}{(1 + \tan^2 x)(4 \tan^2 x + 1)} \\
 &= \int_0^{\infty} \frac{dt}{(t^2 + 1)(4t^2 + 1)} \\
 &= \frac{1}{3} \int_0^{\infty} \left(\frac{4}{4t^2 + 1} - \frac{1}{t^2 + 1} \right) dt \\
 &= \frac{1}{3} \int_0^{\infty} \left(\frac{1}{t^2 + (1/2)^2} - \frac{1}{t^2 + 1} \right) dt \\
 &= \frac{1}{3} (2 \tan^{-1}(2t) - \tan^{-1}(t)) \Big|_0^{\infty} \\
 &= \frac{1}{3} \left(2 \frac{\pi}{2} - \frac{\pi}{2} \right) \\
 &= \frac{\pi}{6}
 \end{aligned}$$

29. We have,

$$\begin{aligned}
 &\int_1^2 \frac{dx}{(2x+1)\sqrt{x^2+x}} \\
 \text{Let } (2x+1) &= \frac{1}{t} \Rightarrow 2 dx = -\frac{1}{t^2} dt \\
 \text{When } x &= 1, \text{ then } t = 1/3 \\
 \text{and } x &= 2, \text{ then } t = 1/5 \\
 \text{Also, } x^2 + x &= \frac{1}{4} \left(\frac{1-t}{t} \right)^2 + \frac{1}{2} \left(\frac{1-t}{t} \right) \\
 &= \frac{1}{4} \left(\frac{1-t}{t} \right) \left(\frac{1+t}{t} \right) = \frac{1}{4} \left(\frac{1-t^2}{t^2} \right) \\
 &= -\int_{1/3}^{1/5} \frac{dt}{\sqrt{1-t^2}} \\
 &= \cos^{-1}(t) \Big|_{1/3}^{1/5} \\
 &= \left(\cos^{-1} \left(\frac{1}{5} \right) - \cos^{-1} \left(\frac{1}{3} \right) \right)
 \end{aligned}$$

30. We have,

$$\begin{aligned}
 &\int_0^4 \frac{dx}{x + \sqrt{x}} \\
 \text{Let } x &= t^2 \\
 \Rightarrow dx &= 2t dt \\
 \text{When } x &= 0, t = 0 \\
 \text{and } x &= 4, t = 2 \\
 &= \int_0^2 \frac{2t dt}{t^2 + 1} \\
 &= \int_0^2 \frac{2 dt}{t + 1}
 \end{aligned}$$

$$= 2(\log t + 1)|_0^2$$

$$= 2\log 3$$

31. We have,

$$\int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} dx$$

Put $\tan^{-1}(x) = t$

$$\Rightarrow \frac{dx}{1+x^2} = dt$$

$$= \int_0^{\pi/4} \frac{t \tan t}{\sec t} dt$$

$$= \int_0^{\pi/4} (t \sin t) dt$$

$$= (t \int \sin t dt)_0^{\pi/4} + \int_0^{\pi/4} \cos t dt$$

$$= (-t \cos t + \sin t)|_0^{\pi/4}$$

$$= -\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \left(1 - \frac{\pi}{4}\right)$$

32. We have,

$$\int_0^{\pi/2} \frac{\cos \theta}{(1 + \sin \theta)(2 + \sin \theta)} d\theta$$

$$= \int_0^1 \frac{dt}{(t+1)(t+2)}, \quad \text{Let } t = \sin \theta$$

$$\Rightarrow dt = \cos \theta d\theta$$

$$= \int_0^1 \left(\frac{1}{t+1} - \frac{1}{t+2} \right) dt$$

$$= \left(\log \left| \frac{t+1}{t+2} \right| \right)_0^1$$

$$= \log \left(\frac{4}{3} \right)$$

33. We have,

$$\int_0^{\pi/2} (\sqrt{\tan x} + \sqrt{\cot x}) dx$$

$$= \int_0^{\pi/2} \left(\frac{\tan x + 1}{\sqrt{\tan x}} \right) dx$$

$$= 2 \int_0^{\infty} \left(\frac{t^2 + 1}{t^4 + 1} \right) dt, \quad \text{Let } \tan x = t^2$$

$$\Rightarrow \sec^2 x dx = 2t dt$$

$$= 2 \int_0^{\infty} \left(\frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} \right) dt$$

$$= 2 \int_0^{\infty} \left(\frac{1 + \frac{1}{t^2}}{\left(t - \frac{1}{t}\right)^2 + 2} \right) dt$$

$$= \frac{2}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} \left(t - \frac{1}{t} \right) \right) \Big|_0^{\infty}$$

$$= \frac{2}{\sqrt{2}} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right]$$

$$= \pi\sqrt{2}$$

34. We have,

$$\int_0^{\pi/2} \frac{dx}{4 \sin^2 x + 5 \cos^2 x}$$

$$= \int_0^{\pi/2} \left(\frac{\sec^2 x dx}{4 \tan^2 x + 5} \right)$$

$$= \int_0^{\infty} \frac{dt}{4t^2 + 5}, \quad \text{Let } t = \tan x$$

$$= \frac{1}{4} \int_0^{\infty} \frac{dt}{t^2 + \left(\frac{\sqrt{5}}{2} \right)^2}$$

$$= \frac{1}{4} \times \frac{2}{\sqrt{5}} \tan^{-1} \left(\frac{2t}{\sqrt{5}} \right) \Big|_0^{\infty}$$

$$= \frac{1}{2\sqrt{5}} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi}{4\sqrt{5}}$$

35. We have,

$$\int_0^{\pi/4} \left(\frac{\sin x + \cos x}{9 + 16 \sin 2x} \right) dx$$

$$= \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{25 - 16(\sin x - \cos x)^2} \right) dx$$

$$= \int_{-1}^0 \frac{dt}{25 - 16t^2} \quad \text{Let } \sin x - \cos x = t$$

$$(\sin x + \cos x) dx = dt$$

$$= -\frac{1}{16} \int_{-1}^0 \frac{dt}{t^2 - \left(\frac{5}{4} \right)^2}$$

$$= -\frac{1}{16} \times \frac{1}{2 \left(\frac{5}{4} \right)} \log \left| \frac{5t - 4}{5t + 4} \right|_{-1}^0$$

$$= \frac{1}{40} \log 9$$

36. We have,

$$\int_0^{\pi/2} \left(\frac{\sin 2x}{\sin^4 x + \cos^4 x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{2 \sin x \cos x}{\sin^4 x + \cos^4 x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{2 \tan x \sec^2 x}{\tan^4 x + 1} \right) dx$$

$$= \int_0^{\infty} \frac{dt}{t^2 + 1},$$

$$\text{Let } \tan^2 x = t$$

$$= \tan^{-1}(t) \Big|_0^{\infty}$$

$$= \frac{\pi}{2}$$

$$37. \text{ We have, } \int_0^1 \left(\sqrt{\frac{1-x}{1+x}} \right) dx.$$

$$= \int_0^1 \left(\frac{1-x}{\sqrt{1+x^2}} \right) dx$$

$$= \int_0^1 \left(\frac{dx}{\sqrt{1-x^2}} \right) - \int_0^1 \left(\frac{x dx}{1-x^2} \right)$$

$$= \int_0^1 \left(\frac{dx}{\sqrt{1-x^2}} \right) + \int_0^1 dt, \quad \text{Let } (1-x^2) = t^2$$

$$= \sin^{-1}(x) \Big|_0^1 + (t) \Big|_0^1$$

$$= \frac{\pi}{2} - 1$$

$$38. \text{ We have,}$$

$$\int_{-1}^4 (2x - 3) dx$$

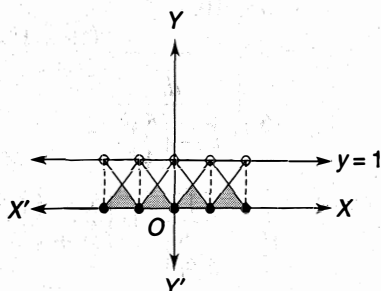
$$= (x^2 - 3x) \Big|_{-1}^4$$

$$= (16 - 12) - (1 + 3)$$

$$= 0$$

$$39. \text{ We have,}$$

$$\int_{-2}^2 f(x) dx$$

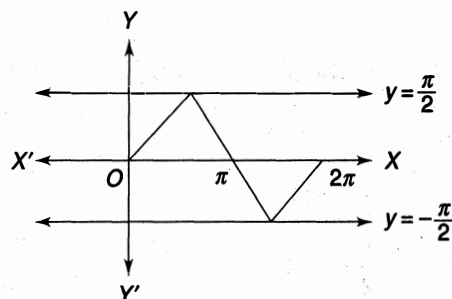


$$\text{Area of } \Delta = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$\text{Hence, the required area} = 4 \times \frac{1}{2} = 2 \text{ sq.u.}$$

$$40. \text{ We have,}$$

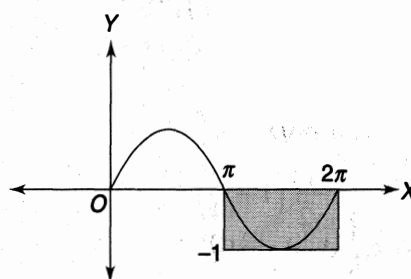
$$\int_0^{2\pi} f(x) dx$$



$$\text{Hence, the required area} = 2 \times \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi^2}{2}$$

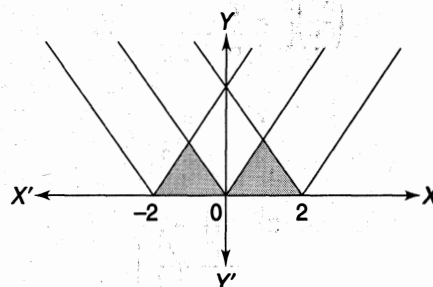
$$41. \text{ We have,}$$

$$\int_0^{2\pi} [\sin x] dx$$



$$\text{Hence, the required area} = \pi \times -1 = -\pi$$

$$42. \text{ We have,}$$

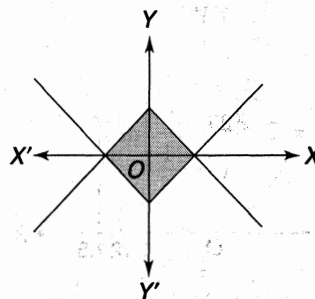


$$\int_{-2}^2 f(x) dx$$

$$\text{Hence, the required area} = 2 \times \frac{1}{2} \times 2 \times 1 = 2 \text{ sq.u.}$$

$$43. \text{ We have,}$$

$$\int_{-1}^1 g(x) dx$$



Hence, the required area

$$\begin{aligned}
 &= 4 \times \Delta \\
 &= 4 \times \frac{1}{2} \times 1 \times 1 \\
 &= 2 \text{ sq.u.}
 \end{aligned}$$

44. Here, $\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\Rightarrow \frac{1}{2} < \frac{2x}{\pi} < 1$$

$$\Rightarrow \left\lfloor \frac{2x}{\pi} \right\rfloor = 0$$

Thus, $\int_{\pi/4}^{\pi/2} \left[\sin x + \left\lfloor \frac{2x}{\pi} \right\rfloor \right] dx = 0$

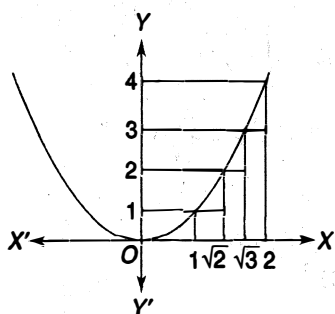
45. Here, $0 < x < 1$

$$\Rightarrow [x] = 0$$

Thus,

$$\begin{aligned}
 &\int_0^1 (x d(x - [x])) \\
 &= \int_0^1 x dx \\
 &= \left(\frac{x^2}{2} \right) \Big|_0^1 = \frac{1}{2}
 \end{aligned}$$

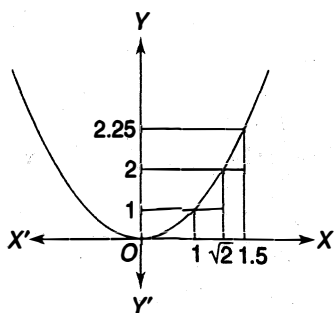
46.



Hence, the required area

$$\begin{aligned}
 &= 1 \cdot (\sqrt{2} - 1) + 2(\sqrt{3} - \sqrt{2}) + 3(\sqrt{3} - 2) \\
 &= (5 - \sqrt{2} - \sqrt{3}) \text{ sq.u.}
 \end{aligned}$$

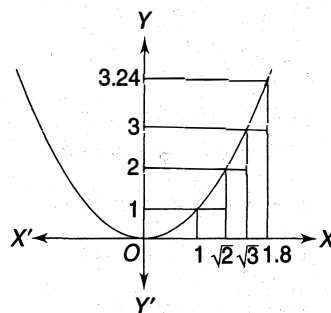
47.



Hence, the required area

$$\begin{aligned}
 &= 1 \cdot (\sqrt{2} - 1) + 2(1.5 - \sqrt{2}) \\
 &= (2 - \sqrt{2}) \text{ sq.u.}
 \end{aligned}$$

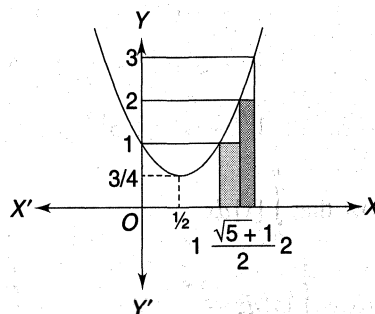
48.



Hence, the required area

$$\begin{aligned}
 &= 1 \cdot (\sqrt{2} - 1) + 2 \cdot (\sqrt{3} - \sqrt{2}) + 3(1.8 - \sqrt{3}) \\
 &= (4.4 - \sqrt{2} - \sqrt{3}) \text{ sq. u.}
 \end{aligned}$$

49.



Hence, the required area

$$\begin{aligned}
 &= 0 + 1 \cdot \left(\frac{\sqrt{5} + 1}{2} - 1 \right) + 2 \cdot \left(2 - \frac{\sqrt{5} + 1}{2} \right) \\
 &= 3 - \left(\frac{\sqrt{5} + 1}{2} \right) \\
 &= \left| \frac{5 - \sqrt{5}}{2} \right|
 \end{aligned}$$

50. Now, $2x^2 - 3 = 0 \Rightarrow x = \sqrt{\frac{3}{2}}$

$$2x^2 - 3 = 1 \Rightarrow x = \sqrt{2}$$

$$2x^2 - 3 = 2 \Rightarrow x = \sqrt{\frac{5}{2}}$$

$$2x^2 - 3 = 3 \Rightarrow x = \sqrt{3}$$

$$2x^2 - 3 = 4 \Rightarrow x = \sqrt{\frac{7}{2}}$$

and $2x^2 - 3 = 5 \Rightarrow x = 2$

Hence, the required area

$$\begin{aligned}
 &= (-1) \left(\sqrt{\frac{3}{2}} - 1 \right) + 0 + 1 \cdot \left(\sqrt{\frac{5}{2}} - \sqrt{2} \right) \\
 &\quad + 2 \cdot \left(\sqrt{3} - \sqrt{\frac{5}{2}} \right) + 3 \cdot \left(\sqrt{\frac{7}{2}} - \sqrt{3} \right)
 \end{aligned}$$

$$+ 4 \cdot \left(2 - \sqrt{\frac{7}{2}} \right)$$

$$= \left(9 - \left(\sqrt{\frac{3}{2}} + \sqrt{2} + \sqrt{\frac{5}{2}} + \sqrt{3} + \sqrt{\frac{7}{2}} \right) \right) \text{ sq.u.}$$

51. Now, $x^3 - 1 = 0 \Rightarrow x = 1$

$$x^3 - 1 = 1 \Rightarrow x = (2)^{1/3}$$

$$x^3 - 1 = 2 \Rightarrow x = (3)^{1/3}$$

$$x^3 - 1 = 3 \Rightarrow x = (4)^{1/3}$$

$$x^3 - 1 = 4 \Rightarrow x = (5)^{1/3}$$

$$x^3 - 1 = 5 \Rightarrow x = (6)^{1/3}$$

$$x^3 - 1 = 6 \Rightarrow x = (7)^{1/3}$$

and $x^3 - 1 = 7 \Rightarrow x = 2$

Hence, the required area

$$= 0 + 1(3^{1/3} - 2^{1/3}) + 2 \cdot (4^{1/3} - 3^{1/3})$$

$$+ 3 \cdot (5^{1/3} - 4^{1/3}) + 4 \cdot (6^{1/3} - 5^{1/3})$$

$$+ 5 \cdot (6^{1/3} - 7^{1/3}) + 6 \cdot (2^{1/3} - 7^{1/3})$$

$$= (12 - (2^{1/3} + 3^{1/3} + 4^{1/3} + 5^{1/3} + 6^{1/3} + 7^{1/3}))$$

52. We know that $\int_0^n [x] dx$

$$= \int_0^1 [x] dx + \int_1^2 [x] dx + \int_2^3 [x] dx + \dots + \int_{n-1}^n [x] dx$$

$$= \int_0^1 (0) dx + \int_1^2 (1) dx + \int_2^3 (2) dx + \dots + \int_{n-1}^n (n-1) dx$$

$$= 0 + 1(2-1) + 2(3-2) + \dots + (n-1)n - (n-1)$$

$$= 1 + 2 + 3 + 4 + \dots + (n-1)$$

$$= \frac{n(n-1)}{2}$$

53. $\int_0^{10} [x] dx$

$$= \frac{10(10-1)}{2}$$

$$= 45$$

54. $\int_0^{10} [x+2] dx$

$$= \int_0^{10} [x] dx + \int_0^{10} 2 dx$$

$$= \frac{10(10-1)}{2} + 2(10-0)$$

$$= 45 + 20$$

$$= 65.$$

55. $\int_0^2 [2x+3] dx$

$$= \int_0^2 [2x+3] dx$$

$$= \int_0^2 [2x] dx + \int_0^2 (3) dx$$

$$= \int_0^{1/2} [2x] dx + \int_{1/2}^1 [2x] dx + \int_1^{3/2} [2x] dx + \int_{3/2}^2 [2x] dx$$

$$+ \int_0^2 [3x] dx$$

$$= \int_0^{1/2} (0) dx + \int_{1/2}^1 (1) dx + \int_1^{3/2} (2) dx + \int_{3/2}^2 (3) dx + 6$$

$$= 0 + \left(1 - \frac{1}{2} \right) + 2 \left(\frac{3}{2} - 1 \right) + 3 \left(2 - \frac{3}{2} \right) + 6$$

$$= \frac{1}{2} + 2 \left(\frac{1}{2} \right) + 3 \left(\frac{1}{2} \right) + 6$$

$$= (1 + 2 + 6)$$

$$= 9$$

56. $\int_0^2 [3x-2] dx$

$$= \int_0^2 [3x] dx - \int_0^2 (2) dx$$

$$= \frac{1}{3} (1 + 2 + 3 + 4 + 5) - 4$$

$$= \left(\frac{15}{3} - 4 \right)$$

$$= (5 - 4)$$

$$= 1$$

57. $\int_0^n \{x\} dx$

$$= \int_0^n \{x\} dx$$

$$= \int_0^n (x - [x]) dx$$

$$= \int_0^n (x) dx - \int_0^n [x] dx$$

$$= \left(\frac{n^2}{2} - \frac{n(n-1)}{2} \right)$$

$$= \left(\frac{n}{2} \right)$$

$$58. \int_0^2 (x[x]) dx =$$

$$\begin{aligned} & \int_0^1 x[x] dx + \int_1^2 x[x] dx \\ &= \int_0^1 x \cdot 0 dx + \int_1^2 x \cdot 1 dx \\ &= \int_1^2 x dx \\ &= \left(\frac{x^2}{2} \right)_1^2 \\ &= \left(2 - \frac{1}{2} \right) = \frac{3}{2} \end{aligned}$$

$$59. \int_1^{e^6} \left[\frac{\log x}{3} \right] dx$$

$$\begin{aligned} &= \int_1^{e^3} \left[\frac{\log x}{3} \right] dx + \int_{e^3}^{e^6} \left[\frac{\log x}{3} \right] dx \\ &= \int_1^{e^3} 0 \cdot dx + \int_{e^3}^{e^6} 1 \cdot dx \\ &= (e^6 - e^3). \end{aligned}$$

$$60. \int_0^{\pi} [2 \sin x] dx$$

$$\begin{aligned} &= \int_0^{\pi/6} [2 \sin x] dx + \int_{\pi/6}^{5\pi/6} [2 \sin x] dx + \int_{5\pi/6}^{\pi} [2 \sin x] dx \\ &= \int_0^{\pi/6} (0) dx + \int_{\pi/6}^{5\pi/6} (1) dx + \int_{5\pi/6}^{\pi} (0) dx \\ &= \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) \cdot 1 \\ &= \frac{2\pi}{3} \end{aligned}$$

$$61. \int_0^{2\pi} [2 \sin x] dx$$

$$\begin{aligned} &= \int_0^{\pi/6} (0) dx + \int_{\pi/6}^{5\pi/6} (1) dx + \int_{5\pi/6}^{\pi} (0) dx \\ &\quad + \int_{\pi}^{7\pi/6} (-1) dx + \int_{7\pi/6}^{11\pi/6} (-2) dx + \int_{11\pi/6}^{2\pi} (-1) dx \\ &= \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) + \left(\frac{7\pi}{6} - \pi \right) (-1) \\ &\quad + \left(\frac{11\pi}{6} - \frac{7\pi}{6} \right) (-2) + \left(2\pi - \frac{11\pi}{6} \right) (-1) \\ &= \frac{2\pi}{3} - \frac{\pi}{6} - \frac{8\pi}{6} - \frac{\pi}{6} \end{aligned}$$

$$= \frac{2\pi}{3} - \frac{10\pi}{6}$$

$$= \frac{4\pi}{6} - \frac{10\pi}{6}$$

$$= -\frac{6\pi}{6} = -\pi$$

$$62. \int_0^{\pi/3} [\sqrt{3} \tan x] dx$$

$$\begin{aligned} &= \int_0^{\pi/6} [\sqrt{3} \tan x] dx + \int_{\pi/6}^{\tan^{-1}(2\sqrt{3})} [\sqrt{3} \tan x] dx \\ &\quad + \int_{\tan^{-1}(2\sqrt{3})}^{\pi/3} [\sqrt{3} \tan x] dx \\ &= \int_0^{\pi/6} (0) dx + \int_{\pi/6}^{\tan^{-1}(2\sqrt{3})} (1) dx + \int_{\tan^{-1}(2\sqrt{3})}^{\pi/3} (2) dx \\ &= \left(\frac{2\pi}{3} - \tan^{-1} \left(\frac{2}{\sqrt{3}} \right) - 1 \right) \end{aligned}$$

$$63. \int_1^{1.5} (\operatorname{sgn}[x]) dx$$

$$\begin{aligned} &= \int_1^{3/2} (1) dx \\ &= \left(\frac{3}{2} - 1 \right) \cdot 1 \\ &= \frac{1}{2} \end{aligned}$$

$$64. \int_0^{102} [\tan^{-1} x] dx$$

$$\begin{aligned} &= \int_0^{\tan(1)} [\tan^{-1} x] dx + \int_{\tan(1)}^{102} [\tan^{-1} x] dx \\ &= \int_0^{\tan(1)} (0) dx + \int_{\tan(1)}^{102} (1) dx \\ &= (102 - \tan(1)) \end{aligned}$$

$$65. \int_0^{2n\pi} [\sin x + \cos x] dx$$

Since it is a periodic function with period 2π .

$$\begin{aligned} &= n \int_0^{2\pi} [\sin x + \cos x] dx \\ &= n \int_0^{2\pi} \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right] dx \\ &= n \int_{\pi/4}^{9\pi/4} [\sqrt{2} \sin x] dx \end{aligned}$$

$$\begin{aligned}
 &= n \left[\left(\frac{3\pi}{4} - \frac{\pi}{4} \right) \cdot 1 + \left(\frac{5\pi}{4} - \pi \right) (-1) + \left(\frac{7\pi}{4} - \frac{5\pi}{4} \right) (-2) \right. \\
 &\quad \left. + \left(2\pi - \frac{3\pi}{4} \right) \cdot (-1) \right] \\
 &= n \left(\frac{2\pi}{4} - \frac{\pi}{4} - \frac{\pi}{4} - \frac{4\pi}{4} \right) \\
 &= -n\pi
 \end{aligned}$$

66. We have,

$$\begin{aligned}
 &\int_0^{5\pi/12} [\tan^{-1} x] dx \\
 \text{Let } I &= \int_0^{5\pi/12} [\tan x] dx \\
 &= \int_0^1 [\tan^{-1} x] dx + \int_1^2 [\tan^{-1} x] dx \\
 &\quad + \int_2^3 [\tan^{-1} x] dx + \int_3^{5\pi/12} [\tan^{-1} x] dx \\
 &= 0 + (\tan^{-1} 2 - \tan^{-1} 1) \cdot 1 + (\tan^{-1} 3 - \tan^{-1} 2) \cdot 2 \\
 &\quad + \left(\frac{5\pi}{12} - \tan^{-1} 3 \right) \cdot 3 \\
 &= \frac{5\pi}{4} - \frac{\pi}{4} - (\tan^{-1} 3 + \tan^{-1} 2)
 \end{aligned}$$

$$\begin{aligned}
 67. \int_{-2}^1 \left[\left(1 + \cos \left(\frac{\pi x}{2} \right) \right) \right] dx \\
 &= \int_{-2}^{-1} (1) dx + \int_{-1}^0 (0) dx + \int_0^1 (1) dx \\
 &= 1 + 0 + 1 \\
 &= 2
 \end{aligned}$$

68. We have,

$$\begin{aligned}
 &\int_0^3 \left([x] + \left[x + \frac{1}{2} \right] + \left[x + \frac{2}{3} \right] \right) dx \\
 \text{Let } I &= \int_0^3 \left([x] + \left[x + \frac{1}{2} \right] + \left[x + \frac{2}{3} \right] \right) dx \\
 &= \int_0^{1/3} 0 \cdot dx + \int_{1/3}^{2/3} 1 \cdot dx + \int_{2/3}^1 2 \cdot dx \\
 &\quad + \int_1^{4/3} 3 \cdot dx + \int_{4/3}^{5/3} 4 \cdot dx + \int_{5/3}^2 5 \cdot dx \\
 &\quad + \int_2^{7/3} 6 \cdot dx + \int_{7/3}^{8/3} 7 \cdot dx + \int_{8/3}^3 8 \cdot dx \\
 &= \frac{1}{3} (1 + 2 + 3 + 4 + 5 + 6 + 7 + 8) \\
 &= \frac{1}{3} \times 8 \times 9 \\
 &= 12
 \end{aligned}$$

$$\begin{aligned}
 69. \int_0^{10\pi} ([\sec^{-1} x] + [\tan^{-1} x]) dx \\
 &= \int_1^{\sec 1} ([\sec^{-1} x] + [\cot^{-1} x]) dx \\
 &\quad + \int_{\sec 1}^{10\pi} ([\sec^{-1} x] + [\cot^{-1} x]) dx \\
 &= \int_1^{\sec 1} (0 + 0) dx + \int_{\sec 1}^{10\pi} (1 + 0) dx \\
 &= (10\pi - \sec 1)
 \end{aligned}$$

$$\begin{aligned}
 70. \int_0^{\pi/4} [\sin x + [\cos x + [\tan x + [\sec x]]]] dx \\
 \text{Let } I &= \int_0^{\pi/4} [\sin x + [\cos x + \tan x + [\sec x]]] dx \\
 \text{Since } 0 < x &< \sqrt{2} \\
 \Rightarrow 1 &< \sec x < \sqrt{2} \\
 \Rightarrow [\sec x] &= 1 \\
 \Rightarrow [\tan x + [\sec x]] &= [\tan x + 1] \\
 &= [\tan x] + 1 \\
 &= 0 + 1 = 1
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow [\cos x + [\tan x + [\sec x]]] \\
 &= [\cos x + 1] = [\cos x] + 1 = 0 + 1 = 1 \\
 \Rightarrow [\sin x + [\cos x + [\tan x + [\sec x]]]] \\
 &= [\sin x + 1] = [\sin x] + 1 = 0 + 1 = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus, } I &= \int_0^{\pi/4} [\sin x + [\cos x + \tan x + [\sec x]]] dx \\
 &= \int_0^{\pi/4} 1 \cdot dx \\
 &= \frac{\pi}{4}
 \end{aligned}$$

71. We have,

$$\begin{aligned}
 &\int_{\pi/4}^{\pi/2} \left[\sin x + \left[\frac{2x}{\pi} \right] \right] dx \\
 \text{Here, } \frac{\pi}{4} &< x < \frac{\pi}{2} \\
 \Rightarrow \frac{\pi}{2} &< 2x < \pi \\
 \Rightarrow \frac{1}{2} &< \frac{2x}{\pi} < 1 \\
 \Rightarrow \left[\frac{2x}{\pi} \right] &= 0
 \end{aligned}$$

$\therefore$ The given integral reduces to

$$\int_{\pi/4}^{\pi/2} \left[\sin x + \left[\frac{2x}{\pi} \right] \right] dx$$

$$= \int_{\pi/4}^{\pi/2} [\sin x + 0] dx$$

$$= 0$$

72. Here, $a = 0, b = 3$

$$\therefore nh = b - a = 3 - 0 = 3,$$

$$f(x) = x$$

$$f(a + rh) = f(rh) = rh$$

Now, $\int_1^3 x dx = \lim_{h \rightarrow 0} h \sum_{r=1}^n f(rh)$

$$= \lim_{h \rightarrow 0} h \cdot \sum_{r=1}^n rh$$

$$= \lim_{h \rightarrow 0} h^2 \sum_{r=1}^n r$$

$$= \lim_{h \rightarrow 0} h^2 \left(\frac{n(n+1)}{2} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{nh(nh+h)}{2} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{3(3+h)}{2} \right)$$

$$= 9/2$$

73. Here, $a = 0, b = 2$

$$nh = b - a = 2, f(x) = x^2$$

Also, $f(a + rh) = f(rh) = (rh)^2$

Now, $\int_0^2 x^2 dx$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^n f(rh)$$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^n (rh)^2$$

$$= \lim_{h \rightarrow 0} h^3 \sum_{r=0}^n r^2$$

$$= \lim_{h \rightarrow 0} h^3 \left(\frac{n(n+1)(2n+1)}{6} \right)$$

$$= \lim_{h \rightarrow 0} h^3 \left(\frac{nh(nh+h)(2nh+1)}{6} \right)$$

$$= \lim_{h \rightarrow 0} h^3 \left(\frac{2(2+h)(4+h)}{6} \right)$$

$$= \frac{8}{3}$$

74. Here, $a = 1, b = 3$

$$nh = b - a = 2$$

$$f(x) = (x^2 + x + 2)$$

$$f(a + rh) = f(1 + rh)$$

$$(1 + rh)^2 + (1 + rh) + 2 = (rh)^2 + 3(rh) + 4$$

Now, $\int_1^3 (x^2 + x + 2) dx$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^n f(a + rh)$$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^n ((rh)^2 + 3(rh) + 4)$$

$$= \lim_{h \rightarrow 0} h \left(\frac{h^2(n(n+1)(2n+1))}{6} \right)$$

$$+ 3h \left(\frac{n(n+1)}{2 + 4h} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{nh(nh+h)(2nh+h)}{6} \right)$$

$$\left(+ 3 \left(\frac{nh(nh+h)}{2} \right) + 4nh \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{2(2+h(4+h))}{6} + \left(\frac{2(2+h)}{2} \right) + 8 \right)$$

$$= \frac{8}{3} + 6 + 8$$

$$= 50/3$$

75. Here $a = 0, b = 1$

$$nh = b - a = 1 - 0 = 1.$$

Also $f(x) = e^{mx}, f(a + rh) = f(rh) = e^{mrh}$

Now, $\int_0^1 e^{mx} dx$

$$= \lim_{h \rightarrow 0} h \sum_{r=1}^n f(a + rh)$$

$$= \lim_{h \rightarrow 0} h \sum_{r=0}^n e^{mrh}$$

$$= \lim_{h \rightarrow 0} h(1 + e^{2mh} + \dots + e^{nmh})$$

$$= \lim_{h \rightarrow 0} h \left(\frac{e^{nmh} - 1}{e^{mh} - 1} \right)$$

$$= \lim_{h \rightarrow 0} h \left(\frac{em - 1}{e^{mh} - 1} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{mh}{e^{mh} - 1} \right) \times \left(\frac{e^m - 1}{m} \right)$$

$$= \left(\frac{e^m - 1}{m} \right)$$

$$76. \text{ Here, } a = 0, b = \frac{\pi}{2}$$

$$nh = \pi/2 - 0 = \pi/2, f(x) = \sin x$$

$$\text{Also } f(a + rh) = f(rh) = \sin(rh)$$

$$\begin{aligned} \text{Now, } \int_0^{\pi/2} \sin x \, dx &= \lim_{h \rightarrow 0} h \sum_{r=1}^n f(rh) \\ &= \lim_{h \rightarrow 0} h \sum_{r=1}^n \sin(rh) \\ &= \lim_{h \rightarrow 0} (\sin h + \sin 2h + \sin 3h + \dots + \sin nh) \\ &= \lim_{h \rightarrow 0} h \left(\frac{\sin(rh/2)}{\sin(h/2)} \times \frac{\sin(n+1)h}{2} \right) \\ &= \lim_{h \rightarrow 0} h \left[\frac{\sin(\pi/4)}{\sin(h/2)} \times \sin\left(\frac{\pi}{2} + h\right) \right] \\ &= \lim_{h \rightarrow 0} h \left[\frac{2}{\sqrt{2}} \times \frac{h/2}{\sin(h/2)} \times \sin\left(\frac{\pi}{4} + \frac{h}{2}\right) \right] \\ &= \sqrt{2} \times \sin\left(\frac{\pi}{4}\right) \\ &= 1 \end{aligned}$$

77. Do yourself

$$87. \lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right]$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{n+r} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n}{n+r} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1}{1+(r/n)} \right) \\ &= \int_0^1 \frac{dx}{1+x} \\ &= \log|1+x|_0^1 \\ &= \log 2 \end{aligned}$$

$$\begin{aligned} 88. \lim_{n \rightarrow \infty} \left[\frac{n}{n^2} + \frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} \right. \\ \left. + \dots + \frac{n}{n^2+(n-1)^2} \right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(\frac{n}{n^2+r^2} \right) \end{aligned}$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \left(\frac{n^2}{n^2+r^2} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \left(\frac{1}{1+(r/n)^2} \right) \\ &= \int_0^1 \frac{dx}{1+x^2} \\ &= \tan^{-1}(x) \Big|_0^1 \\ &= \frac{\pi}{4} \end{aligned}$$

$$89. \lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \dots + \frac{n}{n^2+(2n-1)^2} \right]$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \sum_{r=0}^{2n-1} \left(\frac{n}{n^2+r^2} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \left(\frac{n^2}{n^2+r^2} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{2n-1} \left(\frac{1}{1+(r/n)^2} \right) \\ &= \int_0^2 \frac{dx}{1+x^2} \\ &= \tan^{-1}(x) \Big|_0^2 \\ &= \tan^{-1} 2 \end{aligned}$$

$$90. \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{1+\frac{1}{n}} + \sqrt{1+\frac{2}{n}} + \dots + \sqrt{1+\frac{n}{n}} \right)$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \left(\sqrt{1+\frac{r}{n}} \right) \\ &= \int_0^1 \sqrt{1+x} \, dx \\ &= \frac{2}{3} (1+x)^{3/2} \Big|_0^1 \\ &= \frac{2}{3} [(2)^{3/2} - 1] \end{aligned}$$

$$91. \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \frac{1}{\sqrt{4n^2-9}} + \dots + \frac{1}{\sqrt{3n^2}} \right)$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{\sqrt{4n^2-r^2}} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n}{\sqrt{4n^2-r^2}} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1}{\sqrt{4-(r/n)^2}} \right) \end{aligned}$$

$$= \int_0^1 \frac{dx}{\sqrt{4-x^2}}$$

$$= \sin^{-1}\left(\frac{x}{2}\right)\Big|_0^1$$

$$= \frac{\pi}{6}$$

$$92. \lim_{n \rightarrow \infty} \left(\frac{1^m + 2^m + 3^m + \dots + n^m}{n^{m+1}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{1^m + 2^m + 3^m + \dots + n^m}{n^m} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\left(\frac{1}{n}\right)^m + \left(\frac{2}{n}\right)^m + \left(\frac{3}{n}\right)^m + \dots + \left(\frac{n}{n}\right)^m \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^m$$

$$= \int_0^1 x^m dx = \frac{x^{m+1}}{m+1} \Big|_0^1 = \frac{1}{m+1}$$

$$93. \lim_{n \rightarrow \infty} \left(\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \dots + \frac{1}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \dots + \frac{n+n}{n^2+n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{n+r}{n^2+r^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n^2+nr}{n^2+r^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1+(r/n)}{1+(r/n)^2} \right)$$

$$= \int_0^1 \left(\frac{1+x}{1+x^2} \right) dx$$

$$= \tan^{-1}(x) + \frac{1}{2} \log|x^2+1| \Big|_0^1$$

$$= \left(\frac{\pi}{4} + \frac{1}{2} \log 2 \right)$$

$$94. \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \sum_{r=1}^n \left(\frac{1}{\sqrt{r}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n}{\sqrt{n}\sqrt{r}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1}{\sqrt{r/n}} \right)$$

$$= \int_0^1 \frac{dx}{\sqrt{x}}$$

$$= 2\sqrt{x} \Big|_0^1$$

$$= 2(1-0) = 2$$

$$95. \text{ Let } A = \lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{1/n}$$

$$\Rightarrow \log A = \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{n!}{n^n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{1.2.3 \dots n}{n^n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left(\frac{1}{n}\right) \cdot \left(\frac{2}{n}\right) \cdot \left(\frac{3}{n}\right) \dots \left(\frac{n}{n}\right) \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\log \left(\frac{1}{n}\right) + \log \left(\frac{2}{n}\right) + \dots + \log \left(\frac{n}{n}\right) \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\log \left(\frac{r}{n}\right) \right)$$

$$= \int_0^1 \log x dx$$

$$= (x \log x - x) \Big|_0^1$$

$$= (0-1) - \lim_{x \rightarrow 0^+} (x \log x) + 0$$

$$= -1 + 0 = -1$$

$$\Rightarrow \log L = -1$$

$$\Rightarrow L = e^{-1} = \frac{1}{e}$$

$$96. \text{ Let } L = \lim_{n \rightarrow \infty} \left[1 + \frac{1}{n^2} \left(1 + \frac{2^2}{n^2} \right) \left(1 + \frac{3^2}{n^2} \right) \dots \left(1 + \frac{n^2}{n^2} \right) \right]^{1/n}$$

$$\Rightarrow \log A = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \log \left(1 + \frac{1}{n^2} \right) \left(1 + \frac{2^2}{n^2} \right) \dots \right)$$

$$\log \left(1 + \frac{n^2}{n^2} \right) + \log \left(1 + \frac{n^2}{n^2} \right) \dots$$

$$\Rightarrow \log A = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(1 + \frac{r^2}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(1 + \left(\frac{r}{n}\right)^2 \right)$$

$$= \int_0^1 \log(1+x^2) dx$$

$$= (x \log|1+x^2| - 2x + 2 \tan^{-1} x) \Big|_0^1$$

$$= \log 2 + \frac{\pi-4}{2}$$

$$\Rightarrow \log A - \log 2 = \frac{\pi-4}{2}$$

$$\Rightarrow \log \left(\frac{A}{2} \right) = \pi - \frac{4}{2}$$

$$\Rightarrow A = 2e^{(\pi-4)/2}$$

$$\begin{aligned}
 97. \quad & \lim_{n \rightarrow \infty} \left(\frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{4n} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{1}{n+0} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+3n} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^{3n} \left(\frac{1}{n+r} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{3n} \left(\frac{n}{n+r} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^{3n} \left(\frac{1}{1+(r/n)} \right) \\
 &= \int_0^3 \frac{dx}{1+x} \\
 &= \log(1+x) \Big|_0^3 \\
 &= \log 4 \\
 &= 2 \log 2
 \end{aligned}$$

$$\begin{aligned}
 98. \quad & \lim_{n \rightarrow \infty} \left(\frac{1}{n} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{1}{8n} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{n^2}{(n+0)^3} + \frac{n^2}{(n+1)^3} + \frac{n^2}{(n+2)^3} + \dots + \frac{n^2}{(n+n)^3} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{n^2}{(n+r)^3} \right) \\
 &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=0}^n \left(\frac{n^3}{(n+r)^3} \right) \right] \\
 &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=0}^n \left(\frac{1}{(1+(r/n))^3} \right) \right] \\
 &= \int_0^1 \left(\frac{dx}{(1+x)^3} \right) \\
 &= \left(-\frac{1}{2(x+1)^2} \right) \Big|_0^1 \\
 &= \left| \frac{1}{2} - \frac{1}{8} \right| = \frac{3}{8}
 \end{aligned}$$

$$\begin{aligned}
 99. \quad & \lim_{n \rightarrow \infty} \left(\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \frac{n+3}{n^2+3^2} + \dots + \frac{3}{5n} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+2^2} + \frac{n+3}{n^2+3^2} + \dots + \frac{n+2n}{n^2+(2n)^2} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^{2n} \left(\frac{n+r}{n^2+r^2} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \left(\frac{n^2+rn}{n^2+r^2} \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{1 + \left(\frac{r}{n} \right)}{1 + \left(\frac{r}{n} \right)^2} \right) \\
 &= \int_0^2 \left(\frac{1+x}{1+x^2} \right) dx \\
 &= \left(\tan^{-1} x + \frac{1}{2} \log |1+x^2| \right) \Big|_0^2 \\
 &= \left| \tan^{-1} 2 + \frac{1}{2} \log 5 \right|
 \end{aligned}$$

$$\begin{aligned}
 100. \quad & \lim_{n \rightarrow \infty} \left(\frac{n^2}{(n^2+1)^{3/2}} + \frac{n^2}{(n^2+2^2)^{3/2}} + \dots \right. \\
 & \quad \left. + \frac{n^2}{(n^2+(n-1)^2)^{3/2}} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \left(\frac{n^2}{(n^2+r^2)^{3/2}} \right) \\
 &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=0}^{n-1} \left(\frac{n^3}{(n^2+r^2)^{3/2}} \right) \right] \\
 &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=0}^{n-1} \left(\frac{1}{\left(1 + \left(\frac{r}{n} \right)^2 \right)^{3/2}} \right) \right] \\
 &= \int_0^1 \frac{dx}{(1+x^2)^{3/2}} \\
 &= \int_0^{\pi/4} \left(\frac{\sec^2 \theta}{\sec^3 \theta} \right) d\theta \\
 &= \int_0^{\pi/4} (\cos \theta) d\theta \\
 &= (\sin \theta) \Big|_0^{\pi/4} \\
 &= \frac{1}{\sqrt{2}}
 \end{aligned}$$

$$\begin{aligned}
 101. \quad & \lim_{n \rightarrow \infty} \left(\frac{1^2}{n^3+1^3} + \frac{2^2}{n^3+2^3} + \frac{3^2}{n^3+3^3} + \dots + \frac{1}{2n} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{1^2}{n^3+1^3} + \frac{2^2}{n^3+2^3} + \frac{3^2}{n^3+3^3} + \dots + \frac{n^2}{n^3+n^3} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{r^2}{n^3+r^3} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \left(\frac{nr^2}{n^3+r^3} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \left(\frac{\left(\frac{r}{n} \right)^2}{1 + \left(\frac{r}{n} \right)^3} \right)
 \end{aligned}$$

$$= \int_0^1 \frac{x^2}{1+x^3} dx$$

$$= \left(\frac{1}{3} \log |1+x^3| \right)_0^1$$

$$= \frac{1}{3} \log |2|$$

$$102. \lim_{n \rightarrow \infty} \left(\frac{2^k + 4^k + 6^k + \dots + (2n)^k}{n^{k+1}} \right), k \neq -1$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{2^k + 4^k + 6^k + \dots + (2n)^k}{n^k} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\left(\frac{2}{n} \right)^k + \left(\frac{4}{n} \right)^k + \left(\frac{6}{n} \right)^k + \dots + \left(\frac{2n}{n} \right)^k \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{2r}{n} \right)^k$$

$$= \int_0^1 2^k (x)^k dx$$

$$= 2k \left(\frac{x^{k+1}}{k+1} \right)_0^1$$

$$= \left(\frac{2^k}{k+1} \right)$$

$$103. \lim_{n \rightarrow \infty} \left(\frac{\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + 2\sqrt{n}}{n\sqrt{n}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \sqrt{\frac{3}{n}} + \dots + \sqrt{\frac{4n}{n}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \left(\sqrt{\frac{r}{n}} \right)$$

$$= \int_0^4 \sqrt{x} dx$$

$$= \left(\frac{2}{3} x^{3/2} \right)_0^4$$

$$= \frac{2}{3} \times 4^{3/2} = \frac{2}{3} \times 8 = \frac{16}{3}$$

$$104. \lim_{n \rightarrow \infty} \left(\frac{1}{na} + \frac{1}{na+1} + \frac{1}{na+2} + \dots + \frac{1}{na+n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{n}{na} + \frac{n}{na+1} + \frac{n}{na+2} + \dots + \frac{n}{na+n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \left(\frac{n}{na+r} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^n \left(\frac{1}{a+(r/n)} \right)$$

$$= \int_0^1 \frac{dx}{a+x}$$

$$= (\log |a+x|)_0^1$$

$$= \log \left(\frac{a+1}{a} \right)$$

$$105. \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}\sqrt{2n}} \right)$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\sqrt{n}}{n\sqrt{n+1}} + \frac{\sqrt{n}}{n\sqrt{n+2}} + \frac{\sqrt{n}}{n\sqrt{n+3}} \right.$$

$$\left. + \dots + \frac{\sqrt{n}}{n\sqrt{n+n}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{\sqrt{n}}{\sqrt{n+r}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\sqrt{\frac{1}{1+(r/n)}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{1}{\sqrt{1+(r/n)}} \right)$$

$$= \int_0^1 \frac{dx}{\sqrt{1+x}}$$

$$= \left(\frac{2}{3} (1+x)^{3/2} \right)_0^1$$

$$= \left| \frac{2}{3} (2^{3/2} - 1) \right|$$

$$106. \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \left(\frac{1}{n+r} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \left(\frac{n}{n+r} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \left(\frac{1}{1+(r/n)} \right)$$

$$= \int_0^2 \frac{dx}{1+x}$$

$$= (\log |1+x|)_0^2 = \log 3$$

$$107. \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \left(\frac{1}{\sqrt{n^2-r^2}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \left(\frac{n}{\sqrt{n^2-r^2}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \left(\frac{1}{\sqrt{1-(r/n)^2}} \right)$$

$$= \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

$$= [\sin^{-1}(x)]_0^1$$

$$= \frac{\pi}{2}$$

$$108. \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{\sqrt{n^2+3rn}} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{\sqrt{n^2 + 3rn}} \right)$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{n}{\sqrt{n^2 + 3rn}} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{1}{\sqrt{1 + 3(r/n)}} \right) \right]$$

$$= \int_0^1 \frac{dx}{\sqrt{1 + 3x}}$$

$$= \left(\frac{2}{9} (1 + 3x)^{3/2} \right)_0^1$$

$$= \frac{2}{9} (8 - 1) = \frac{14}{9}$$

$$109. \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sum_{r=1}^n \left(\frac{n\sqrt{n}}{\sqrt{r}(\sqrt{r} + 4\sqrt{n})^2} \right) \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sum_{r=1}^n \left(\frac{1}{\sqrt{(r/n)}(\sqrt{(r/n)} + 4)^2} \right) \right)$$

$$= \int_0^1 \frac{dx}{\sqrt{x}(\sqrt{x} + 4)^2}$$

$$= \int_4^5 \left(\frac{2dt}{t^2} \right)$$

$$= -\left(\frac{2}{t} \right)_4^5$$

$$= -2 \left(\frac{1}{5} - \frac{1}{4} \right) = -2 \times -\frac{1}{20} = \frac{1}{10}$$

$$110. \lim_{n \rightarrow \infty} \sum_{r=1}^{10n} \left(\frac{1}{\sqrt{r}\sqrt{n}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{10n} \left(\frac{n}{\sqrt{r}\sqrt{n}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{10n} \left(\frac{\sqrt{n}}{\sqrt{r}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{10n} \left(\frac{1}{\sqrt{(r/n)}} \right)$$

$$= \int_0^{10} \frac{dx}{\sqrt{x}}$$

$$= \left(\frac{2}{3} (x^{3/2}) \right)_0^{10} = \frac{2}{3} (10)^{3/2}$$

$$111. \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{n}{(r+n)\sqrt{r^2 + 2nr}} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{n^2}{(r+n)\sqrt{r^2 + 2nr}} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{(1 + (r/n))\sqrt{(r/n)^2 + 2(r/n)}} \right)$$

$$= \int_0^1 \frac{dx}{(1+x)\sqrt{x^2 + 2x}}$$

$$= \int_0^1 \frac{dx}{(1+x)\sqrt{(x+1)^2 - 1}}$$

$$= \sec^{-1}(2) - \sec^{-1}(1)$$

$$= \left(\frac{\pi}{3} - 0 \right) = \frac{\pi}{3}$$

$$112. \text{ Let } I = \int_0^4 |x - 2| dx$$

$$= \int_0^2 |x - 2| dx + \int_2^4 |x - 2| dx$$

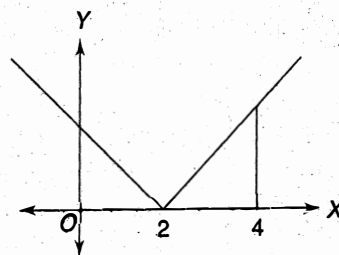
$$= -\int_0^2 (x - 2) dx + \int_2^4 (x - 2) dx$$

$$= \left(2x - \frac{x^2}{2} \right)_0^2 + \left(\frac{x^2}{2} - 2x \right)_2^4$$

$$= (4 - 2) + (8 - 8) - (2 - 4)$$

$$= 2 + 2 = 4$$

Alternate Method

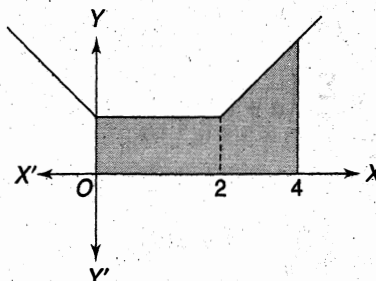


$$I = \int_0^4 |x - 2| dx$$

$$= \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 2 \times 2$$

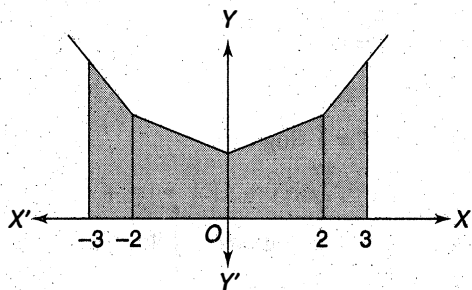
$$= 4 \text{ sq.u.}$$

113.



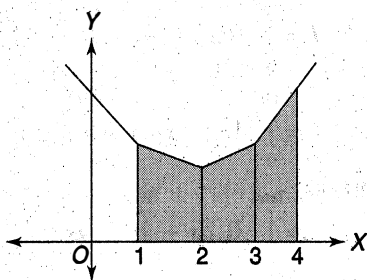
$$\begin{aligned} \text{Let } I &= \int_0^4 (|x| + |x-2|) dx \\ &= 2 \times 2 + \frac{1}{2}(2+6) \times 2 = 4 + 8 = 12 \end{aligned}$$

$$114. \text{ Let } I = \int_{-3}^3 (|x+2| + |x| + |x-2|) dx$$



$$\begin{aligned} &= 2 \times \frac{1}{2} \times (9+6) + 2 \times \frac{1}{2} \times (4+6) \times 2 \\ &= 15 + 20 \\ &= 35 \text{ sq.u.} \end{aligned}$$

$$115. \text{ Let } I = \int_1^4 (|x-1| + |x-2| + |x-3|) dx$$



$$\begin{aligned} &= 2 \times \frac{1}{2} \times (3+2) \times 1 + \frac{1}{2} \times (3+6) \times 1 \\ &= 5 + \frac{9}{2} \\ &= 19/2 \text{ sq.u.} \end{aligned}$$

$$\begin{aligned} 116. \text{ Let } I &= \int_0^3 |x^2 - 3x + 2| dx \\ &= \int_0^1 |x^2 - 3x + 2| dx + \int_1^2 |x^2 - 3x + 2| dx \\ &\quad + \int_2^3 |x^2 - 3x + 2| dx \\ &= -\int_0^1 (x^2 - 3x + 2) dx + \int_1^2 (x^2 - 3x + 2) dx \\ &\quad - \int_2^3 (x^2 - 3x + 2) dx \end{aligned}$$

$$\begin{aligned} &= -\left(\frac{x^3}{3} - \frac{3}{2}x^2 + 2x\right)\Big|_0^1 + \left(\frac{x^3}{3} - \frac{3}{2}x^2 + 2x\right)\Big|_1^2 \\ &\quad + \left(\frac{x^3}{3} - \frac{3}{2}x^2 + 2x\right)\Big|_2^3 \\ &= -\frac{5}{6} + \frac{21}{6} - \frac{7}{6} \\ &= \frac{9}{6} \\ &= \frac{3}{2} \end{aligned}$$

$$\begin{aligned} 117. \text{ Let } I &= \int_{-4}^3 |x^2 - 4| dx \\ &= \int_{-4}^{-2} |x^2 - 4| dx + \int_{-2}^2 |x^2 - 4| dx + \int_2^3 |x^2 - 4| dx \\ &= \int_{-4}^{-2} (x^2 - 4) dx - \int_{-2}^2 (x^2 - 4) dx + \int_2^3 (x^2 - 4) dx \\ &= \left(\frac{x^3}{3} - 4x\right)\Big|_{-4}^{-2} - \left(\frac{x^3}{3} - 4x\right)\Big|_{-2}^2 + \left(\frac{x^3}{3} - 4x\right)\Big|_2^3 \\ &= \left(8 - \frac{8}{3}\right) - \left(-\frac{64}{3} + 16\right) + 2\left(8 - \frac{8}{3}\right) \\ &\quad + \left(\frac{27}{3} - 12\right) - \left(\frac{8}{3} - 8\right) \\ &= 71/3 \text{ sq.u.} \end{aligned}$$

$$\begin{aligned} 118. \text{ Let } I &= \int_0^{\pi} |\sin x| dx \\ &= \int_0^{\pi} \sin x dx \\ &= (-\cos x)\Big|_0^{\pi} \\ &= -(\cos \pi - \cos 0) \\ &= -(-1 - 1) \\ &= 2 \end{aligned}$$

Note: The values of

$$1. \int_0^{2\pi} |\sin x| dx = 4$$

$$2. \int_0^{\pi} |\cos x| dx = 2$$

$$3. \int_0^{2\pi} |\cos x| dx = 4.$$

119. We have,

$$|\sin x| - |\cos x| = 0$$

$$\Rightarrow |\sin x| = |\cos x|$$

$$\Rightarrow |\tan x| = 1$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\begin{aligned} \text{Let } I &= \int_0^{\pi} |\sin x - \cos x| dx \\ &= 2 \int_0^{\pi/2} |\sin x - \cos x| dx \\ &= 2 \times \sqrt{2} \int_0^{\pi/2} \left| \frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x \right| dx \\ &= 2\sqrt{2} \int_0^{\pi/2} \left| \sin \left(x - \frac{\pi}{4} \right) \right| dx \\ &= 2\sqrt{2} \int_{-\pi/4}^{\pi/4} |\sin x| dx \\ &= 4\sqrt{2} \int_0^{\pi/4} |\sin x| dx \\ &= 4\sqrt{2} \int_0^{\pi/4} \sin x dx \\ &= 4\sqrt{2} (-\cos x)_0^{\pi/4} \\ &= 4\sqrt{2} \left(1 - \frac{1}{\sqrt{2}} \right) \\ &= 4(\sqrt{2} - 1) \end{aligned}$$

$$\begin{aligned} 120. \text{ Let } I &= \int_{-1}^{3/2} |x \sin \pi x| dx \\ &= \int_{-1}^1 |x \sin \pi x| dx + \int_1^{3/2} |x \sin \pi x| dx \\ &= 2 \int_0^1 |x \sin \pi x| dx + \frac{1}{\pi^2} \int_{\pi}^{3\pi/2} |x \sin x| dx \\ &= 2 \int_0^1 x \sin \pi x dx - \frac{1}{\pi^2} \int_{\pi}^{3\pi/2} x \sin x dx \\ &= 2 \left[x \left(\frac{-\cos \pi x}{\pi} \right) + \frac{\sin \pi x}{\pi^2} \right]_0^1 \\ &\quad - \frac{1}{\pi^2} [x(-\cos x) + \sin x]_{\pi}^{3\pi/2} \\ &= 2 \left(\frac{1}{\pi} \right) - \frac{1}{\pi^2} \{ (0 - 1) - \pi \} \\ &= \frac{3}{\pi} + \frac{1}{\pi^2} = \left(\frac{3\pi + 1}{\pi^2} \right) \end{aligned}$$

$$121. \int_{-\pi/2}^{2\pi} \sin^{-1}(\sin x) dx$$

$$= \int_{-\pi/2}^0 \sin^{-1}(\sin x) dx$$

$$\begin{aligned} &+ \int_0^{\pi} \sin^{-1}(\sin x) dx + \int_{\pi}^{2\pi} \sin^{-1}(\sin x) dx \\ &= -\left(\frac{1}{2} \cdot \frac{\pi}{2} \cdot \frac{\pi}{2} \right) + \left(\frac{1}{2} \cdot \pi \cdot \frac{\pi}{2} \right) - \left(\frac{1}{2} \cdot \pi \cdot \frac{\pi}{2} \right) \\ &= -\frac{\pi^2}{8} + \frac{\pi^2}{4} - \frac{\pi^2}{4} \\ &= -\frac{\pi^2}{8} \end{aligned}$$

$$\begin{aligned} 122. \text{ Let } I &= \int_{1/e}^e |\log x| dx \\ &= \int_{1/e}^1 |\log x| dx + \int_1^e |\log x| dx \\ &= \int_{1/e}^1 (-\log x) dx + \int_1^e (\log x) dx \\ &= -(x \log x - x) \Big|_{1/e}^1 + (x \log x - x) \Big|_1^e \\ &= \left(-1 + \frac{2}{e} \right) + 1 \\ &= \left(2 - \frac{2}{e} \right) \end{aligned}$$

$$\begin{aligned} 123. \text{ Let } I &= \int_0^{100} [\tan^{-1} x] dx \\ &= \int_0^{\tan 1} [\tan^{-1} x] dx + \int_{\tan 1}^{100} [\tan^{-1} x] dx \\ &= \int_0^{\tan 1} 0 \cdot dx + \int_{\tan 1}^{100} 1 \cdot dx \\ &= 0 + 100 - \tan 1 \\ &= (100 - \tan 1) \end{aligned}$$

$$\begin{aligned} 124. \text{ Let } I &= \int_0^{1/2} [2 \sin^{-1} x] dx \\ &= \int_0^{\sin(1/2)} [2 \sin^{-1} x] dx + \int_{\sin(1/2)}^{1/2} [2 \sin^{-1} x] dx \\ &= \int_0^{\sin(1/2)} 0 \cdot dx + \int_{\sin(1/2)}^{1/2} 1 \cdot dx \\ &= 0 + 1 \cdot \left(\frac{1}{2} - \left(\frac{1}{2} \right) \right) \\ &= \frac{1}{2} - \sin \left(\frac{1}{2} \right) \end{aligned}$$

$$\begin{aligned} 125. \text{ Let } I &= \int_{-2}^1 \left[x \left[1 + \cos \left(\frac{\pi x}{2} \right) \right] + 1 \right] dx \\ &= \frac{2}{\pi} \int_{-\pi}^{\pi/2} \left[\frac{2x}{\pi} [1 + \cos x] + 1 \right] dx \end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{\pi} \int_{-\pi}^{-\pi/2} \left[\frac{2x}{\pi} [1 + \cos x] + 1 \right] dx \\
 &\quad + \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \left[\frac{2x}{\pi} [1 + \cos x] + 1 \right] dx \\
 &= \frac{2}{\pi} \int_{-\pi}^{-\pi/2} \left[\frac{2x}{\pi} \cdot 0 + 1 \right] dx + \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \left[\frac{2x}{\pi} + 1 \right] dx \\
 &= \frac{2}{\pi} \int_{-\pi}^{-\pi/2} 1 \cdot dx + \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} [x + 1] dx \\
 &= \frac{2}{\pi} \left(-\frac{\pi}{2} + \pi \right) + \int_{-1}^1 [x + 1] dx \\
 &= \frac{2}{\pi} \left(-\frac{\pi}{2} + \pi \right) + \int_{-1}^1 [x] dx + \int_{-1}^1 1 dx \\
 &= 1 + \int_{-1}^0 [x] dx + \int_0^1 1 dx + 2 \\
 &= 3 + (-1) \cdot 1 + 0 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 126. \text{ Let } I &= \int_0^{\sqrt{3}} \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx \\
 &= \int_0^1 \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx + \int_1^{\sqrt{3}} \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx \\
 &= \int_0^1 2 \tan^{-1} x dx + \int_1^{\sqrt{3}} (-\pi + 2 \tan^{-1} x) dx \\
 &= 2 \int_0^1 \tan^{-1} x dx - \int_1^{\sqrt{3}} (x) dx + \int_1^{\sqrt{3}} (2 \tan^{-1} x) dx \\
 &= 2 \int_0^{\sqrt{3}} \tan^{-1} x dx - \int_1^{\sqrt{3}} (\pi) dx \\
 &= 2 \left[(x \tan^{-1} x) \Big|_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \left(\frac{x}{1+x^2} \right) dx \right] \pi - (x) \Big|_1^{\sqrt{3}} \\
 &= 2 \left[(\sqrt{3} \tan^{-1}(\sqrt{3}) - 0) - \frac{1}{2} \log(1+x^2) \Big|_0^{\sqrt{3}} \right] \\
 &\quad - \pi(\sqrt{3} - 1) \\
 &= \frac{2\pi\sqrt{3}}{3} - \log 4 - \pi(\sqrt{3} - 1) \\
 &= \left[\pi \left(1 - \frac{1}{\sqrt{3}} \right) - \log 4 \right] \text{ sq. u}
 \end{aligned}$$

Note: $\tan^{-1} \left(\frac{2x}{1-x^2} \right) = \begin{cases} \pi + 2 \tan^{-1} x, & x < -1 \\ 2 \tan^{-1} x, & -1 < x < 1 \\ -\pi + 2 \tan^{-1} x, & x > 1 \end{cases}$

$$\begin{aligned}
 127. \text{ Let } I &= \int_{-\pi/2}^{\pi/2} (\cos |x| + \sin |x|) dx \\
 &= 2 \int_0^{\pi/2} (\cos |x| + \sin |x|) dx \\
 &= 2 \int_0^{\pi/2} (\cos x + \sin x) dx \\
 &= 2(\sin x - \cos x) \Big|_0^{\pi/2} \\
 &= 2((1 + 0) - (0 - 1)) \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 128. \text{ Let } I &= \int_0^2 \left| \cos \left(\frac{\pi x}{2} \right) \right| dx \\
 &= \int_0^1 \left| \cos \left(\frac{\pi x}{2} \right) \right| dx + \int_1^2 \left| \cos \left(\frac{\pi x}{2} \right) \right| dx \\
 &= \int_0^1 \left| \cos \left(\frac{\pi x}{2} \right) \right| dx - \int_1^2 \left| \cos \left(\frac{\pi x}{2} \right) \right| dx \\
 &= \frac{2}{\pi} \left(\sin \left(\frac{\pi x}{2} \right) \right) \Big|_0^1 - \frac{2}{\pi} \left(\sin \left(\frac{\pi x}{2} \right) \right) \Big|_1^2 \\
 &= \frac{2}{\pi} (1 - 0) - \frac{2}{\pi} (0 - 1) \\
 &= \frac{4}{\pi}
 \end{aligned}$$

$$\begin{aligned}
 129. \text{ Let } I &= \int_{-2}^2 f(x) dx \\
 &= \int_{-2}^0 f(x) dx + \int_0^2 f(x) dx \\
 &= \int_{-2}^0 (x - [x]) dx + \int_0^2 (x + |x|) dx \\
 &= \int_{-2}^0 (x) dx - \int_{-2}^0 [x] dx + \int_0^2 (2x) dx \\
 &= \int_{-2}^0 (x) dx - \int_{-2}^{-1} [x] dx - \int_{-1}^0 [x] dx + \int_0^2 (2x) dx \\
 &= -2 + 4 - \int_{-2}^{-1} [x] dx - \int_{-1}^0 [x] dx \\
 &= 2 - [-2(-1 + 2) + (-1)(0 + 1)] \\
 &= -2 + 2 + 1 + 4 \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 130. \text{ Let } I &= \int_{-1}^1 [x[1 + \sin \pi x] + 1] dx \\
 &= \int_{-1}^0 [x[1 + \sin \pi x] + 1] dx
 \end{aligned}$$

$$\begin{aligned}
 & + \int_0^1 [x[1 + \sin \pi x] + 1] \\
 & = \int_{-1}^0 1 dx + \int_0^1 [x + 1] dx \\
 & = \int_{-1}^0 1 dx + \int_0^1 [x] dx + \int_0^1 1 dx \\
 & = \int_{-1}^0 1 dx + \int_0^1 [x] dx \\
 & = \int_{-1}^0 1 dx + \int_0^1 0 dx \\
 & = 1[1 - (-1)] \\
 & = 2 \text{ sq.u.}
 \end{aligned}$$

Notes: 1. If $-1 < x < 0 \Rightarrow [1 + \sin \pi x] = 0$

2. If $0 < x < 1 \Rightarrow [1 + \sin \pi x] = 1$

131. We have, $f(x) = \int_1^x \frac{\ln t}{1+t} dt$... (i)

Now, $f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\ln t}{1+t} dt$

Let $t = \frac{1}{y} \Rightarrow dt = -\frac{1}{y^2} dy$

The given integral reduces to

$$\begin{aligned}
 & = \int_1^y \frac{\ln\left(\frac{1}{y}\right)}{1+\left(\frac{1}{y}\right)} \times -\frac{dy}{y^2} \\
 & = \int_1^y \frac{-\ln(y)}{1+y} \times -\frac{dy}{y} \\
 & = \int_1^y \frac{\ln(y)}{y(1+y)} dy \\
 & = \int_1^t \frac{\ln(t)}{t(1+t)} dt \quad \dots (ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 f(x) + f\left(\frac{1}{x}\right) & = \int_1^x \frac{\ln(t)}{(1+t)} dt + \int_1^x \frac{\ln(t)}{t(1+t)} dt \\
 & = \int_1^x \frac{(1+t) \ln(t)}{t(1+t)} dt \\
 & = \int_1^x \frac{\ln(t)}{t} dt
 \end{aligned}$$

$$= \frac{(\ln t)^2}{2} \Big|_1^x = \frac{(\ln x)^2}{2}$$

Thus, $f(e) + f\left(\frac{1}{e}\right) = \left(\frac{\ln e}{2}\right)^2 = \frac{1}{2}$

132. Let

$$\begin{aligned}
 f(x) & = \int_0^{\sin^2 x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\cos^2 x} \cos^{-1}(\sqrt{t}) dt \\
 \Rightarrow f'(x) & = \sin^{-1}(\sqrt{\sin^2 x}) \sin 2x \\
 & \quad + \cos^{-1}(\sqrt{\cos^2 x}) (-\sin 2x) \\
 & = (\sin^{-1}(|\sin x|) - \cos^{-1}(|\cos x|)) \sin 2x \\
 & = (\sin^{-1}(\sin x) - \cos^{-1}(\cos x)) \sin 2x \\
 & = 0
 \end{aligned}$$

$\Rightarrow f'(x) = 0$

$\Rightarrow f(x) = c$

$\Rightarrow c = \int_c^{\cos x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\sin^2 x} \cos^{-1}(\sqrt{t}) dt$

Let $x = \pi/4$, we get

$$\begin{aligned}
 c & = \int_c^{1/2} \sin^{-1}(\sqrt{t}) dt + \int_0^{1/2} \cos^{-1}(\sqrt{t}) dt \\
 & = \int_c^{1/2} (\sin^{-1}(\sqrt{t}) + \cos^{-1}(\sqrt{t})) dt \\
 & = \int_c^{1/2} \left(\frac{\pi}{2}\right) dt \\
 & = \left(\frac{\pi}{4}\right)
 \end{aligned}$$

133. We have,

$$\begin{aligned}
 & \int_0^2 |x^2 + 2x - 3| dx \\
 & = \int_0^1 |x^2 + 2x - 3| dx + \int_1^0 |x^2 + 2x - 3| dx \\
 & = -\int_0^1 (x^2 + 2x - 3) dx + \int_1^2 (x^2 + 2x - 3) dx \\
 & = -\left(\frac{x^3}{3} + x^2 - 3x\right)_0^1 + \left(\frac{x^3}{3} + x^2 - 3x\right)_1^2 \\
 & = -\left(\frac{1}{3} + 1 - 3\right) + \left(\frac{8}{3} + 4 - 6\right) - \left(\frac{1}{3} + 1 - 3\right) \\
 & = \left(\frac{8}{3} + 4 - 6\right) - 2\left(\frac{1}{3} + 1 - 3\right) \\
 & = \left(\frac{8}{3} - 2\right) - 2\left(\frac{1}{2} - 2\right)
 \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{8}{3} - \frac{2}{3}\right) + (4 - 2) \\
 &= 2 + 2 \\
 &= 4
 \end{aligned}$$

1 34. Do yourself

$$\begin{aligned}
 1 \ 35. & \int_1^{10\pi} ([\sec^{-1}x] + [\cot^{-1}x]) dx \\
 &= \int_0^{\sec 1} ([\sec^{-1}x] + [\cot^{-1}x]) dx \\
 &\quad + \int_{\sec 1}^{10\pi} ([\sec^{-1}x] + [\cot^{-1}x]) dx \\
 &= \int_0^{\sec 1} (0 + 0) dx + \int_{\sec 1}^{10\pi} (1 + 0) dx \\
 &= (10\pi - \sec 1)
 \end{aligned}$$

$$\begin{aligned}
 1 \ 36. & \int_{-\pi/2}^{2\pi} ([\cot^{-1}x]) dx \\
 &= \int_{-\pi/2}^{\cot 2} ([\cot^{-1}x]) dx + \int_{\cot 2}^{\cot 1} ([\cot^{-1}x]) dx + \int_{\cot 1}^{2\pi} ([\cot^{-1}x]) dx \\
 &= \int_{-\pi/2}^{\cot 2} (2) dx + \int_{\cot 2}^{\cot 1} (1) dx + \int_{\cot 1}^{2\pi} (0) dx \\
 &= (\pi + \cot 1 + \cot 2).
 \end{aligned}$$

137. We have,

$$\int_1^2 [2x^2 - 3] dx$$

$$\text{Now, } 2x^2 - 3 = 0 \Rightarrow x = \sqrt{\frac{3}{2}}$$

$$2x^2 - 3 = 1 \Rightarrow x = \sqrt{2}$$

$$2x^2 - 3 = 2 \Rightarrow x = \sqrt{\frac{5}{2}}$$

$$2x^2 - 3 = 3 \Rightarrow x = \sqrt{3}$$

$$2x^2 - 3 = 4 \Rightarrow x = \sqrt{\frac{7}{2}}$$

$$\text{and } 2x^2 - 3 = 5 \Rightarrow x = 2$$

Thus, the given integral can be written as

$$\begin{aligned}
 &\int_1^{\sqrt{3/2}} [2x^2 - 3] dx + \int_{\sqrt{3/2}}^{\sqrt{2}} [2x^2 - 3] dx \\
 &\quad + \int_{\sqrt{2}}^{\sqrt{5/2}} [2x^2 - 3] dx + \int_{\sqrt{5/2}}^{\sqrt{3}} [2x^2 - 3] dx \\
 &\quad + \int_{\sqrt{3}}^{\sqrt{7/2}} [2x^2 - 3] dx + \int_{\sqrt{7/2}}^2 [2x^2 - 3] dx
 \end{aligned}$$

$$\begin{aligned}
 &= \int_1^{\sqrt{3/2}} (-1) dx + \int_{\sqrt{3/2}}^{\sqrt{2}} (0) dx + \int_{\sqrt{2}}^{\sqrt{5/2}} (1) dx \\
 &= \int_{\sqrt{5/2}}^{\sqrt{3}} (2) dx + \int_{\sqrt{3}}^{\sqrt{7/2}} (3) dx + \int_{\sqrt{7/2}}^2 (4) dx \\
 &= (-1) \left(\sqrt{\frac{3}{2}} - 1 \right) + 0 + 1 \cdot \left(\sqrt{\frac{5}{2}} - \sqrt{2} \right) \\
 &\quad + 2 \cdot \left(\sqrt{3} - \sqrt{\frac{5}{2}} \right) + 3 \cdot \left(\sqrt{\frac{7}{2}} - \sqrt{3} \right) \\
 &\quad + 4 \cdot \left(2 - \sqrt{\frac{7}{2}} \right) \\
 &= \left(9 - \left(\sqrt{\frac{3}{2}} + \sqrt{2} + \sqrt{\frac{5}{2}} + \sqrt{3} + \sqrt{\frac{7}{2}} \right) \right) \text{ sq. u.}
 \end{aligned}$$

138. Do yourself

139. We have,

$$\begin{aligned}
 &\int_0^1 [x^2 - x + 1] dx \\
 &= \int_0^1 [x^2 - x + 1] dx + \int_1^{\left(\frac{\sqrt{5}+1}{2}\right)} [x^2 - x + 1] dx \\
 &\quad + \int_{\left(\frac{\sqrt{5}+1}{2}\right)}^2 [x^2 - x + 1] dx \\
 &= \int_0^1 (0) dx + \int_1^{\left(\frac{\sqrt{5}+1}{2}\right)} (1) dx + \int_{\left(\frac{\sqrt{5}+1}{2}\right)}^2 (2) dx \\
 &= \left(\frac{\sqrt{5}+1}{2} \right) + 2 \left(2 - \frac{\sqrt{5}+1}{2} \right) \\
 &= \left| 3 - \frac{\sqrt{5}+1}{2} \right| \\
 &= \left| \frac{5-\sqrt{5}}{2} \right|
 \end{aligned}$$

140. Do yourself

141. Do yourself

142. We have,

$$\begin{aligned}
 &\frac{\int_0^n [x] dx}{\int_0^n \{x\} dx} \\
 &= \frac{\frac{n(n-1)}{2}}{\frac{n^2}{2} - \frac{n(n-1)}{2}} \\
 &= \frac{\frac{n(n-1)}{2}}{\frac{n}{2}} \\
 &= n - 1
 \end{aligned}$$

143. We have,

$$\begin{aligned} \sum_{n=1}^{1000} \int_0^n (e^{x-[x]}) dx \\ &= \int_0^{1000} (e^{x-[x]}) dx \\ &= 1000 \int_0^1 (e^{x-[x]}) dx \\ &= 1000 \int_0^1 (e^{x-0}) dx \\ &= (1000(e^x))_0^1 \\ &= 1000(e - 1) \end{aligned}$$

144. We have,

$$\begin{aligned} \int_{\frac{1}{e}}^{\tan x} \frac{1}{(1+t^2)} dt + \int_{\frac{1}{e}}^{\cot x} \frac{dt}{(1+t^2)} \\ &= \int_{\frac{1}{e}}^{\tan x} \frac{1}{(1+t^2)} dt + \int_e^{\tan x} \frac{dy}{y^2 \left(1 + \frac{1}{y^2}\right)}, \\ &\quad \text{Let } t = \frac{1}{y} \\ &= \int_{\frac{1}{e}}^{\tan x} \frac{1}{(1+t^2)} dt - \int_e^{\tan x} \frac{dy}{(y^2+1)} \\ &= \int_{\frac{1}{e}}^{\tan x} \frac{1}{(1+t^2)} dt - \int_e^{\tan x} \frac{dt}{(t^2+1)} \\ &= \int_{\frac{1}{e}}^{\tan x} \frac{1}{(1+t^2)} dt + \int_{\tan x}^e \frac{dt}{(t^2+1)} \\ &= \int_{\frac{1}{e}}^e \frac{1}{(1+t^2)} dt \\ &= (\tan^{-1}(t))_{1/e}^e \\ &= \tan^{-1}(e) - \tan^{-1}\left(\frac{1}{e}\right) \end{aligned}$$

145. Let $I = \int_0^{\pi/2} \left(\frac{\sin^n x}{\sin^n x + \cos^n x} \right) dx \quad \dots(i)$

Also $I = \int_0^{\pi/2} \left(\frac{\cos^n x}{\cos^n x + \sin^n x} \right) dx \quad \dots(ii)$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} \Rightarrow 2I &= \int_0^{\pi/2} \left(\frac{\sin^n x + \cos^n x}{\cos^n x + \sin^n x} \right) dx \\ &= \int_0^{\pi/2} dx \\ &= \frac{\pi}{2} \end{aligned}$$

$$\Rightarrow I = \frac{\pi}{4}$$

146. Let $I = \int_4^{10} \left(\frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} \right) dx \quad \dots(i)$

$$\begin{aligned} &= \int_0^{14} \left(\frac{[x^2]}{[14-x]^2 + [x^2]} \right) dx \\ &= \int_0^{14} \left(\frac{[(14-x)^2]}{[(x)^2] + [14-x]^2} \right) dx \quad \dots(ii) \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \int_0^{14} \left(\frac{[x^2] + [(14-x)^2]}{[(x)^2] + [14-x]^2} \right) dx \\ &= \int_0^{14} dx \\ &= 14 \end{aligned}$$

$$\Rightarrow I = 7$$

147. Let $I = \int_0^{\infty} \frac{dx}{(1+x^a)(1+x^2)}$

$$= \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1 + (\tan \theta)^a)(1 + \tan^2 \theta)}$$

Let $x = \tan \theta$

$$\begin{aligned} &= \int_0^{\pi/2} \left(\frac{d\theta}{1 + (\tan \theta)^a} \right) \\ &= \int_0^{\pi/2} \left(\frac{\cos^a \theta}{\cos^a \theta + \sin^a \theta} \right) d\theta \quad \dots(i) \end{aligned}$$

$$= \int_0^{\pi/2} \left(\frac{\sin^a \theta}{\sin^a \theta + \cos^a \theta} \right) d\theta \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \int_0^{\pi/2} \left(\frac{\cos^a \theta + \sin^a \theta}{\sin^a \theta + \cos^a \theta} \right) d\theta \\ &= \int_0^{\pi/2} d\theta \\ &= \frac{\pi}{2} \end{aligned}$$

$$\Rightarrow I = \frac{\pi}{4}$$

148. Let $I = \int_0^{\pi/4} (1 + \tan x) dx$

$$= \int_0^{\pi/4} \left(1 + \tan\left(\frac{\pi}{4} - x\right) \right) dx$$

$$\begin{aligned}
 &= \int_0^{\pi/4} \left(1 + \frac{1 - \tan x}{1 + \tan x} \right) dx \\
 &= \int_0^{\pi/4} \left(\frac{1 + \tan x + 1 - \tan x}{1 + \tan x} \right) dx \\
 &= \int_0^{\pi/4} \left(\frac{2}{1 + \tan x} \right) dx \\
 &= \int_0^{\pi/4} \log 2x dx - \int_0^{\pi/4} \log(1 + \tan x) dx \\
 &= \frac{\pi}{4} \log 2 - I
 \end{aligned}$$

$$\Rightarrow 2I = \frac{\pi}{4} \log 2$$

$$\Rightarrow I = \frac{\pi}{8} \log 2$$

149. Let

$$\begin{aligned}
 I &= \int_0^1 \tan^{-1} \left(\frac{2x-1}{1+x-x^2} \right) dx \\
 &= \int_0^1 \tan^{-1} \left(\frac{x+(x-1)}{1-x(x-1)} \right) dx \\
 &= \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1}(1-x) dx \\
 &= \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1} x dx \\
 &= 2 \int_0^1 \tan^{-1} x dx \\
 &= 2 \left(x \tan^{-1} x - \frac{1}{2} \log |1+x^2| \right) \Big|_0^1 \\
 &= 2 \left(\frac{\pi}{4} - \frac{1}{2} \log 2 \right) \\
 &= \left(\frac{\pi}{2} - \log 2 \right) \text{ sq.u.}
 \end{aligned}$$

150. Let

$$\begin{aligned}
 I &= \int_0^1 \cot^{-1}(1-x+x^2) dx \\
 &= \int_0^1 \tan^{-1} \left(\frac{1}{1-x+x^2} \right) dx \\
 &= \int_0^1 \tan^{-1} \left(\frac{x-(x-1)}{1+x(x-1)} \right) dx \\
 &= \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1}(1-x) dx \\
 &= \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1} x dx \\
 &= 2 \int_0^1 \tan^{-1} x dx
 \end{aligned}$$

$$\begin{aligned}
 &= 2 \left(x \tan^{-1} x - \frac{1}{2} \log |1+x^2| \right) \Big|_0^1 \\
 &= 2 \left(\frac{\pi}{4} - \frac{1}{2} \log 2 \right) \\
 &= \left(\frac{\pi}{2} - \log 2 \right) \text{ sq.u.}
 \end{aligned}$$

151. Let $I = \int_0^{\pi} \left(\frac{dx}{1+2^{\tan x}} \right)$... (i)

$$\begin{aligned}
 &= \int_0^{\pi} \left(\frac{dx}{1+2^{\tan(\pi-x)}} \right) \\
 &= \int_0^{\pi} \left(\frac{dx}{1+2^{-\tan x}} \right) \\
 &= \int_0^{\pi} \left(\frac{2^{\tan x} dx}{1+2^{\tan x}} \right) \quad \dots (ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_0^{\pi} \left(\frac{(1+2^{\tan x})}{(1+2^{\tan x})} \right) dx \\
 &= \int_0^{\pi} dx = \pi
 \end{aligned}$$

$$\Rightarrow I = \frac{\pi}{2}$$

152. Let $I = \int_0^1 \log \left(\frac{1}{x} - 1 \right) dx$... (i)

$$\begin{aligned}
 &= \int_0^1 \log \left(\frac{1-x}{x} \right) dx \\
 &= \int_0^1 \log \left(\frac{1-(1-x)}{1-x} \right) dx \\
 &= \int_0^1 \left(\frac{x}{1-x} \right) dx \quad \dots (ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_0^1 \log \left(\frac{1-x}{x} \right) dx + \int_0^1 \log \left(\frac{x}{1-x} \right) dx \\
 &= \int_0^1 \log \left(\frac{1-x}{x} \times \frac{x}{1-x} \right) dx \\
 &= \int_0^1 \log(1) dx \\
 &= 0
 \end{aligned}$$

$$\Rightarrow I = 0$$

153. Let $I = \int_0^1 \frac{\log(1+x)}{(1+x^2)} dx$

$$\text{Let } x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \frac{\log(1 + \tan \theta)}{(1 + \tan^2 \theta)} \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \log(1 + \tan \theta) d\theta \quad \dots(i)$$

$$= \int_0^{\pi/4} \log\left(1 + \tan\left(\frac{\pi}{4} - \theta\right)\right) d\theta$$

$$= \int_0^{\pi/4} \log\left(1 + \frac{1 - \tan \theta}{1 + \tan \theta}\right) d\theta$$

$$= \int_0^{\pi/4} \left(\frac{2}{1 + \tan \theta}\right) d\theta$$

$$= \int_0^{\pi/4} \log(2) d\theta - \int_0^{\pi/4} \log(1 + \tan \theta) d\theta$$

$$= \int_0^{\pi/4} \log(2) d\theta - I$$

$$\Rightarrow 2I = \int_0^{\pi/4} \log 2 d\theta$$

$$= \frac{\pi}{4} \log 2$$

$$\Rightarrow I = \frac{\pi}{8} \log 2$$

154. Let

$$I = \int_0^1 x(1-x)^{2013} dx$$

$$= \int_0^1 (1-x)(1-(1-x))^{2013} dx$$

$$= \int_0^1 (1-x)(x)^{2013} dx$$

$$= \int_0^1 (x^{2013} - x^{2014}) dx$$

$$= \left(\frac{x^{2014}}{2014} - \frac{x^{2015}}{2015}\right)_0^1$$

$$= \left(\frac{1}{2014} - \frac{1}{2015}\right)$$

$$= \left(\frac{2015 - 2014}{2014 \times 2015}\right)$$

$$= \left(\frac{1}{2014 \times 2015}\right)$$

155. Let

$$I = \int_0^{\pi} \left(\frac{x}{1 + \sin x}\right) dx \quad \dots(i)$$

$$= \int_0^{\pi} \left(\frac{\pi - x}{1 + \sin(\pi - x)}\right) dx$$

$$= \int_0^{\pi} \left(\frac{\pi - x}{1 + \sin x}\right) dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_0^{\pi} \left(\frac{\pi}{1 + \sin x}\right) dx$$

$$= \pi \int_0^{\pi} \left(\frac{dx}{1 + \sin x}\right)$$

$$= \pi \int_0^{\pi} \left(\frac{(1 - \sin x) dx}{1 - \sin^2 x}\right)$$

$$= \pi \int_0^{\pi} \left(\frac{(1 - \sin x)}{\cos^2 x}\right) dx$$

$$= \pi \int_0^{\pi} (\sec^2 x - \sec x \tan x) dx$$

$$= \pi (\tan x - \sec x)_0^{\pi}$$

$$= \pi(1 + 1) = 2\pi$$

$$\Rightarrow I = \pi$$

156. Let

$$I = \int_0^{\pi/2} (\pi \sin^2 x) dx \quad \dots(i)$$

$$= \int_0^{\pi/2} \cos\left(\pi \sin^2\left(\frac{\pi}{2} - x\right)\right) dx$$

$$= \int_0^{\pi/2} \cos(\pi \cos^2 x) dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_0^{\pi/2} \cos(\pi \sin^2 x) dx + \int_0^{\pi/2} \cos(\pi \cos^2 x) dx$$

$$= \int_0^{\pi/2} (\cos(\pi \sin^2 x) + \cos(\pi \cos^2 x)) dx$$

$$= \int_0^{\pi/2} 2 \cos\left(\frac{\pi}{2}\right) \cos\left(\frac{\pi}{2} \cos 2x\right) dx$$

$$= 0$$

$$\Rightarrow I = 0$$

157. We have,

$$M = \int_0^1 \left(\frac{e^t}{1+t}\right) dt$$

$$= \left(e^t \int \frac{dt}{1+t}\right)_0^1 - \int_0^1 e^t \log(1+t) dt$$

$$= (e^t \log|1+t|)_0^1 - N$$

$$\Rightarrow M + N = (e \times \log 2)$$

158. We have,

$$a_n = \int_0^{\pi/2} \frac{\cos^2 nx}{\sin x} dx$$

$$\text{Then } a_{n+1} = \int_0^{\pi/2} \left(\frac{\cos^2(n+1)x}{\sin x} \right) dx$$

$$\begin{aligned} \text{Thus, } a_{n+1} - a_n &= \int_0^{\pi/2} \left(\frac{\cos^2(n+1)x - \cos^2 nx}{\sin x} \right) dx \\ &= \frac{1}{2} \int_0^{\pi/2} \left(\frac{2\cos^2(n+1)x - 2\cos^2 nx}{\sin x} \right) dx \\ &= \frac{1}{2} \int_0^{\pi/2} \left(\frac{\cos(2n+2)x - \cos 2nx}{\sin x} \right) dx \\ &= -\frac{1}{2} \int_0^{\pi/2} \left(\frac{2\sin(2n+1)x \cdot \sin x}{\sin x} \right) dx \\ &= -\int_0^{\pi/2} (\sin(2n+1)x) dx \\ &= \left(\frac{\cos(2n+1)x}{2n+1} \right)_0^{\pi/2} \\ &= -\frac{1}{2n+1} \end{aligned}$$

159. Similiar to 145

160. Similiar to 145

161. Similiar to 145

$$\begin{aligned} 162. \text{ Let } I &= \int_0^{\theta} \log(1 + \tan \theta \cdot \tan x) dx, \theta \in \left(0, \frac{\pi}{2}\right) \\ &= \int_0^{\theta} \log(1 + \tan \theta \cdot \tan(\theta - x)) dx \\ &= \int_0^{\theta} \log \left(1 + \tan \theta \left(\frac{\tan \theta - \tan x}{1 + \tan \theta \cdot \tan x} \right) \right) dx \\ &= \int_0^{\theta} \log \left(\frac{1 + \tan^2 \theta}{1 + \tan \theta \cdot \tan x} \right) dx \\ &= \int_0^{\theta} \log(1 + \tan^2 \theta) dx - \int_0^{\theta} \log(1 + \tan \theta \cdot \tan x) dx \\ &= \int_0^{\theta} \log(\sec^2 \theta) dx - I \end{aligned}$$

$$\Rightarrow 2I = 2 \int_0^{\theta} \log(\sec \theta) dx$$

$$\Rightarrow I = \theta \log(\sec \theta)$$

$$\begin{aligned} 163. \text{ Let } I &= \int_0^{\pi/4} \log(1 + \tan x) dx \\ &= \int_0^{\pi/4} \log \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right) dx \\ &= \int_0^{\pi/4} \log \left(\frac{2}{1 + \tan x} \right) dx \end{aligned}$$

$$= \int_0^{\pi/4} \log(2) dx - \int_0^{\pi/4} \log(1 + \tan x) dx$$

$$= \frac{\pi}{4} \log(2) - I$$

$$\Rightarrow 2I = \frac{\pi}{4} \log(2)$$

$$\Rightarrow I = \frac{\pi}{8} \log(2)$$

$$\text{But } k = \int_0^{\pi} \log(\sin x) dx = -\pi \log(2)$$

$$\text{Thus, } I = -\frac{k}{8}$$

$$\begin{aligned} 164. \int_0^{\pi/2} \log \left(\frac{4 + 3 \sin x}{4 + 3 \cos x} \right) dx \\ &= \int_0^{\pi/2} \log(4 + 3 \sin x) dx - \int_0^{\pi/2} \log(4 + 3 \cos x) dx \\ &= \int_0^{\pi/2} \log(4 + 3 \cos x) dx - \int_0^{\pi/2} \log(4 + 3 \cos x) dx \end{aligned}$$

$$\text{where } \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$= 0$$

$$165. \text{ Let } I = \int_0^{\pi/2} \left(\frac{\sin x - \cos x}{1 + \sin x \cdot \cos x} \right) dx \quad \dots(i)$$

$$= \int_0^{\pi/2} \left(\frac{\cos x - \sin x}{1 + \sin x \cos x} \right) dx \quad \dots(ii)$$

Adding (i) and (ii), we get,

$$2I = \int_0^{\pi/2} \left(\frac{\sin x - \cos x + \cos x - \sin x}{1 + \sin x \cos x} \right) dx$$

$$2I = 0$$

$$I = 0$$

$$\int_0^{\pi/2} \left(\frac{\sin x - \cos x}{1 + \sin x \cos x} \right) dx = 0$$

$$166. \int_0^1 x(1-x)^n dx$$

$$= \int_0^1 x((1-x)x^n) dx$$

$$= \int_0^1 (x^n - x^{n+1}) dx$$

$$= \left(\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2} \right)_0^1$$

$$= \frac{1}{(n+1)(n+2)}$$

$$167. \text{ Let } I = \int_0^1 \log \left(\frac{1}{x} - 1 \right) dx$$

$$= \int_0^1 \log\left(\frac{1-x}{x}\right) dx \quad \dots(i)$$

$$= \int_0^1 \log\left(\frac{x}{1-x}\right) dx \quad \dots(ii)$$

Adding (i) and (ii), we get,

$$2I = \int_0^1 \log\left(\frac{1-x}{x} \times \frac{x}{1-x}\right) dx$$

$$2I = \int_0^1 \log(1) dx$$

$$2I = 0$$

$$I = 0$$

$$\int_0^1 \log\left(\frac{1}{x} - 1\right) dx = 0$$

$$\begin{aligned} 168. \text{ Let } I &= \int_0^{\pi} \left(\frac{x \sin x}{1 + \cos^2 x} \right) dx \\ &= \int_0^{\pi} \left(\frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} \right) dx \\ &= \int_0^{\pi} \left(\frac{(\pi - x) \sin(x)}{1 + \cos^2(x)} \right) dx \\ &= \pi \int_0^{\pi} \left(\frac{\sin(x)}{1 + \cos^2(x)} \right) dx - \int_0^{\pi} \left(\frac{x \sin(x)}{1 + \cos^2(x)} \right) dx \\ &= \pi \int_0^{\pi} \left(\frac{\sin(x)}{1 + \cos^2(x)} \right) dx - I \\ \Rightarrow 2I &= \pi \int_0^{\pi} \left(\frac{\sin(x)}{1 + \cos^2(x)} \right) dx \\ &= -\pi (\tan^{-1}(\cos x))_0^{\pi} \\ &= -\pi (\tan^{-1}(-1) - \tan^{-1}(1)) \\ &= -\pi \left(-\frac{\pi}{4} - \frac{\pi}{4} \right) = -\pi \left(-\frac{\pi}{2} \right) \\ \Rightarrow I &= \frac{\pi^2}{4} \end{aligned}$$

$$\begin{aligned} 169. \text{ Let } I &= \int_0^{\pi} \left(\frac{x}{1 + \sin x} \right) dx \\ &= \int_0^{\pi} \left(\frac{(\pi - x)}{1 + \sin(\pi - x)} \right) dx \\ &= \int_0^{\pi} \left(\frac{(\pi - x)}{1 + \sin x} \right) dx \\ &= \pi \int_0^{\pi} \left(\frac{dx}{1 + \sin x} \right) - \int_0^{\pi} \left(\frac{x}{1 + \sin x} \right) dx \\ &= \pi \int_0^{\pi} \left(\frac{dx}{1 + \sin x} \right) - I \end{aligned}$$

$$\begin{aligned} \Rightarrow 2I &= \pi \int_0^{\pi} \left(\frac{dx}{1 + \sin x} \right) \\ &= \pi \int_0^{\pi} \left(\frac{1 - \sin x}{\cos^2 x} \right) dx \\ &= \pi \int_0^{\pi} ((\sec^2 x - \sec x \cdot \tan x) dx) \\ &= \pi (\tan x - \sec x)_0^{\pi} \\ &= \pi [0 - (-1) - (0 - 1)] = 2\pi \\ \Rightarrow I &= \pi \end{aligned}$$

$$\begin{aligned} 170. \text{ Let } I &= \int_0^{\frac{\pi}{2}} \left(\frac{x \sin x \cdot \cos x}{\sin^4 x + \cos^4 x} \right) dx \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{\left(\frac{\pi}{2} - x \right) \sin x \cdot \cos x}{\sin^4 x + \cos^4 x} \right) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\sin x \cdot \cos x}{\sin^4 x + \cos^4 x} \right) dx - \int_0^{\frac{\pi}{2}} \left(\frac{x \sin x \cdot \cos x}{\sin^4 x + \cos^4 x} \right) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\sin x \cdot \cos x}{\sin^4 x + \cos^4 x} \right) dx - I \\ \Rightarrow 2I &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \left(\frac{\sin x \cdot \cos x}{\sin^4 x + \cos^4 x} \right) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \left(\frac{2 \tan x \cdot \sec^2 x}{\tan^4 x + 1} \right) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \left(\frac{2 \tan x \cdot \sec^2 x}{\tan^4 x + 1} \right) dx \\ &= \frac{\pi}{8} \tan^{-1}(\tan^2 x)_0^{\frac{\pi}{2}} \\ \Rightarrow I &= \frac{\pi}{8} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi^2}{16} \end{aligned}$$

171. We have,

$$\begin{aligned} &\int_0^1 \cot^{-1}(1 - x + x^2) dx \\ &= \int_0^1 \tan^{-1} \left(\frac{1}{1 - x + x^2} \right) dx \\ &= \int_0^1 \tan^{-1} \left(\frac{x + 1(1 - x)}{1 - x(1 - x^2)} \right) dx \\ &= \int_0^1 (\tan^{-1}(x) + \tan^{-1}(1 - x)) dx \\ &= \int_0^1 (\tan^{-1}(x)) dx + \int_0^1 (\tan^{-1}(1 - x)) dx \end{aligned}$$

$$= \int_0^1 (\tan^{-1}(x)) dx + \int_0^1 (\tan^{-1}(x)) dx$$

$$= 2 \int_0^1 \tan^{-1}(x) dx$$

$$= 2 \left(x \tan^{-1}(x) - \frac{1}{2} \log|1 + x^2| \right)_0^1$$

$$= 2 \left(x \tan^{-1}(1) - \frac{1}{2} \log(2) \right)$$

$$= 2 \left(\frac{\pi}{4} - \frac{1}{2} \log(2) \right)$$

$$= \left| \frac{\pi}{4} - \log(2) \right|$$

172. We have,

$$\int_0^1 \tan^{-1}(1 - x + x^2) dx$$

$$= \int_0^1 \left(\frac{\pi}{2} - \cot^{-1}(1 - x + x^2) \right) dx$$

$$= \int_0^1 \left(\frac{\pi}{2} - \cot^{-1}(1 - x + x^2) \right) dx$$

$$= \int_0^1 \left(\frac{\pi}{2} \right) dx - \int_0^1 (\cot^{-1}(1 - x + x^2)) dx$$

$$= \frac{\pi}{2} - \left(\frac{\pi}{2} - \log 2 \right)$$

$$= \log 2$$

173. We have,

$$\int_0^{\infty} \frac{dx}{(1 + x^a)(1 + x^2)}, (a > 0)$$

Put $x = \tan \theta$

$$= \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1 + \tan^a \theta) \sec^2 \theta}$$

$$= \int_0^{\pi/2} \frac{d\theta}{(1 + \tan^a \theta)}$$

$$= \int_0^{\pi/2} \frac{\cos^a \theta}{(\cos^a \theta + \sin^a \theta)} d\theta$$

$$= \frac{\pi}{4}$$

174. Let $I = \int_0^{\pi/2} \frac{dx}{\sqrt{\tan x} - \sqrt{\cot x}}$

$$= \int_0^{\pi/2} \frac{\sqrt{\tan x}}{(\tan x - 1)} dx \quad \dots(i)$$

$$= \int_0^{\pi/2} \frac{\sqrt{\cot(x)}}{(\cot(x) - 1)} dx \quad \dots(ii)$$

$$= - \int_0^{\pi/2} \frac{\sqrt{\tan(x)}}{(\tan(x) - 1)} dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{\sqrt{\tan(x)}}{(\tan(x) - 1)} dx - \int_0^{\pi/2} \frac{\sqrt{\tan(x)}}{(\tan(x) - 1)} dx$$

$$\Rightarrow 2I = 0$$

$$\Rightarrow I = 0$$

175. Let $I = \int_0^{\pi} \left(\frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} \right) dx$

$$= \int_0^{\pi} \left(\frac{(\pi - x)}{a^2 \cos^2 x + b^2 \sin^2 x} \right) dx$$

$$= \pi \int_0^{\pi} \left(\frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} \right) - I$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \left(\frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} \right)$$

$$= \pi \int_0^{\pi} \left(\frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x} \right)$$

$$= 2\pi \int_0^{\pi/2} \left(\frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x} \right)$$

$$\Rightarrow I = \int_0^{\pi/2} \left(\frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x} \right)$$

$$= \pi \left(\frac{1}{ab} \tan^{-1}(b \tan x) \right)_0^{\pi/2}$$

$$= \frac{\pi}{ab} \times \frac{\pi}{2} = \frac{\pi^2}{2ab}$$

176. Do yourself

177. Let $I = \int_0^{\pi} \frac{2^{\sin x} \times \cos x}{2^{[\sin x]}} dx$

$$= \int_0^{\pi} \frac{2^{\sin x}}{2^{[\sin x]}} \cos(\pi - x) dx$$

$$= - \int_0^{\pi} \frac{2^{\sin x}}{2^{[\sin x]}} \cos(x) dx$$

$$= -I$$

$$\Rightarrow 2I = 0$$

$$\Rightarrow I = 0$$

178. Let $I = \int_0^{\pi} x \log(\sin x) dx$

$$= \int_0^{\pi} (\pi - x) \log[\sin(\pi - x)] dx$$

$$= \pi \int_0^{\pi} \log[\sin(x)] dx - \int_0^{\pi} x \log[\sin(x)] dx$$

$$= \pi \int_0^{\pi} \log[\sin(x)] dx - I$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \log(\sin(x)) dx$$

$$= 2\pi \int_0^{\pi/2} \log(\sin(x)) dx$$

$$\Rightarrow I = \pi \int_0^{\pi/2} \log(\sin(x)) dx$$

$$= \pi \left(\frac{1}{2} \log\left(\frac{1}{2}\right) \right)$$

$$= \left(\frac{\pi}{2} \log\left(\frac{1}{2}\right) \right)$$

$$179. \text{ For } n > 0, \int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$\text{Let } I = \int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$= \int_0^{2\pi} \frac{(2\pi - x) \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$= 2\pi \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx - \int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$= 2\pi \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx - I$$

$$\Rightarrow 2I = 2\pi \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$\Rightarrow I = \pi \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$= 4\pi \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$= 4\pi \times \left(\frac{\pi}{4} \right)$$

$$= \pi^2$$

180. We have,

$$\int_0^{\pi/2} \frac{\cos x}{1 + \sin x + \cos x} dx$$

$$= \int_0^{2\pi} \left(\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2) + 2 \tan(x/2) + 1 - \tan^2(x/2)} \right)$$

$$= \int_0^1 \left(\frac{1 - t^2}{2(1+t)} \times \frac{2dt}{1+t^2} \right), \quad \text{Let } \tan(x/2) = t$$

$$= -\int_0^1 \left(\frac{t-1}{(1+t^2)} \right) dt$$

$$= -\left(\frac{1}{2} \log|1+t^2| - \tan^{-1}(t) \right)_0^1$$

$$= \left| \frac{\pi}{4} - \frac{1}{2} \log 2 \right|$$

$$181. \text{ Given } I = \int_0^{\pi/2} \frac{dx}{1 + \sin x},$$

we have,

$$\int_0^{\pi} \left(\frac{x^2 \cos x}{(1 + \sin x)^2} \right) dx$$

$$= \left[x^2 \int \left(\frac{\cos x}{(1 + \sin x)^2} \right) dx \right]_0^{\pi} + \int_0^{\pi} \frac{2x}{1 + \sin x} dx$$

$$= \left(-\frac{x^2}{(1 + \sin x)} \right)_0^{\pi} + \int_0^{\pi} \frac{x}{1 + \sin x} dx$$

$$= (-\pi^2) + 2 \int_0^{\pi} \frac{x}{1 + \sin x} dx \quad \dots(i)$$

$$= -\pi^2 + 2 \int_0^{\pi} \frac{(\pi - x)}{1 + \sin x} dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I = -2\pi^2 + 2 \int_0^{\pi} \frac{\pi}{1 + \sin x} dx$$

$$\Rightarrow I = -\pi^2 + \pi \int_0^{\pi} \frac{dx}{1 + \sin x}$$

$$= -\pi^2 + \pi I$$

$$= \pi I - \pi^2$$

182. We have,

$$\lim_{t \rightarrow a} \frac{\int_a^t f(t) dt - \left(\frac{t-a}{2} \right) (f(t) + f(a))}{(t-a)^3} = 0$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{f(t) - \frac{1}{2} (f(t) + f(a)) - \frac{(t-a)}{2} f'(a)}{3(t-a)^2} = 0$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{2f(t) - [f(t) + f(a)] - (t-a)f'(a)}{6(t-a)^2} = 0$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{[f(t) - f(a)] - (t-a)f'(a)}{6(t-a)^2} = 0$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{f'(t) - f'(a)}{12(t-a)} = 0$$

$$\Rightarrow \lim_{t \rightarrow a} \frac{f''(t)}{12} = 0$$

$$\Rightarrow f''(a) = 0$$

$$\Rightarrow f'(a) = c$$

$$\Rightarrow f(a) = cx + k$$

Clearly, $f(x)$ is of almost degree 1.

$$183. \text{ Let } I = \int_a^b \left(\frac{f(x)}{f(x) + f(a+b-x)} \right) dx \quad \dots(i)$$

$$\text{Also } I = \int_a^b \left(\frac{f(a+b-x)}{f(a+b-x) + f(x)} \right) dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \int_a^b \left(\frac{f(x) + f(a+b-x)}{f(a+b-x) + f(x)} \right) dx \\ &= \int_a^b dx \\ &= (b-a) \end{aligned}$$

$$\Rightarrow I = \left(\frac{b-a}{2} \right)$$

$$184. \text{ Let } I = \int_3^5 \left(\frac{[x^2]}{[x^2 - 16x + 64] + [x^2]} \right) dx \quad \dots(i)$$

$$\begin{aligned} &= \int_3^5 \left(\frac{[x^2]}{[8-x]^2 + [x^2]} \right) dx \\ &= \int_3^5 \left(\frac{[(8-x)^2]}{[(x)^2] + [(8-x)^2]} \right) dx \quad \dots(ii) \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \int_3^5 \left(\frac{[x^2] + [(8-x)^2]}{[(x)^2] + [(8-x)^2]} \right) dx \\ &= \int_3^5 dx \\ &= (5-3) = 2 \end{aligned}$$

$$\Rightarrow I = 1$$

$$185. \text{ Let } I = \int_{-2}^2 \left(\frac{dx}{1+3^x} \right) \quad \dots(i)$$

$$\text{Also } I = \int_{-2}^2 \left(\frac{dx}{1+3^x} \right) \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get,

$$\begin{aligned} 2I &= \int_{-2}^2 \left(\frac{1+3^x}{1+3^x} \right) dx \\ &= \int_{-2}^2 dx \\ &= |2 + 2| = 4 \end{aligned}$$

$$\Rightarrow I = 2$$

$$186. \text{ Let } I = \int_{\pi/6}^{5\pi/6} \left(\frac{dx}{1+e^{\tan x}} \right) \quad \dots(i)$$

$$\begin{aligned} &= \int_{\pi/6}^{5\pi/6} \left(\frac{dx}{1+e^{-\tan x}} \right) \\ &= \int_{\pi/6}^{5\pi/6} \left(\frac{e^{\tan x}}{e^{\tan x} + 1} \right) dx \quad \dots(ii) \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \left(\frac{e^{\tan x} + 1}{e^{\tan x} + 1} \right) dx = \int_{\pi/6}^{5\pi/6} dx \\ &= \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) = \frac{2\pi}{3} \end{aligned}$$

$$\Rightarrow I = \frac{\pi}{3}$$

$$187. \text{ Let } I = \int_{-\pi}^{\pi} \left(\frac{\cos^2 x}{1+a^x} \right) dx \quad \dots(i)$$

$$\begin{aligned} &= \int_{-\pi}^{\pi} \left(\frac{\cos^2(-x)}{1+a^{-x}} \right) dx \\ &= \int_{-\pi}^{\pi} \left(\frac{a^2 \cos^2(x)}{1+a^x} \right) dx \quad \dots(ii) \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \int_{-\pi}^{\pi} \left(\frac{(1+a^x) \cos^2(x)}{(1+a^x)} \right) dx \\ &= \int_{-\pi}^{\pi} \cos^2(x) dx \\ &= 2 \int_{-\pi}^{\pi} \cos^2(x) dx \\ &= \int_0^{\pi} (2 \cos^2(x)) dx \\ &= \int_0^{\pi} (1 + \cos 2x) dx \\ &= \left(x + \frac{\sin 2x}{2} \right)_0^{\pi} \\ &= \pi \end{aligned}$$

$$\Rightarrow I = \pi/2$$

$$188. \text{ Let } I = \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \cos^{-1} \left(\frac{2x}{1+x^2} \right) \right) dx$$

$$= \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \cos^{-1} \left(\frac{2(-x)}{1+x^2} \right) \right) dx$$

$$\begin{aligned}
 &= \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \cos^{-1} \left(-\frac{2(x)}{1+x^2} \right) \right) dx \\
 &= \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \cos^{-1} \left(\pi - \cos^{-1} \left(\frac{2(x)}{1+x^2} \right) \right) \right) dx \\
 &= \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{\pi x^4}{1-x^4} - \frac{x^4}{1-x^4} \cos^{-1} \left(\frac{2(x)}{1+x^2} \right) \right) dx \\
 &= \int_{-1/\sqrt{3}}^{1/\sqrt{3}} (\pi x^4 / 1 - x^4) dx \\
 &\quad - \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \cos^{-1} \left(\frac{2(x)}{1+x^2} \right) \right) dx
 \end{aligned}$$

$$= \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{\pi x^4}{1-x^4} \right) dx - I$$

$$\Rightarrow 2I = \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \left(\frac{\pi x^4}{1-x^4} \right) dx$$

$$= \int_0^{1/\sqrt{3}} \left(\frac{\pi x^4}{1-x^4} \right) dx$$

$$\Rightarrow I = \int_0^{1/\sqrt{3}} \left(\frac{\pi x^4}{1-x^4} \right) dx$$

$$= \int_0^{1/\sqrt{3}} \left(\frac{x^4}{1-x^4} \right) dx$$

$$\Rightarrow k = \pi$$

$$\begin{aligned}
 189. \text{ Given } I_1 &= \int_{1-k}^k x \cdot f(x(1-x)) dx \\
 &= \int_1^k (1-x) f[(1-x)x] dx \\
 &= \int_1^k f[(1-x)x] dx - \int_1^k x f[(1-x)x] dx \\
 &= I_2 - I_1
 \end{aligned}$$

$$\Rightarrow I_1 = I_2 - I_1$$

$$\Rightarrow 2I_1 = I_2$$

$$\Rightarrow I_1/I_2 = 1/2$$

$$190. \text{ Let } I = \int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \left(\frac{x \sin(x^2)}{\sin x^2 + \sin(\ln(6-x^2))} \right) dx$$

$$\text{Let } x^2 = t \Rightarrow x dx = 1/2 dt$$

$$= \frac{1}{2} \int_{\ln 2}^{\ln 3} \left(\frac{\sin t}{\sin t + \sin(\ln 6 - t)} \right) dt \quad \dots(i)$$

$$= \frac{1}{2} \int_{\ln 2}^{\ln 3} \left(\frac{\sin(\ln 6 - t)}{\sin(\ln 6 - t) + \sin t} \right) dt \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_{\ln 2}^{\ln 3} \left(\frac{\sin(t) + \sin(\ln 6 - t)}{\sin(\ln 6 - t) + \sin(t)} \right) dt$$

$$= \frac{1}{2} \int_{\ln 2}^{\ln 3} dt$$

$$= \frac{1}{2} (\ln 3 - \ln 2)$$

$$= \frac{1}{2} \left(\ln \left(\frac{3}{2} \right) \right)$$

$$\Rightarrow I = \frac{1}{4} \left(\ln \left(\frac{3}{2} \right) \right)$$

$$191. \text{ Let } I_n = \int_{-\pi}^{\pi} \left(\frac{\sin nx}{(1 + \pi^x) \sin x} \right) dx$$

$$= \int_{-\pi}^{\pi} \left(\frac{\sin nx}{(1 + \pi^x) \sin x} \right) dx \quad \dots(i)$$

$$= \int_{-\pi}^{\pi} \left(\frac{\sin n(-x)}{(1 + \pi^{-x}) \sin(-x)} \right) dx$$

$$= \int_{-\pi}^{\pi} \left(\frac{\sin n(x)}{(1 + \pi^{-x}) \sin(x)} \right) dx$$

$$= \int_{-\pi}^{\pi} \left(\frac{\pi x \sin n(x)}{(1 + \pi^x) \sin(x)} \right) dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I_n = \int_{-\pi}^{\pi} \left(\frac{(1 + \pi^x) \sin n(x)}{(1 + \pi^x) \sin(x)} \right) dx$$

$$= \int_{-\pi}^{\pi} \left(\frac{\sin n(x)}{\sin(x)} \right) dx$$

$$= 2 \int_0^{\pi} \left(\frac{\sin n(x)}{\sin(x)} \right) dx$$

$$\Rightarrow I_n = \int_0^{\pi} \left(\frac{\sin n(x)}{\sin(x)} \right) dx$$

$$I_{n+2} - I_n = \int_0^{\pi} \left(\frac{\sin(n+2)x - \sin nx}{\sin x} \right) dx$$

$$= \int_0^{\pi} \left(\frac{2 \cos(n+1)x \times \sin x}{\sin x} \right) dx$$

$$= \int_0^{\pi} 2 \cos(n+1)x dx$$

$$= 2 \left(\frac{\sin(n+1)x}{n+1} \right)_0^\pi = 0$$

$$\Rightarrow I_{n+2} = I_n$$

$$\text{Now, } I_0 = 0 \text{ and } I_1 = \int_0^\pi \left(\frac{\sin x}{\sin x} \right) dx = \pi$$

$$\text{Since } I_{n+2} = I_n, \text{ so}$$

$$I_1 = I_3 = I_5 = \dots = I_{2n-1} = \pi$$

$$\text{and } I_0 = I_2 = I_4 = \dots = I_{2n} = 0$$

$$\text{Thus, } \sum_{m=1}^{10} I_{2m+1} = I_3 + I_5 + \dots + I_{21} = 10\pi$$

$$\text{and } \sum_{m=1}^{10} I_{2m} = I_2 + I_4 + \dots + I_{20} = 0$$

192. Let $I = \int_0^1 \left(\frac{x^4(1-x)^4}{1+x^2} \right) dx$

$$= \int_0^1 \left(\frac{x^4(x^4 - 4x^3 + 6x^2 - 4x + 1)}{x^2 + 1} \right) dx$$

$$= \int_0^1 \left(\frac{x^8 - 4x^7 + 6x^6 - 4x^5 + x^4}{x^2 + 1} \right) dx$$

$$= \int_0^1 \left(\frac{x^8 - 6x^6 + x^4}{x^2 + 1} \right) dx$$

$$= \int_0^1 \left(\frac{x^6(x^2 + 1) + x^4(x^2 + 1) + 4x^6}{x^2 + 1} \right) dx$$

$$= \int_0^1 \left(\frac{(x^2(x^2 + 1)(x^2 + 1) + 4x^6)}{x^2 + 1} \right) dx$$

$$= \int_0^1 \left((x^4 + x^2) + \frac{4(x^6 + 1 - 1)}{x^2 + 1} \right) dx$$

$$= \int_0^1 \left((x^4 + x^2) - \frac{4}{x^2 + 1} + 4(x^4 - x^2 + 1) \right) dx$$

$$= \int_0^1 \left((5x^4 - 3x^2 + 4) - \frac{4}{x^2 + 1} \right) dx$$

$$= (x^5 - x^3 + 4x - 4 \tan^{-1} x)_0^1$$

$$= [4 - 4 \tan^{-1}(1)] = (4 - \pi)$$

193. Given $f(a+b-x) = f(x)$,
then prove that

$$\int_a^b xf(x) = \frac{a+b}{2} \int_a^b f(x) dx$$

Let $I = \int_a^b xf(x) dx$... (i)

$$= \int_a^b (a+b-x)f(a+b-x) dx$$

$$= \int_a^b (a+b-x)f(x) dx \quad \dots (ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_a^b (a+b)f(x) dx$$

$$\Rightarrow I = \int_a^b \left(\frac{a+b}{2} \right) f(x) dx$$

194. Let $I = \int_4^{10} \frac{[x^2]}{[x^2 - 28x + 196] + [x^2]} dx$... (i)

$$= \int_4^{10} \left(\frac{[x^2]}{[(14-x)^2 + x^2]} \right) dx$$

$$= \int_4^{10} \left(\frac{[14-x]^2}{[(14-x)^2 + x^2]} \right) dx$$

$$= \int_4^{10} \left(\frac{[14-x]^2 + [x^2]}{[(14-x)^2 + x^2]} \right) dx \quad \dots (ii)$$

$$\Rightarrow 2I = \int_4^{10} \left(\frac{[14-x]^2 + [x^2]}{[(14-x)^2 + x^2]} \right) dx$$

$$= \int_4^{10} dx$$

$$= (10 - 4) = 6$$

$$\Rightarrow I = 3$$

195. Let $I = \int_{\pi/4}^{3\pi/4} \left(\frac{dx}{1 + \cos x} \right)$... (i)

$$= \int_{\pi/4}^{3\pi/4} \left(\frac{dx}{1 + \cos(\pi - x)} \right)$$

$$= \int_{\pi/4}^{3\pi/4} \left(\frac{dx}{1 + \cos(\pi - x)} \right) \quad \dots (ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_{\pi/4}^{3\pi/4} \left(\frac{1}{1 + \cos(x)} + \frac{1}{1 - \cos(x)} \right) dx$$

$$= \int_{\pi/4}^{3\pi/4} \left(\frac{2}{1 + \cos^2(x)} \right) dx$$

$$\Rightarrow I = \int_{\pi/4}^{3\pi/4} \left(\frac{dx}{1 - \cos^2(x)} \right)$$

$$\begin{aligned}
 &= \int_{\pi/4}^{3\pi/4} (\operatorname{cosec}^2 x) dx \\
 &= (-\cot x) \Big|_{\pi/4}^{3\pi/4} = -(-1-1) = 2
 \end{aligned}$$

196. Let $I = \int_{-\pi/2}^{\pi/2} \left(\frac{\cos x}{1+e^x} \right) dx$... (i)

$$\begin{aligned}
 &= \int_{-\pi/2}^{\pi/2} \left(\frac{\cos(-x)}{1+e^{-x}} \right) dx \\
 &= \int_{-\pi/2}^{\pi/2} \left(\frac{\cos(x)}{1+e^{-x}} \right) dx \\
 &= \int_{-\pi/2}^{\pi/2} \left(\frac{e^x \cos(x)}{1+e^x} \right) dx \quad \dots (ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 \Rightarrow 2I &= \int_{-\pi/2}^{\pi/2} \left(\frac{(1+e^x)\cos(x)}{1+e^x} \right) dx \\
 &= \int_{-\pi/2}^{\pi/2} (\cos(x)) dx \\
 &= (\sin x) \Big|_{-\pi/2}^{\pi/2}
 \end{aligned}$$

$$\Rightarrow I = 1$$

197. Let $I = \int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x} + 1}$... (i)

$$\begin{aligned}
 &= \int_{-\pi/2}^{\pi/2} \frac{dx}{e^{-\sin x} + 1} \\
 &= \int_{-\pi/2}^{\pi/2} \left(\frac{e^{\sin x}}{e^{\sin x} + 1} \right) dx \quad \dots (ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_{-\pi/2}^{\pi/2} \left(\frac{e^{\sin x} + 1}{e^{\sin x} + 1} \right) dx \\
 &= \int_{-\pi/2}^{\pi/2} dx \\
 &= \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = \pi
 \end{aligned}$$

$$\Rightarrow I = \left(\frac{\pi}{2} \right)$$

198. Let $I = \int_{50}^{100} \frac{\ln x}{\ln x + \ln(150-x)} dx$... (i)

$$= \int_{50}^{100} \frac{\ln(150-x)}{\ln(150-x) + \ln(x)} dx \quad \dots (ii)$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_{50}^{100} \frac{\ln(150-x) + \ln(x)}{\ln(150-x) + \ln(x)} dx \\
 &= \int_{50}^{100} dx \\
 &= (100 - 50) \\
 &= 50
 \end{aligned}$$

$$\Rightarrow I = 25$$

199. Given $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} xf(x(2-x)) dx$... (i)

$$= \int_{\sin^2 t}^{1+\cos^2 t} (2-x)f((2-x)x) dx \quad \dots (ii)$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_{\sin^2 t}^{1+\cos^2 t} (2-x+x)f((2-x)x) dx \\
 &= \int_{\sin^2 t}^{1+\cos^2 t} 2f((2-x)x) dx
 \end{aligned}$$

$$\Rightarrow I_1 = \int_{\sin^2 t}^{1+\cos^2 t} f((2-x)x) dx = I_2$$

$$= I_2$$

$$\Rightarrow \frac{I_1}{I_2} = 1$$

200. Given $I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\} dx$... (i)

Also $= \int_{f(-a)}^{f(a)} (1-x)g\{(1-x)x\} dx$... (ii)

Adding Eqs (i) and (ii), we get,

$$2I = \int_{f(-a)}^{f(a)} g\{(1-x)x\} dx$$

$$\Rightarrow 2I_1 = I_2$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{1}{2}$$

201. Do yourself

202. Do yourself.

203. Let $f(x) = \sin^5 x$

$$\Rightarrow f(-x) = \sin^5(-x) = -\sin^5 x$$

$\therefore f(x)$ is an odd function.

Thus, the value of the given integral

$$= \int_{-\pi/2}^{\pi/2} \sin^5 x dx = 0$$

204. Let $f(x) = x^3 \tan^4 x$

$$\Rightarrow f(-x) = (-x)^3 \tan^4(-x)$$

$$= -x^3 \tan^4 x$$

$\therefore f(x)$ is an odd function.

Thus, the value of the given integral

$$= \int_{-\pi/4}^{\pi/4} x^3 \tan^4 x dx = 0$$

205. Let $f(x) = \log\left(\frac{2 - \sin x}{2 + \sin x}\right)$

$$\Rightarrow f(-x) = \log\left(\frac{2 + \sin x}{2 - \sin x}\right)$$

$$= -\log\left(\frac{2 - \sin x}{2 + \sin x}\right) = -f(x)$$

$\therefore f(x)$ is an odd function.

Thus, the value of the given integral is

$$= \int_{-2}^2 \log\left(\frac{2 - \sin x}{2 + \sin x}\right) dx = 0$$

206. Let $I = \int_0^2 x(x-1)(x-2) dx$

$$= \int_{-1}^1 (t-1)t(t+1) dt, \text{ where } x-1 = t$$

$$= \int_{-1}^1 t(t^2-1) dt$$

$$= \int_{-1}^1 x(x^2-1) dx$$

$$= 0, \text{ since it is an odd function.}$$

207. Let

$$I = \int_{2010}^{2014} [(x-2010)(x-2011)(x-2012)(x-2013)(x-2014)] dx$$

Let $x - 2012 = t$

$$= \int_{-2}^2 (t+2)(t+1)(t-1)(t-2) dt$$

$$= \int_{-2}^2 (t^2-4)(t^2-1) dt$$

$$= \int_{-2}^2 x(x^2-4)(x^2-1) dx$$

$$= 0, \quad (\because \text{it is an odd function.})$$

208. Let

$$I = \int_{-\pi/4}^{\pi/4} \left(\frac{x^{2013} - 3x^{2011} + 5x^{2009} - 7x^{2007} + 1007}{\cos^2 x} \right) dx$$

$$= \int_{-\pi/4}^{\pi/4} \left(\frac{x^{2013} - 3x^{2011} + 5x^{2009} - 7x^{2007}}{\cos^2 x} \right) dx$$

$$+ \int_{-\pi/4}^{\pi/4} \left(\frac{1007}{\cos^2 x} \right) dx$$

$$= 0 + \int_{-\pi/4}^{\pi/4} \left(\frac{1007}{\cos^2 x} \right) dx$$

$$= 2 \int_0^{\pi/4} \left(\frac{1007}{\cos^2 x} \right) dx$$

$$= 2014 \int_0^{\pi/4} \sec^2 x dx$$

$$= 2014 \times (\tan x)_0^{\pi/4}$$

$$= 2014 \times (1 - 0) = 2014$$

209. Given $f(x) = \frac{\sin^{2013} x}{x^{2014} - x^{2012} + 1}$

$$\Rightarrow f(-x) = \frac{-\sin^{2013} x}{x^{2014} - x^{2012} + 1}$$

$$= -f(x)$$

$\therefore f(x)$ is an odd function.

Hence, the value of the given integral

$$= \int_{-2010}^{2010} f(x) dx = 0$$

210. Let

$$I = \int_{-10}^{10} \left(\tan^{2013} x + \left(\frac{2013 \sqrt{x}}{1 + x^{2014}} \right) + 1007 \right) dx$$

$$= \int_{-10}^{10} (0 + 0 + 1007) dx$$

$$= 2 \int_0^{10} (2007) dx$$

$$= 2014 \times (10 - 0)$$

$$= 20140$$

$$211. \text{ Let } I = \int_{\sqrt{\ln(1/2)}}^{\sqrt{\ln(2)}} x^3 \left(\frac{x^2}{e^{x^2} - 1} + \frac{x^2}{2} + 1 \right) dx$$

$$= \frac{1}{2} \int_{\ln(1/2)}^{\ln(2)} t \left(\frac{t}{e^t - 1} + \frac{t}{2} + 1 \right) dt$$

$$= \frac{1}{2} \int_{-\ln(2)}^{\ln 2} x \left(\frac{x}{e^x - 1} + \frac{x}{2 + 1} \right) dx$$

$$\text{Let } f(x) = x \left(\frac{x}{e^x - 1} + \frac{x}{2 + 1} \right)$$

$$\Rightarrow f(-x) = -x \left(\frac{-x}{e^{-x} - 1} - \frac{x}{2 + 1} \right)$$

$$= -x \left(\frac{x}{1 - e^{-x}} - \frac{x}{2 + 1} \right)$$

$$= -x \left(\frac{x e^x}{e^x - 1} - \frac{x}{2} + 1 \right)$$

$$= -x \left(\left(\frac{x e^x}{e^x - 1} - x \right) + \frac{x}{2} + 1 \right)$$

$$= -x \left(\left(\frac{x e^x - x e^x + x}{e^x - 1} \right) + \frac{x}{2} + 1 \right)$$

$$= -x \left(\frac{x}{e^x - 1} + \frac{x}{2} + 1 \right)$$

$$= -f(x)$$

$\therefore f(x)$ is an odd function.

$\Rightarrow$ The value of the given Integral is zero.

$$I = 0, (\because \text{it is an odd function})$$

212. The given integral can be written as

$$\int_{-2}^0 ((x+1)^3 + 2 + (x+1)\cos(x+1)) dx$$

$$\text{Let } (x+1) = t$$

$$= \int_{-1}^2 (t^3 + 2 + t \cos t) dt$$

$$= \int_{-1}^1 (t^3 + t \cos t) dt + \int_{-1}^1 2 dt$$

$$= 0 + \int_{-1}^1 2 dt$$

$$= 2[1 - (-1)] = 4$$

213. We have,

$$f(x) = \begin{cases} 1-x & : 0 \leq x < 1 \\ x-1 & : 1 \leq x < 2 \end{cases}$$

and f is periodic with period 2.

Also, since $[-x] = 1 - [x]$ for non integral values of x

So, f is an even function.

$$\text{Thus, } \int_{-10}^{10} f(x) \cos(\pi x) dx$$

$$= 2 \int_0^{10} f(x) \cos(\pi x) dx$$

$$= 2.5 \int_0^2 f(x) \cos(\pi x) dx$$

$$= 10 \int_0^2 f(x) \cos(\pi x) dx$$

$$= 10(I + J),$$

$$\text{Let } I = \int_0^1 f(x) \cos(\pi x) dx$$

$$\text{and } J = \int_1^2 f(x) \cos(\pi x) dx$$

$$\text{Now, } I = \int_0^1 f(x) \cos(\pi x) dx$$

$$= \int_0^1 (1-x) \cos(\pi x) dx$$

$$= \int_0^1 (1-(1-x)) \cos(\pi(1-x)) dx$$

$$= -\int_0^1 x \cos(\pi x) dx$$

$$\text{Also, } J = \int_1^2 (x-1) \cos(\pi x) dx$$

$$= \int_0^1 t \cos(\pi(t+1)) dt, \text{ where } x-1 = t$$

$$= -\int_0^1 t \cos(\pi t) dt$$

$$= -\int_0^1 x \cos(\pi x) dx$$

$$= I$$

$$\text{Thus, } I = J = -\int_0^1 x \cos(\pi x) dx$$

$$= \left(-\frac{x}{\pi} \sin(\pi x) \right)_0^1 + \frac{1}{\pi} \int_0^1 \sin(\pi x) dx$$

$$= -\frac{1}{\pi^2} (\cos \pi x)_0^1$$

$$= \frac{2}{\pi^2}$$

$$\begin{aligned}\text{Therefore, } & \int_{-10}^{10} f(x) \cos(\pi x) dx \\ &= 10(I + J) \\ &= 20I \\ &= 20 \times \frac{2}{\pi^2}\end{aligned}$$

$$\text{Now, } \frac{\pi^2}{10} \times \int_{10}^{10} f(x) \cos(\pi x) dx = \frac{\pi^2}{10} \times \frac{40}{\pi^2} = 4$$

214. Let

$$\begin{aligned}I &= \int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5) dx \\ &= \int_{-2}^2 (t+2)(t+1)t(t-1)(t-2) dt \\ &= \int_{-2}^2 t(t^2-4)(t^2-1) dt \\ &= 0, \quad (\because \text{it is an odd function.})\end{aligned}$$

$$\begin{aligned}\text{215. Let } I &= \int_{-\pi}^{\pi} \left(\frac{2x(1+\sin x)}{1+\cos^2 x} \right) dx \\ &= \int_{-\pi}^{\pi} \left(\frac{2x}{1+\cos^2 x} \right) dx + \int_{-\pi}^{\pi} \left(\frac{2x \sin x}{1+\cos^2 x} \right) dx \\ &= 0 + 4 \int_0^{\pi} \left(\frac{x \sin x}{1+\cos^2 x} \right) dx \\ &= 4 \int_0^{\pi} \left(\frac{(\pi-x) \sin(\pi-x)}{1+\cos^2(\pi-x)} \right) dx \\ &= 4 \int_0^{\pi} \left(\frac{(\pi-x) \sin x}{1+\cos^2 x} \right) dx \\ &= 4\pi \int_0^{\pi} \left(\frac{\sin x}{1+\cos^2 x} \right) dx - 4 \int_0^{\pi} \left(\frac{x \sin x}{1+\cos^2 x} \right) dx \\ &= 4\pi \int_0^{\pi} \left(\frac{\sin x}{1+\cos^2 x} \right) dx - I \\ \Rightarrow 2I &= 4\pi \int_0^{\pi} \left(\frac{\sin x}{1+\cos^2 x} \right) dx \\ \Rightarrow I &= 2\pi \int_0^{\pi} \left(\frac{\sin x}{1+\cos^2 x} \right) dx \\ &= 4\pi \int_0^{\pi/2} \left(\frac{\sin x}{1+\cos^2 x} \right) dx\end{aligned}$$

$$\begin{aligned}&= -4\pi \int_1^0 \frac{dt}{1+t^2} \\ &= -4\pi (\tan^{-1} t)_1^0 \\ &= -4\pi \left(-\frac{\pi}{4} \right) \\ &= 4 \times \frac{\pi^2}{4} \\ &= \pi^2\end{aligned}$$

216. Let

$$\begin{aligned}I &= \int_{-\pi/3}^{\pi/3} \left(\frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} \right) dx \\ &= \int_{-\pi/3}^{\pi/3} \left(\frac{\pi}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} \right) dx \\ &\quad + \int_{-\pi/3}^{\pi/3} \left(\frac{4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} \right) dx \\ &= \int_{-\pi/3}^{\pi/3} \left(\frac{\pi}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} \right) dx + 0 \\ &= 2 \int_0^{\pi/3} \left(\frac{\pi}{2 - \cos\left(x + \frac{\pi}{3}\right)} \right) dx \\ &= 2\pi \int_0^{\pi/3} \left(\frac{dx}{2 - \cos\left(x + \frac{\pi}{3}\right)} \right) \\ &= 2\pi \int_{\pi/3}^{2\pi/3} \left(\frac{dt}{2 - \cos t} \right), \quad t = x + \pi/3 \\ &= 2\pi \int_{\pi/3}^{2\pi/3} \left(\frac{dt}{1 + (1 - \cos t)} \right) \\ &= 2\pi \int_{\pi/3}^{2\pi/3} \left(\frac{dt}{1 + 2\sin^2(t/2)} \right) \\ &= 4\pi \int_{\pi/6}^{\pi/3} \left(\frac{dy}{1 + 2\sin^2 y} \right), \quad \text{Let } t/2 = y \\ &= 4\pi \int_{\pi/6}^{\pi/3} \left(\frac{\sec^2 y dy}{\sec^2 y + 2\tan^2 y} \right) \\ &= 4\pi \int_{\pi/6}^{\pi/3} \left(\frac{\sec^2 y}{1 + 3\tan^2 y} \right) dy \\ &= \frac{4\pi}{3} \int_{\pi/6}^{\pi/3} \left(\frac{\sec^2 y}{(1/\sqrt{3})^2 + \tan^2 y} \right) dy\end{aligned}$$

$$\begin{aligned}
 &= \frac{4\pi}{3} \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{dv}{(1/\sqrt{3})^2 + v^2} \\
 &= \frac{4\pi}{3} \times \sqrt{3} (\tan^{-1}(v\sqrt{3}))_{1/\sqrt{3}}^{\sqrt{3}} \\
 &= \frac{4\pi}{\sqrt{3}} \left(\tan^{-1}(3) - \frac{\pi}{4} \right)
 \end{aligned}$$

217. Let

$$\begin{aligned}
 I &= \int_{-1}^1 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right) + \tan^{-1}\left(\frac{1+x^2}{x}\right) \right) dx \\
 &= \int_{-1}^0 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right) + \tan^{-1}\left(\frac{1+x^2}{x}\right) \right) dx \\
 &\quad + \int_0^1 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right) + \tan^{-1}\left(\frac{1+x^2}{x}\right) \right) dx \\
 &= \int_{-1}^0 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right) + \cot^{-1}\left(\frac{x}{1+x^2}\right) \right) dx \\
 &\quad + \int_0^1 \left(\tan^{-1}\left(\frac{x}{1+x^2}\right) + \cot^{-1}\left(\frac{x}{1+x^2}\right) \right) dx \\
 &= \int_{-1}^0 \left(-\pi + \frac{\pi}{2} \right) dx + \int_0^1 \left(\frac{\pi}{2} \right) dx \\
 &= \int_{-1}^0 \left(-\frac{\pi}{2} \right) dx + \int_0^1 \left(\frac{\pi}{2} \right) dx \\
 &= \left(-\frac{\pi}{2} \right) (0 - (-1)) + \left(\frac{\pi}{2} \right) (1 - 0) \\
 &= \left| -\frac{\pi}{2} + \frac{\pi}{2} \right| \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 237. \text{ Let } I &= \int_0^{\pi} \sin x \, dx \\
 &= 2 \int_0^{\pi/2} \sin x \, dx \quad [\because \sin(\pi - x) = \sin x] \\
 &= 2(-\cos x)_0^{\pi/2} \\
 &= 2[-(0 - 1)] \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 238. \text{ Let } I &= \int_0^{\pi} \cos x \, dx \\
 &= 0 \quad [\because \cos(\pi - x) = -\cos x]
 \end{aligned}$$

$$\begin{aligned}
 239. \text{ Let } I &= \int_0^{\pi} \cos^{2103} x \, dx \\
 &= 0 \quad [\because \cos(\pi - x) = -\cos x]
 \end{aligned}$$

$$\begin{aligned}
 240. \text{ Let } I &= \int_0^{\pi} (\sin^3 x + \cos^5 x + \tan^7 x) dx \\
 &= \int_0^{\pi} (\sin^3 x) dx + \int_0^{\pi} (\cos^5 x) dx + \int_0^{\pi} (\tan^7 x) dx \\
 &= \int_0^{\pi} (\sin^3 x) dx + 0 + 0 \\
 &= \int_0^{\pi} (\sin^3 x) dx \\
 &= \frac{1}{4} \int_0^{\pi} (4 \sin^3 x) dx \\
 &= \frac{1}{2} \int_0^{\pi/2} (3 \sin x - \sin^3 x) dx \\
 &= \frac{1}{2} \left(-3 \cos x + \frac{\cos^3 x}{3} \right)_0^{\pi/2} \\
 &= \frac{1}{2} \left(-3 \left(0 - \left(1 + \frac{1}{3} \right) \right) \right) \\
 &= \frac{1}{2} \left(-3 \times -\frac{4}{3} \right) \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 241. \text{ Let } I &= \int_0^{\pi/2} \log \sin x \, dx \quad \dots(i) \\
 &= \int_0^{\pi/2} \log \sin \left(\frac{\pi}{2} - x \right) dx \\
 &= \int_0^{\pi/2} \log \cos x \, dx \quad \dots(ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_0^{\pi/2} \log \sin x \, dx + \int_0^{\pi/2} \log \cos x \, dx \\
 &= \int_0^{\pi/2} (\log \sin x + \log \cos x) dx \\
 &= \int_0^{\pi/2} (\log \sin x \cos x) dx \\
 &= \int_0^{\pi/2} \left(\log \left(\frac{2 \sin x \cos x}{2} \right) \right) dx \\
 &= \int_0^{\pi/2} \left(\log \left(\frac{\sin 2x}{2} \right) \right) dx \\
 &= \int_0^{\pi/2} (\log(\sin 2x)) dx - \int_0^{\pi/2} (\log 2) dx
 \end{aligned}$$

 Let $2x = t$

$$= \frac{1}{2} \int_0^{\pi} [\log(\sin t)] dt - \int_0^{\pi/2} [\log 2] dx$$

$$= \frac{1}{2} \int_0^{\pi} [\log(\sin x)] dx - \int_0^{\pi/2} [\log 2] dx$$

$$= \frac{1}{2} \int_0^{\pi} [\log(\sin x)] dx - \frac{\pi}{2} \log 2$$

$$= \frac{2}{2} \int_0^{\pi/2} [\log(\sin x)] dx - \frac{\pi}{2} \log 2$$

$$= I - \frac{\pi}{2} \log 2$$

$$\Rightarrow 2I = I - \frac{\pi}{2} \log 2$$

$$\Rightarrow I = -\frac{\pi}{2} \log 2 = \frac{1}{2} \log \left(\frac{1}{2} \right)$$

Notes

- $\int_0^{\pi/2} \log \cos x dx = \frac{1}{2} \log \left(\frac{1}{2} \right)$
- $\int_0^{\pi/2} \log \sec x dx = -\frac{1}{2} \log \left(\frac{1}{2} \right)$
- $\int_0^{\pi/2} \log \operatorname{cosec} x dx = -\frac{1}{2} \log \left(\frac{1}{2} \right)$
- $\int_0^{\pi/2} \log \left(\frac{\sin x \cos x}{\cot x} \right) dx = -\log \left(\frac{1}{2} \right)$
- $\int_0^{\pi/2} \log \left(\frac{\sin x \cos x}{\tan x} \right) dx = -\log \left(\frac{1}{2} \right)$

242. Let $I = \int_0^{\pi} x \log(\sin x) dx$

$$= \int_0^{\pi} (\pi - x) \log(\sin(\pi - x)) dx$$

$$= \int_0^{\pi} (\pi - x) \log(\sin x) dx$$

$$= \int_0^{\pi} \pi \log(\sin x) dx - \int_0^{\pi} x \log(\sin x) dx$$

$$\Rightarrow I = \pi \int_0^{\pi} \log(\sin x) dx - I$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \log(\sin x) dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \log(\sin x) dx$$

$$\Rightarrow I = 2 \times \frac{\pi}{2} \int_0^{\pi/2} \log(\sin x) dx$$

$$\Rightarrow I = 2 \times \frac{\pi}{2} \left(\frac{\pi}{2} \log \left(\frac{1}{2} \right) \right)$$

$$\Rightarrow I = \left| \frac{\pi^2}{2} \log \left(\frac{1}{2} \right) \right|$$

243. Let $I = \int_0^{\pi/2} \sin 2x \log(\tan x) dx$... (i)

$$= \int_0^{\pi/2} \sin 2 \left(\frac{\pi}{2} - x \right) \log \left(\tan \left(\frac{\pi}{2} - x \right) \right) dx$$

$$= \int_0^{\pi/2} \sin(\pi - 2x) \log \left(\tan \left(\frac{\pi}{2} - x \right) \right) dx$$

$$= \int_0^{\pi/2} \sin(\pi - 2x) \log(\cot x) dx$$

$$= \int_0^{\pi/2} \sin 2x \log(\cot x) dx$$
 ... (ii)

Adding Eqs (i) and (ii), we get

$$2I = \int_0^{\pi/2} \sin 2x \log(\tan x) dx + \int_0^{\pi/2} \sin 2x \log(\cot x) dx$$

$$= \int_0^{\pi/2} \sin 2x [\log(\tan x) + \log(\cot x)] dx$$

$$= \int_0^{\pi/2} \sin 2x [\log(\tan x \cdot \cot x)] dx$$

$$= \int_0^{\pi/2} \sin 2x (\log 1) dx$$

$$= 0$$

$$\Rightarrow I = 0$$

244. Let $I = \int_0^{\pi} \left(\frac{x dx}{1 + \cos \alpha \cdot \sin x} \right)$

$$= \int_0^{\pi} \left(\frac{(\pi - x) dx}{1 + \cos \alpha \cdot \sin(\pi - x)} \right)$$

$$= \int_0^{\pi} \left(\frac{(\pi - x) dx}{1 + \cos \alpha \cdot \sin x} \right)$$

$$= \int_0^{\pi} \left(\frac{\pi dx}{1 + \cos \alpha \cdot \sin x} \right) - \int_0^{\pi} \left(\frac{x dx}{1 + \cos \alpha \cdot \sin x} \right)$$

$$= \int_0^{\pi} \left(\frac{\pi dx}{1 + \cos \alpha \cdot \sin x} \right) - I$$

$$\Rightarrow 2I = \int_0^{\pi} \left(\frac{\pi dx}{1 + \cos \alpha \cdot \sin x} \right)$$

$$= 2 \int_0^{\pi/2} \left(\frac{\pi dx}{1 + \cos \alpha \cdot \sin x} \right)$$

$$\Rightarrow I = \int_0^{\pi/2} \left(\frac{\pi dx}{1 + k \sin x} \right), \quad (\text{where } k = \cos \alpha)$$

$$= \int_0^{\pi/2} \left(\frac{\pi dx}{1 + k \left(\frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \right)} \right)$$

$$= \int_0^{\pi/2} \left(\frac{\pi(1 + \tan^2(x/2)) dx}{(1 + \tan^2(x/2)) + k(2 \tan(x/2))} \right)$$

$$= \pi \int_0^{\pi/2} \left(\frac{\sec^2(x/2) dx}{(1 + \tan^2(x/2)) + k(2 \tan(x/2))} \right)$$

$$= 2\pi \int_0^1 \left(\frac{dt}{(1 + t^2) + 2kt} \right)$$

$$= 2\pi \int_0^1 \left(\frac{dt}{(t + k)^2 + (1 - k)^2} \right)$$

$$= 2\pi \times \frac{1}{\sqrt{1 - k^2}} \tan^{-1} \left(\frac{t + k}{\sqrt{1 - k^2}} \right)_0^1$$

$$= 2\pi \times \frac{1}{\sqrt{1 - k^2}} \left(\tan^{-1} \left(\frac{1 + \cos \alpha}{\sin \alpha} \right) - \tan^{-1} \left(\frac{\cos \alpha}{\sin \alpha} \right) \right)$$

$$= 2\pi \times \frac{1}{\sqrt{1 - k^2}} \left[\tan^{-1} \left(\frac{2 \cos^2(\alpha/2)}{2 \sin(\alpha/2) \cos(\alpha/2)} \right) \right.$$

$$\left. - \left(\tan^{-1} \left(\frac{\cos \alpha}{\sin \alpha} \right) \right) \right]$$

$$= \frac{2\pi}{\sqrt{1 - k^2}} \left[\tan^{-1} \left(\cot \left(\frac{\alpha}{2} \right) \right) - \tan^{-1}(\cot \alpha) \right]$$

$$= \frac{2\pi}{\sqrt{1 - k^2}} \left[\left(\frac{\pi}{2} - \frac{\alpha}{2} \right) - \left(\frac{\pi}{2} - \alpha \right) \right]$$

$$= \frac{2\pi}{\sqrt{1 - k^2}} \times \frac{\alpha}{2}$$

$$= \frac{\pi \alpha}{\sin \alpha}$$

$$245. \text{ Let } I = \int_{-\pi/4}^{\pi/4} \log(\sin x + \cos x) dx$$

$$= \int_{-\pi/4}^{\pi/4} \log(\sin x + \cos x) dx$$

$$= \int_{-\pi/4}^{\pi/4} \log \left(\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right) dx$$

$$= \int_{-\pi/4}^{\pi/4} \log(\sqrt{2}) dx + \int_{-\pi/4}^{\pi/4} \log \left(\sin \left(x + \frac{\pi}{4} \right) \right) dx$$

$$= 2 \int_{-0}^{\pi/4} \log(\sqrt{2}) dx + \int_{-\pi/4}^{\pi/4} \log \left(\sin \left(x + \frac{\pi}{4} \right) \right) dx$$

$$= 2 \times \frac{1}{2} \times \frac{\pi}{4} \times \log 2 + \int_{-\pi/4}^{\pi/4} \log \left(\sin \left(x + \frac{\pi}{4} \right) \right) dx$$

$$= \frac{\pi}{4} \times \log 2 + \int_{-\pi/4}^{\pi/4} \log \left(\sin \left(x + \frac{\pi}{4} \right) \right) dx$$

$$\text{Let } \left(x + \frac{\pi}{4} \right) = t \Rightarrow dx = dt$$

$$= \frac{\pi}{4} \times \log 2 + \int_0^{\pi/2} \log(\sin t) dt$$

$$= \frac{\pi}{4} \times \log 2 + \int_0^{\pi/2} \log(\sin x) dx$$

$$= \frac{\pi}{4} \times \log 2 - \frac{\pi}{2} \times \log 2$$

$$= -\frac{\pi}{2} \times \log 2$$

$$246. \text{ Let } I = \int_0^{\pi} \frac{dx}{3 \sin^2 x + 4 \cos^2 x}$$

$$= 2 \int_0^{\pi/2} \frac{dx}{3 \cos^2 x + 4 \sin^2 x}$$

$$= 2 \int_0^{\pi/2} \left(\frac{\sec^2 x dx}{3 + 4 \tan^2 x} \right)$$

$$= 2 \int_0^{\infty} \frac{dt}{3 + 4t^2} \quad \text{Let } t = \tan x \Rightarrow dt = \sec^2 x dx$$

$$= \frac{1}{2} \int_0^{\infty} \left(\frac{dt}{t^2 + \left(\frac{\sqrt{3}}{2} \right)^2} \right)$$

$$= \frac{1}{2} \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t}{\sqrt{3}} \right)_0^{\infty}$$

$$= \frac{1}{2} \times \frac{2}{\sqrt{3}} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi}{2\sqrt{3}}$$

247. Let

$$\begin{aligned}
 I &= \int_0^{\pi} \left(\frac{x \sin x}{1 + \cos^2 x} \right) dx \\
 &= \int_0^{\pi} \left(\frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} \right) dx \\
 &= \int_0^{\pi} \left(\frac{(\pi - x) \sin x}{1 + \cos^2 x} \right) dx \\
 &= \int_0^{\pi} \left(\frac{\pi \sin x}{1 + \cos^2 x} \right) dx - \int_0^{\pi} \left(\frac{x \sin x}{1 + \cos^2 x} \right) dx \\
 &= \int_0^{\pi} \left(\frac{\pi \sin x}{1 + \cos^2 x} \right) dx - I
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow 2I &= \int_0^{\pi} \left(\frac{\pi \sin x}{1 + \cos^2 x} \right) dx \\
 &= 2\pi \int_0^{\pi/2} \left(\frac{\sin x}{1 + \cos^2 x} \right) dx
 \end{aligned}$$

$$\Rightarrow I = \pi \int_0^{\pi/2} \left(\frac{\sin x}{1 + \cos^2 x} \right) dx$$

$$\begin{aligned}
 &= -\pi \int_0^{\infty} \left(\frac{dt}{1 + t^2} \right) \quad \text{Let } \cos x = t \\
 &= -\pi (\tan^{-1} t)_0^{\infty} \\
 &= -\pi \left(\frac{\pi}{2} - 0 \right) \\
 &= -\frac{\pi^2}{2}
 \end{aligned}$$

248. Let

$$\begin{aligned}
 I &= \int_0^{\infty} \log \left(x + \frac{1}{x} \right) \frac{dx}{1 + x^2} \\
 &= \int_0^{\pi/2} \left(\frac{\log(\tan \theta + \cot \theta) \sec^2 \theta}{(1 + \tan^2 \theta)} \right) d\theta \\
 &= \int_0^{\pi/2} \log(\tan \theta + \cot \theta) d\theta \\
 &= \int_0^{\pi/2} \log \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \right) d\theta \\
 &= \int_0^{\pi/2} \log \left(\frac{1}{\sin \theta \cos \theta} \right) d\theta \\
 &= \int_0^{\pi/2} \log \left(\frac{2}{2 \sin \theta \cos \theta} \right) d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= \int_0^{\pi/2} \log \left(\frac{2}{\sin 2\theta} \right) d\theta \\
 &= \int_0^{\pi/2} \log 2 d\theta - \int_0^{\pi/2} \log(\sin 2\theta) d\theta \\
 &= \frac{\pi}{2} \log 2 - \frac{1}{2} \int_0^{\pi} \log(\sin t) dt \\
 &= \frac{\pi}{2} \log 2 - \frac{1}{2} \int_0^{\pi} \log(\sin x) dx \\
 &= \frac{\pi}{2} \log 2 - \int_0^{\pi/2} \log(\sin x) dx \\
 &= \frac{\pi}{2} \log 2 - \left(-\frac{\pi}{2} \log 2 \right) \\
 &= \pi \log 2
 \end{aligned}$$

$$249. \text{ We have, } I_2 = \int_0^1 x^{1006} (1 - x^{2014})^{1006} dx$$

$$= \int_0^1 x^{1006} [1 - (x^{1007})^2]^{1007} dx$$

$$\text{Let } x^{1007} = t \Rightarrow 1007 \times x^{1006} dx = dt$$

$$\begin{aligned}
 &= \frac{1}{1007} \int_0^1 (1 - t^2)^{1007} dt \\
 &= \frac{1}{1007} \int_0^1 [1 - (1 - t)^2]^{1007} dt \\
 &= \frac{1}{1007} \int_0^1 [t(2 - t)]^{1007} dt \\
 &= \frac{1}{1007} \int_0^1 t^{1007} (2 - t)^{1007} dt
 \end{aligned}$$

$$\text{Let } t = 2y \Rightarrow 2 dt = 2 dy$$

$$\begin{aligned}
 &= \frac{2^{2014}}{1007} \int_0^1 (y)^{1007} (1 - y)^{1007} dy \\
 &= \frac{2^{2014}}{1007} \int_0^1 (x)^{1007} (1 - x)^{1007} dy \\
 &= \frac{2^{2014}}{1007} \times I_1
 \end{aligned}$$

Now,

$$2^{2014} \times \left(\frac{I_2}{I_1} \right)$$

$$= 2^{2014} \times \left(\frac{2^{2014}}{1007} \times I_1 \times \frac{1}{I_1} \right)$$

$$= \frac{2^{4028}}{1007}$$

250. Do yourself.

251. Let $I = \int_0^1 \frac{\log x}{\sqrt{1-x^2}} dx$

Let $x = \sin \theta$
 $\Rightarrow dx = \cos \theta d\theta$

$$= \int_0^{\pi/2} \frac{\log(\sin \theta) \cdot \cos \theta d\theta}{\cos \theta}$$

$$= \int_0^{\pi/2} \log(\sin \theta) d\theta$$

$$= \left| -\frac{\pi}{2} \log 2 \right|$$

$$= \frac{\pi}{2} \log \left(\frac{1}{2} \right)$$

252. Let $I = \int_0^{\pi} \log(1 - \cos x) dx$

$$= \int_0^{\pi} \log \left(2 \sin^2 \left(\frac{x}{2} \right) \right) dx$$

$$= \int_0^{\pi} \left(\log 2 + \log \left(\sin^2 \left(\frac{x}{2} \right) \right) \right) dx$$

$$= |\pi \log(2)| + \int_0^{\pi} \left(\log \left(\sin^2 \left(\frac{x}{2} \right) \right) \right) dx$$

Let $I_1 = \int_0^{\pi} \left(\log \left(\sin^2 \left(\frac{x}{2} \right) \right) \right) dx \quad \dots(i)$

$$= 2 \int_0^{\pi} \left(\log \left(\sin \left(\frac{x}{2} \right) \right) \right) dx$$

$$= 2 \int_0^{\pi} \left(\log \left(\sin \left(\frac{\pi - x}{2} \right) \right) \right) dx$$

$$= 2 \int_0^{\pi} \left(\log \left(\cos \left(\frac{x}{2} \right) \right) \right) dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I_1 = 2 \int_0^{\pi} \left(\log \left(\sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) \right) \right) dx$$

$$\Rightarrow I_1 = \int_0^{\pi} \left(\log \left(\sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) \right) \right) dx$$

$$= \int_0^{\pi} \left(\log \left\{ \frac{1}{2} \left(2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) \right) \right\} \right) dx$$

$$= \pi \log \left(\frac{1}{2} \right) + \int_0^{\pi} (\log(\sin x)) dx$$

$$= \pi \log \left(\frac{1}{2} \right) + 2 \int_0^{\pi} (\log(\sin x)) dx$$

$$= \pi \log \left(\frac{1}{2} \right) + 2 \left(-\frac{\pi}{2} \log(2) \right)$$

$$= \pi \log \left(\frac{1}{2} \right) + 2 \left(\frac{\pi}{2} \log \left(\frac{1}{2} \right) \right)$$

$$= \pi \log \left(\frac{1}{2} \right) + \left(\pi \log \left(\frac{1}{2} \right) \right)$$

$$= 2\pi \log \left(\frac{1}{2} \right)$$

Thus, $I = \pi \log|2| + 2\pi \log \left(\frac{1}{2} \right)$

$$= \pi \log(2) - 2\pi \log|2|$$

$$= -\pi \log|2|$$

253. Let $I = \int_0^{\infty} \log \left(x + \frac{1}{x} \right) \frac{dx}{1+x^2}$

$$= \int_0^{\pi/2} \log(\tan \theta + \cot \theta) d\theta, \text{ where } x = \tan \theta$$

$$= \int_0^{\pi/2} \log \left(\frac{1}{\sin \theta \cos \theta} \right) d\theta$$

$$= \int_0^{\pi/2} \log \left(\frac{2}{\sin 2\theta} \right) d\theta$$

$$= \left(\frac{\pi}{2} \log 2 \right) - \int_0^{\pi/2} \log(\sin(2\theta)) d\theta$$

$$= \left(\frac{\pi}{2} \log 2 \right) - \frac{1}{2} \int_0^{\pi} \log(\sin(t)) d\theta$$

$$= \left(\frac{\pi}{2} \log 2 \right) - \int_0^{\pi} \log(\sin(t)) d\theta$$

$$= \left(\frac{\pi}{2} \log 2 \right) - \left(-\frac{\pi}{2} \log|2| \right)$$

$$= 2 \left(\frac{\pi}{2} \log 2 \right)$$

$$= |\pi \log(2)|$$

254. Let $I = \int_0^{\pi} \frac{x}{1 + \cos^2 x} dx \quad \dots(i)$

$$= \int_0^{\pi} \frac{(\pi - x)}{1 + \cos^2 x} dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned} 2I &= \pi \int_0^{\pi} \frac{dx}{1 + \cos^2 x} \\ &= 2\pi \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x} \\ \Rightarrow I &= \pi \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x} \\ &= \pi \int_0^{\pi/2} \frac{\sec^2 x dx}{\tan^2 x + 2} \\ &= \frac{\pi}{\sqrt{2}} \left(\tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) \right)_0^{\pi/2} \\ &= \frac{\pi}{\sqrt{2}} \times \frac{\pi}{2} \\ &= \frac{\pi^2}{2\sqrt{2}} \end{aligned}$$

$$\begin{aligned} 255. \text{ Let } I &= \int_0^{400\pi} \sqrt{1 - \cos 2x} dx \\ &= \int_0^{400\pi} (\sqrt{2\sin^2 x}) dx \\ &= \int_0^{400\pi} (\sqrt{2} |\sin x|) dx \\ &= \sqrt{2} \int_0^{400\pi} |\sin x| dx \\ &= \sqrt{2} \times 400 \int_0^{\pi} |\sin x| dx \\ &= 400 \times \sqrt{2} \int_0^{\pi} \sin x dx \\ &= 400 \times \sqrt{2} (-\cos x)_0^{\pi} \\ &= 400 \times \sqrt{2} - (-1 - 1) \\ &= 800\sqrt{2} \end{aligned}$$

$$\begin{aligned} 256. \text{ Let } I &= \int_0^{1000} e^{x-[x]} dx \\ &= 1000 \int_0^1 e^{x-[x]} dx \\ &= 1000 \int_0^1 e^{x-0} dx \end{aligned}$$

$$\begin{aligned} &= 1000(e^x)_0^1 \\ &= 1000(e^1 - 1) \\ &= 1000(e - 1) \end{aligned}$$

$$\begin{aligned} 257. \text{ Let } I &= \int_0^{2000\pi} \left(\frac{dx}{1 + 5^{\tan x}} \right) \\ &= 1000 \int_0^{2\pi} \left(\frac{dx}{1 + 5^{\sin x}} \right) \\ &= 1000 \times \int_0^{2\pi} \frac{dx}{1 + 5^{\sin x}} \\ &= 1000 \times \int_0^{2\pi} \frac{dx}{1 + 5^{\sin(2\pi - x)}} \\ &= 1000 \times \int_0^{2\pi} \frac{dx}{1 + 5^{-\sin x}} \\ &= 1000 \times \int_0^{2\pi} \frac{5^{\sin x} dx}{1 + 5^{\sin x}} \end{aligned}$$

$$\begin{aligned} \Rightarrow 2I &= 1000 \times \int_0^{2\pi} \left(\frac{1 + 5^{\sin x}}{1 + 5^{\sin x}} \right) dx \\ &= 1000 \times \int_0^{2\pi} dx = 1000 \times 2\pi \end{aligned}$$

$$\Rightarrow I = 1000 \times \pi$$

$$\begin{aligned} 258. \text{ Let } I &= \int_0^{1007} \{2x\} dx, \text{ where } \{.,\} = \text{FIP} \\ &= 2014 \int_0^{1/2} \{2x\} dx, \end{aligned}$$

where the given function is periodic with period $1/2$.

$$\begin{aligned} &= 2014 \int_0^{1/2} (2x - [2x]) dx \\ &= 2014 \int_0^{1/2} (2x - 0) dx \\ &= 2014 \times \left(\frac{2x^2}{2} \right)_0^{1/2} \\ &= 2014 \times \frac{1}{4} \\ &= \frac{1007}{2} \end{aligned}$$

$$\begin{aligned}
 259. \text{ Let } I &= \int_0^{\pi} |\sin x + \cos x| dx \\
 &= \sqrt{2} \int_0^{\pi} \left| \sin \left(x + \frac{\pi}{4} \right) \right| dx \\
 &= \sqrt{2} \int_{\pi/4}^{\pi+\pi/4} |\sin y| dy, \quad \text{Let } x + \frac{\pi}{4} = y \\
 &= \sqrt{2} \int_0^{\pi} |\sin x| dx \\
 &= 2\sqrt{2}, \text{ since } \int_0^{\pi} |\sin x| dx = 2
 \end{aligned}$$

$$\begin{aligned}
 260. \text{ Let } I &= \int_0^{100\pi} (|\sin x| + |\cos x|) dx \\
 &= 200 \int_0^{\pi/2} (|\sin x| + |\cos x|) dx \\
 &= 200 \int_0^{\pi/2} (\sin x + \cos x) dx \\
 &= 200\sqrt{2} \int_0^{\pi/2} \left(\sin \left(x + \frac{\pi}{4} \right) \right) dx \\
 &= 200\sqrt{2} \int_{\pi/4}^{\pi/2+\pi/4} \sin y dy, \quad \text{Let } x + \frac{\pi}{4} = y \\
 &= 200\sqrt{2} \int_0^{\pi/2} \sin x dx \\
 &= 200\sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 261. \text{ Let } I &= \int_0^{100\pi} (|\sin x| - |\cos x|) dx \\
 &= 100 \int_0^{\pi} (|\sin x| - |\cos x|) dx \\
 &= 100 \left(\int_0^{\pi} (\sin x) dx - \int_0^{\pi/2} (\cos x) dx + \int_{\pi/2}^{\pi} (\cos x) dx \right) \\
 &= 100 \left((-\cos x)_0^{\pi} - (\sin x)_0^{\pi/2} + (\sin x)_{\pi/2}^{\pi} \right) \\
 &= 100(2 - (-1) + (-1)) \\
 &= 200
 \end{aligned}$$

$$\begin{aligned}
 262. \text{ Let } I &= \int_{-503\pi}^{504\pi} |\cos x| dx \\
 &= (504 - (-503)) \int_0^{\pi} |\cos x| dx \\
 &= 1007 \times \int_0^{\pi} |\cos x| dx
 \end{aligned}$$

$$\begin{aligned}
 &= 1007 \times \left(\int_0^{\pi/2} \cos x dx - \int_{\pi/2}^{\pi} \cos x dx \right) \\
 &= 1007 \times \left((\sin x)_0^{\pi/2} - (\sin x)_{\pi/2}^{\pi} \right) \\
 &= 1007 \times (1 - (-1)) \\
 &= 2014
 \end{aligned}$$

$$\begin{aligned}
 263. \text{ Let } I &= \int_0^{50\pi/3} \sqrt{\frac{1 - \cos 2x}{2}} dx \\
 &= \int_0^{50\pi/3} \sqrt{\sin^2 x} dx \\
 &= \int_0^{50\pi/3} |\sin x| dx \\
 &= \int_0^{16\pi+2\pi/3} |\sin x| dx \\
 &= \int_0^{16\pi} |\sin x| dx + \int_{16\pi}^{16\pi+2\pi/3} |\sin x| dx \\
 &= \int_0^{16\pi} |\sin x| dx + \int_0^{2\pi/3} |\sin x| dx \\
 &= 16 \int_0^{\pi} |\sin x| dx + \int_0^{2\pi/3} |\sin x| dx \\
 &= 16 \int_0^{\pi} \sin x dx + \int_0^{2\pi/3} \sin x dx \\
 &= 16(-\cos x)_0^{\pi} + (-\cos x)_0^{2\pi/3} \\
 &= 16(-(-1 - 1)) + \left(-\left(-\frac{1}{2} - 1\right) \right) \\
 &= 16 \times 2 + \frac{3}{2} \\
 &= \frac{67}{2}
 \end{aligned}$$

$$264. \text{ Given } \alpha = \int_0^1 \frac{\sin t}{1+t} dt$$

$$\begin{aligned}
 \text{Now, } &\int_{4\pi-2}^{4\pi} \left(\frac{\sin \left(\frac{t}{2} \right)}{4\pi + 2 - t} \right) dt \\
 &= \int_{4\pi-2}^{4\pi} \left(\frac{\sin \left(\frac{t}{2} \right)}{4\pi + 2 - t} \right) dt \\
 &= \int_{4\pi-2}^{4\pi} \left(\frac{\sin \left(\frac{t}{2} \right)}{2 + (4\pi - t)} \right) dt
 \end{aligned}$$

$$= \frac{1}{2} \int_{4\pi-2}^{4\pi} \left(\frac{\sin\left(\frac{t}{2}\right)}{1 + \left(2\pi - \frac{t}{2}\right)} \right) dt$$

$$\text{Let } \left(2\pi - \frac{t}{2}\right) = y \Rightarrow dt = -2dy$$

$$= -\int_1^0 \left(\frac{\sin(2\pi - y)}{1 + y} \right) dy$$

$$= -\int_1^0 \left(\frac{\sin y}{1 + y} \right) dy$$

$$= -\int_0^1 \left(\frac{\sin t}{1 + t} \right) dt$$

$$= -\alpha$$

$$265. \text{ Given } \int_0^1 \left(\frac{e^t}{t+1} \right) dt$$

$$\text{We have } \int_{b-1}^b \left(\frac{e^{-1}}{(t-b-1)} \right) dt$$

$$\text{Let } t - b = -y \Rightarrow dt = -dy$$

$$= -\int_1^0 \left(\frac{e^{-b} \cdot e^y}{-(y+1)} \right) dy$$

$$= -e^{-b} \int_0^1 \left(\frac{e^y}{(y+1)} \right) dy$$

$$= -e^{-b} \int_0^1 \left(\frac{e^t}{(t+1)} \right) dt$$

$$= -a^{e-b}$$

$$266. \text{ Given } F(x) + F\left(\frac{1}{x} + 2\right) = 10 \quad \dots(i)$$

Replacing x by $\left(x + \frac{1}{2}\right)$, we get

$$F\left(x + \frac{1}{2}\right) + F(x+1) = 10 \quad \dots(ii)$$

Subtracting Eq. (ii) from Eq. (i), we get

$$F(x) - F(x+1) = 0$$

$$\Rightarrow F(x) = F(x+1)$$

$\Rightarrow$ Thus, the period of $F(x)$ is 1.

Now,

$$\int_0^{201} F(x) dx$$

$$= \int_0^{201} F(x) dx$$

$$= 201 \times \int_0^1 F(x) dx$$

$$= 201 \times \left(\int_0^{1/2} F(x) dx + \int_{1/2}^1 F(x) dx \right)$$

Putting $x = \left(y + \frac{1}{2}\right) \Rightarrow dx = dy$ in the 2nd integral, we have

$$201 \times \left(\int_0^{1/2} F(x) dx + \int_0^{1/2} F\left(y + \frac{1}{2}\right) dy \right)$$

$$= 201 \times \left(\int_0^{1/2} F(x) dx + \int_0^{1/2} F\left(x + \frac{1}{2}\right) dx \right)$$

$$= 201 \times \int_0^{1/2} \left(F(x) + F\left(x + \frac{1}{2}\right) \right) dx$$

$$= 201 \times 10$$

$$= 2010$$

$$267. \text{ Let } I = \int_0^{50\pi} (\sin^4 x + \cos^4 x) dx$$

$$= \int_0^{100 \times \frac{\pi}{2}} (\sin^4 x + \cos^4 x) dx$$

$$= 100 \times \int_0^{\pi/2} (\sin^4 x + \cos^4 x) dx$$

$$= 100 \times \int_0^{\pi/2} (1 - 2\sin^2 x \cos^2 x) dx$$

$$= 50 \times \int_0^{\pi/2} (2 - 4\sin^2 x \cos^2 x) dx$$

$$= 50 \times \int_0^{\pi/2} (2 - \sin^2 2x) dx$$

$$= 25 \times \int_0^{\pi/2} (4 - 2\sin^2 2x) dx$$

$$= 25 \times \int_0^{\pi/2} [-4(1 - \cos^4 x)] dx$$

$$= 25 \times \int_0^{\pi/2} (3 + \cos^4 x) dx$$

$$= 100 \times \int_0^{\pi/2} \left(\frac{3}{4} + \frac{1}{4} \cos^4 x \right) dx$$

Thus, the value of k is 100.

268. We have,

$$g(x + \pi) = \int_0^{x+\pi} \cos^4 t dt$$

$$= \int_0^{\pi} \cos^4 t dt + \int_{\pi}^{x+\pi} \cos^4 t dt$$

$$= \int_0^{\pi} \cos^4 t dt + \int_0^x \cos^4 t dt$$

$$= g(\pi) + g(x)$$

$$= g(x) + g(\pi)$$

$$269. \int_0^{2014\pi} \sqrt{1 - \cos^2 x} \, dx$$

$$= \int_0^{2014\pi} \sqrt{\sin^2 x} \, dx$$

$$= \int_0^{2014\pi} |\sin x| \, dx$$

$$= 2014 \int_0^{\pi} |\sin x| \, dx$$

$$= 2014 \times 2$$

$$= 4028.$$

$$270. \int_0^{2012} e^{x-[x]} \, dx$$

$$= 2012 \int_0^1 e^{x-[x]} \, dx$$

$$= 2012 \int_0^1 e^x \, dx$$

$$= 2012(e^x)_0^1$$

$$= 2012(e - 1)$$

$$271. \int_0^{4\pi} |\cos x| \, dx$$

$$= 4 \int_0^{\pi} |\cos(x)| \, dx$$

$$= 4 \times 2 = 8$$

$$272. \int_0^{10\pi} |\sin x| \, dx$$

$$= 10 \int_0^{\pi} |\sin(x)| \, dx$$

$$= 10 \times 2$$

$$= 20$$

$$273. \int_0^{[x]} (x - [x]) \, dx, \text{ where } [.] = \text{G.I.F.}$$

$$= [x] \int_0^1 (x - [x]) \, dx$$

$$= [x] \int_0^1 (x - 0) \, dx$$

$$= [x] \int_0^1 (x) \, dx$$

$$= [x] \left(\frac{x^2}{2} \right)_0^1$$

$$= \frac{[x]}{2}$$

$$274. \int_0^{4\pi} [\sin x + \cos x] \, dx$$

$$= 2 \left(\int_0^{2\pi} [\sin x + \cos x] \, dx \right)$$

$$= 2(-\pi)$$

$$= 2(-2\pi)$$

$$275. \int_1^{e^{37}} \frac{\pi \sin(\pi \log x)}{x} \, dx$$

$$= \int_0^{37\pi} \sin(t) \, dt,$$

where $\pi \log(x) = t$

$$= \int_0^{37\pi} \sin(t) \, dt$$

$$= (-\cos(t))_0^{37\pi}$$

$$= -(-1 - 1)$$

$$= 2$$

$$276. \int_0^{\frac{32\pi}{3}} \sqrt{1 + \cos 2x} \, dx$$

$$= \int_0^{\frac{32\pi}{3}} (\sqrt{2} |\cos(x)|) \, dx$$

$$= \sqrt{2} \left[\int_0^{10\pi} |\cos(x)| \, dx + \int_{10\pi}^{\frac{32\pi}{3}} |\cos(x)| \, dx \right]$$

$$= \sqrt{2} \left[\int_0^{\pi} |\cos(x)| \, dx + \int_{\frac{2\pi}{3}}^{\frac{2\pi}{3}} |\cos(x)| \, dx \right]$$

$$= \sqrt{2} \left[10 \times 2 + \int_0^{\frac{\pi}{2}} |\cos(x)| \, dx - \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} |\cos(x)| \, dx \right]$$

$$= \sqrt{2} \left[10 \times 2 + 1 - \left(\frac{\sqrt{3}}{2} - 1 \right) \right]$$

$$= \sqrt{2} \left[22 - \frac{\sqrt{3}}{2} \right]$$

$$= 122\sqrt{2} - \sqrt{6}$$

$$278. \int_0^{10\pi/3} |\sin x| \, dx$$

$$= \int_0^{3\pi} |\sin x| \, dx + \int_{3\pi}^{10\pi/3} |\sin x| \, dx$$

$$= 3 \int_0^{\pi} |\sin x| \, dx + \int_0^{\pi/3} |\sin x| \, dx$$

$$= 3 \times 2 - \left(\frac{1}{2} - 1 \right)$$

$$= 6 + \frac{1}{2} = \frac{13}{2}$$

$$\begin{aligned}
 280. \int_{-\pi}^{199\pi} \sqrt{\frac{1 - \cos^2 x}{2}} dx \\
 &= (199 - (-)) \int_0^{\pi} |\sin(x)| dx \\
 &= 200 \int_0^{\pi} |\sin(x)| dx \\
 &= 200 \times 2 = 400
 \end{aligned}$$

$$\begin{aligned}
 281. \int_{-20\pi}^{20\pi} |\cos x| dx \\
 &= [20 - (-20)] \int_0^{\pi} |\cos(x)| dx \\
 &= 40 \int_0^{\pi} |\cos(x)| dx \\
 &= 40 \times 2 = 80
 \end{aligned}$$

$$\begin{aligned}
 282. \int_0^{100} \sin(x - [x]) dx \\
 &= 100 \int_0^1 \sin(x - [x]) dx \\
 &= 100 \int_0^1 \sin(x - [0]) dx \\
 &= 100 \int_0^1 \sin(x) dx \\
 &= -100[\cos(x)]_0^1 \\
 &= -100[\cos(1) - 1] \\
 &= 100[1 - \cos(1)]
 \end{aligned}$$

$$\begin{aligned}
 283. \int_0^{2014\pi} (|\sin x| + |\cos x|) dx \\
 &= 4028 \int_0^{\pi/2} (|\sin x| + |\cos x|) dx \\
 &= 4028 \int_0^{\pi/2} (\sin x + \cos x) dx \\
 &= 4028(-\cos x + \sin x)_0^{\pi/2} \\
 &= 4028(1 + 1) \\
 &= 8056
 \end{aligned}$$

$$\begin{aligned}
 284. \int_0^{2014\pi} (|\sin x| - |\cos x|) dx \\
 &= 2014 \left(\int_0^{\pi} (|\sin x| - |\cos x|) dx \right) \\
 &= 2014 \left(\int_0^{\pi} |\sin x| dx - \int_0^{\pi} |\cos x| dx \right) \\
 &= 2014(2 - 2) \\
 &= 0.
 \end{aligned}$$

285. We have,

$$\int_0^{n\pi+V} |\sin x| dx$$

$$\begin{aligned}
 &= \int_0^{n\pi} |\sin x| dx + \int_{n\pi}^{n\pi+V} |\sin x| dx \\
 &= \int_0^{n\pi} |\sin x| dx + \int_0^V |\sin x| dx \\
 &= n \int_0^{\pi} |\sin x| dx + \int_0^V (\sin x) dx \\
 &= 2n - (\cos(x))_0^V \\
 &= 2n - (\cos V - 1) \\
 &= (2n + 1) - \cos(V)
 \end{aligned}$$

286. We have,

$$\begin{aligned}
 &\int_0^{2n\pi} [\sin x + \cos x] dx \\
 &= n \int_0^{2\pi} [\sin x + \cos x] dx \\
 &= n \int_0^{2\pi} \left[\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \right] dx \\
 &= n \int_{\pi/4}^{9\pi/4} [\sqrt{2} \sin(t)] dt, \quad \left[\text{Let } t = x + \frac{\pi}{4} \right] \\
 &= n \int_{\pi/4}^{9\pi/4} [\sqrt{2} \sin(x)] dx \\
 &= n(-\pi) \\
 &= -n\pi
 \end{aligned}$$

287. We have,

$$\begin{aligned}
 &\int_0^{2\pi} \sin^2(100x) dx \\
 &= \frac{1}{100} \int_0^{200\pi} \sin^2(t) dt \\
 &= \frac{200}{100} \int_0^{\pi} \sin^2(x) dx \\
 &= \int_0^{\pi} [2 \sin^2(x)] dx \\
 &= \int_0^{\pi} [1 - \cos(2x)] dx \\
 &= \left(x - \frac{\sin(2x)}{2} \right)_0^{\pi} \\
 &= (\pi - 0) \\
 &= \pi
 \end{aligned}$$

288. We have,

$$\begin{aligned}
 &\int_{10\pi+\pi/6}^{10\pi+\pi/3} (\sin x + \cos x) dx \\
 &= \int_{\pi/6}^{\pi/3} (\sin x + \cos x) dx \\
 &= (\sin x - \cos x)_{\pi/6}^{\pi/3}
 \end{aligned}$$

$$\begin{aligned}
 &= \left(\frac{\sqrt{3}}{2} - \frac{1}{2} - \frac{1}{2} + \frac{\sqrt{3}}{2} \right) \\
 &= 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right) \\
 &= \sqrt{3} - 1
 \end{aligned}$$

289. We have,

$$\begin{aligned}
 &\int_0^5 \frac{\tan^{-1}(x - [x])}{1 + (x - [x])^2} dx \\
 &= 5 \int_0^1 \frac{\tan^{-1}(x - [x])}{1 + (x - [x])^2} dx \\
 &= 5 \int_0^1 \frac{\tan^{-1}[x - (0)]}{1 + [x - (0)]^2} dx \\
 &= 5 \int_0^1 \frac{\tan^{-1}(x)}{1 + x^2} dx \\
 &= 5 \left(\frac{(\tan^{-1}x)^2}{2} \right)_0^1 \\
 &= 5 \left(\frac{(\pi/4)^2}{2} \right) \\
 &= \frac{5\pi^2}{32}
 \end{aligned}$$

290. Given $F(x) + F\left(x + \frac{1}{2}\right) = 3$

Replacing x by $\left(x + \frac{1}{2}\right)$, we get

$$F\left(x + \frac{1}{2}\right) + F(x + 1) = 3$$

Subtracting Eq. (ii) from Eq. (i), we get

$$F(x) - F(x + 1) = 0$$

$$\Rightarrow F(x) = F(x + 1)$$

Thus, $F(x)$ is a periodic function with period 1.

Now, $\int_0^{1500} F(x) dx$

$$\begin{aligned}
 &= 1500 \int_0^1 F(x) dx \\
 &= 1500 \left(\int_0^{1/2} F(x) dx + \int_{1/2}^1 F(x) dx \right) \\
 &= 1500 \left(\int_0^{1/2} F(x) dx + \int_0^{1/2} F\left(t + \frac{1}{2}\right) dt \right), x = t + \frac{1}{2} \\
 &= 1500 \left(\int_0^{1/2} F(x) dx + \int_0^{1/2} F\left(x + \frac{1}{2}\right) dx \right) \\
 &= 1500 \left(\int_0^{1/2} F(x) dx + F\left(x + \frac{1}{2}\right) dx \right) \\
 &= 1500 \times 3 \times \frac{1}{2} \\
 &= 2250
 \end{aligned}$$

291. It is given that $\int_0^{n+1} f(x) dx = n^2$, where $n \in \mathbb{N}$

Now, $\int_{-2}^4 f(x) dx$

$$\begin{aligned}
 &= \int_{-2}^{-1} f(x) dx + \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx \\
 &\quad + \int_1^2 f(x) dx + \int_2^3 f(x) dx + \int_3^4 f(x) dx \\
 &= 4 + 1 + 0 + 1 + 4 + 9 \\
 &= 19
 \end{aligned}$$

292. We have $\int_0^{\pi/2} \sin^m x \cos^m x dx = k \int_0^{\pi/2} \sin^m x dx$

$$\Rightarrow \frac{1}{2^m} \int_0^{\pi/2} (2 \sin x \cos x)^m dx = k \int_0^{\pi/2} \sin^m x dx$$

$$\Rightarrow \frac{1}{2^m} \int_0^{\pi/2} [\sin(2x)]^m dx = k \int_0^{\pi/2} \sin^m x dx$$

$$\Rightarrow \frac{1}{2^{m+1}} \int_0^{\pi} [\sin(t)]^m dt = k \int_0^{\pi/2} \sin^m x dx$$

where $t = (2x)$

$$\Rightarrow \frac{1}{2^{m+1}} \int_0^{\pi} (\sin(x))^m dx = k \int_0^{\pi/2} \sin^m x dx$$

$$\Rightarrow \frac{2}{2^{m+1}} \int_0^{\pi/2} (\sin(x))^m dx = k \int_0^{\pi/2} \sin^m x dx,$$

$$\Rightarrow \frac{1}{2^m} \int_0^{\pi/2} [\sin^m(x)] dx = k \int_0^{\pi/2} \sin^m x dx,$$

Thus, $k = \frac{1}{2^m} = 2^{-m}$

293. Given $M = \int_0^{\pi} f(\cos^2 x) dx$

Now, $N = \int_0^{3\pi} f(\cos^2 x) dx$

$$\begin{aligned}
 &= 3 \int_0^{\pi} f(\cos^2 x) dx \\
 &= 3M
 \end{aligned}$$

Hence, the result.

294. Given $\frac{d}{dx}(F(x)) = \frac{e^{\sin x}}{x}$, $x > 0$

$$\Rightarrow F(x) = \int \left(\frac{e^{\sin x}}{x} \right) dx$$

Now, $\int_1^4 \frac{2e^{\sin^2 x}}{x} dx$

$$\begin{aligned}
 &= \int_1^4 \frac{2x e^{\sin^2 x}}{x^2} dx
 \end{aligned}$$

$$\begin{aligned}
 &= \int_1^{16} \left(\frac{e^{\sin t}}{t} \right) dt \\
 &= \int_1^{16} \left(\frac{e^{\sin x}}{x} \right) dx \\
 &= F(16) - F(1)
 \end{aligned}$$

Thus, $k = 16$.

295. At $x = 1$, $y = \int_1^1 \frac{dt}{\sqrt{1+t^2}} = 0$

So, the point is $(1, 0)$.

Also $y = \int_x^3 \frac{dt}{\sqrt{1+t^2}}$

$$\Rightarrow \frac{dy}{dx} = \frac{3x^2}{\sqrt{1+x^6}} - \frac{2x}{\sqrt{1+x^4}}$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{x=1} = \left(\frac{3}{\sqrt{2}} - \frac{2}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}}$$

Hence, the equation of the tangent is

$$y - 0 = \frac{1}{\sqrt{2}}(x - 1)$$

$$\Rightarrow y - y\sqrt{2} - 1 = 0$$

296. We have,

$$f(x) = \int_0^x e^t(t-1)(t-2)dt$$

$$\Rightarrow f'(x) = e^x(x-1)(x-2)$$

By the sign scheme, we can say that $f(x)$ is strictly increasing in the interval $(-\infty, 1) \cup (2, \infty)$.

297. We have,

$$f(x) = \int_0^x t(t+1)(t-2)dt$$

$$\Rightarrow f'(x) = x(x+1)(x-2)$$

By the sign scheme, we can say that the function $f(x)$ has points of minima at $x = -1$ and 2 , and point of maxima at $x = 0$.

298. We have $f(x) = \int_0^x e^t(t-1)^3(t-2)^2dt$

$$\Rightarrow f'(x) = e^x(x-1)^3(x-2)^2$$

By the sign scheme, the function has a point of inflexion at $x = 2$

299. We have

$$y = \int_{1/10}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{1/10}^{\sin^2 x} \cos^{-1} \sqrt{t} dt$$

$$\begin{aligned}
 \Rightarrow \frac{dy}{dx} &= (\sin^{-1}(\sin x) - \cos^{-1}(\cos x)) \sin^2 x \\
 &= (\sin^{-1}(\sin x) - \cos^{-1}(\cos x)) \sin^2 x \\
 &= (x - x) \sin^2 x \\
 &= 0 \text{ for all } x
 \end{aligned}$$

$$\Rightarrow y = \text{constant}$$

$\Rightarrow$ Thus, the curve represents a straight line parallel to x -axis.

Since y is independent of x , so we can consider the value of $x = \pi/4$

$$\begin{aligned}
 \Rightarrow y &= \int_{1/10}^{1/2} \sin^{-1} \sqrt{t} dt + \int_{1/10}^{1/2} \cos^{-1} \sqrt{t} dt \\
 &= \int_{1/10}^{1/2} (\sin^{-1} \sqrt{t} + \cos^{-1} \sqrt{t}) dt \\
 &= \int_{1/10}^{1/2} \frac{\pi}{2} dt \\
 &= \frac{\pi}{2} \left(\frac{1}{2} - \frac{1}{10} \right) = \frac{\pi}{5}
 \end{aligned}$$

300. Given $F(x) = \int_x^{2x} \sqrt{5-3\sin^2 t} dt + \int_0^y \sin t dt$

$$\Rightarrow F'(x) = \sqrt{5-3\sin^2(2x)} - \sqrt{5-3\sin^2 x} + \sin y$$

301. Give,

$$f(x) = \int_1^x \sqrt{2-t^2} dt$$

$$\Rightarrow f'(x) = \sqrt{2-x^2}$$

Also, $x^2 - f'(x) = 0$

$$\Rightarrow x^2 = \sqrt{2-x^2}$$

$$\Rightarrow x^4 = 2 - x^2$$

$$\Rightarrow x^4 + x^2 - 2 = 0$$

$$\Rightarrow (x^2 + 2)(x^2 - 1) = 0$$

$$\Rightarrow (x^2 - 1) = 0, \quad (\because x \text{ is real})$$

$$\Rightarrow x = \pm 1$$

302. Given,

$$f(x) = \int_{1/x^2}^{x^2} \cos \sqrt{t} dt$$

$$\begin{aligned}
 \Rightarrow f'(x) &= 2x(\cos \sqrt{x^2}) + \left(\frac{2}{x^3} \right) \left(\cos \sqrt{\frac{1}{x^2}} \right) \\
 &= 2(\cos(1)) + 2(\cos(1)) \\
 &= 4(\cos(1))
 \end{aligned}$$

303. Given,

$$\int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x, \quad x \in \left(0, \frac{\pi}{2} \right)$$

Now, $-\sin^2 x f(\sin x) \cos x = -\cos x$

$$\Rightarrow \sin^2 x f(\sin x) \cos x = \cos x$$

$$\Rightarrow f(\sin x) = \frac{1}{\sin^2 x}$$

$$\Rightarrow f\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{\left(\frac{1}{\sqrt{3}}\right)^2} = 3$$

304. Given,

$$F(x) = \int_0^x f(t) dt$$

$$\Rightarrow F'(x) = f(x)$$

$$\Rightarrow f(x) = F'(x)$$

$$\text{Also, } F(x^2) = x^2 + x^3$$

$$\Rightarrow F'(x^2)(2x) = 2x + 3x^2$$

$$\Rightarrow F'(x^2) = \frac{2x + 3x^2}{2x}$$

$$= \left(1 + \frac{3}{2}x\right)$$

$$\Rightarrow F(x^2) = F'(x^2) = \left(1 + \frac{3}{2}x\right)$$

$$\Rightarrow f(4) = \left(1 + \frac{3}{2} \times 2\right) = 1 + 3 = 4$$

305. Given,

$$\int_0^t x f(x) dx = \frac{2}{5} t^5$$

$$\Rightarrow t^2 f(t^2) 2t = \frac{2}{5} (5t^4)$$

$$\Rightarrow t^2 f(t^2) 2t = 2t^4$$

$$\Rightarrow 2t^3 f(t^2) = 2t^4$$

$$\Rightarrow f(t^2) = t$$

Putting $t = \frac{2}{5}$, we get

$$\Rightarrow f\left(\frac{4}{25}\right) = \frac{2}{5}$$

$$306. \text{ Let } f(x) = \int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$$

$$\Rightarrow f'(x) = \sin^{-1}(\sqrt{\sin^2 x}) \sin(2x)$$

$$- \cos^{-1}(\sqrt{\cos^2 x}) \sin(2x)$$

Putting $x = \frac{\pi}{4}$, we get

$$\Rightarrow f'\left(\frac{\pi}{4}\right) = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \cdot 1 - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) \cdot 1$$

$$\Rightarrow f'\left(\frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{\pi}{4} = 0$$

$$\Rightarrow f'\left(\frac{\pi}{4}\right) = 0$$

$$\Rightarrow f(x) = c$$

Now,

$$f\left(\frac{\pi}{4}\right) = \int_0^{1/2} \sin^{-1}(\sqrt{t}) dt + \int_0^{1/2} \cos^{-1}(\sqrt{t}) dt$$

$$= \int_0^{1/2} (\sin^{-1}(\sqrt{t}) + \cos^{-1}(\sqrt{t})) dt$$

$$= \int_0^{1/2} \left(\frac{\pi}{4}\right) dt = \frac{\pi}{2} \cdot \frac{1}{2} = \frac{\pi}{4}$$

307. Given,

$$\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$$

$$\Rightarrow f(x) = 1 - x f(x)$$

$$\Rightarrow (1+x) f(x) = 1$$

$$\Rightarrow f(x) = \frac{1}{(1+x)}$$

$$\Rightarrow f(1) = \frac{1}{(1+1)} = \frac{1}{2}$$

308. We have,

$$\lim_{x \rightarrow 0} \frac{1}{x} \left[\int_y^a e^{\sin^2 t} dt - \int_{x+y}^a e^{\sin^2 t} dt \right]$$

$$= \lim_{x \rightarrow 0} \frac{\left[\int_y^a e^{\sin^2 t} dt - \int_{x+y}^a e^{\sin^2 t} dt \right]}{x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{0 + e^{\sin^2(x+y)}}{1} \right)$$

$$= e^{\sin^2 y}$$

309. Do yourself.

$$310. \text{ Let } I = \int_0^{\pi} \sin^{100} x \cos^{99} x dx$$

$$= \int_0^{\pi} \cos^{100} \left(\frac{\pi}{2} - x \right) \sin^{99} \left(\frac{\pi}{2} - x \right) dx$$

$$= - \int_{\pi/2}^{-\pi/2} \cos^{100} y \sin^{99} y dy \quad \text{Let } \frac{\pi}{2} - x = y$$

$$= \int_{-\pi/2}^{\pi/2} \cos^{100} y \sin^{99} y dy$$

$$= \int_{-\pi/2}^{\pi/2} \cos^{100} x \sin^{99} x dx$$

$$= 0, \quad (\because \text{it is an odd function.})$$

311. Do yourself.

$$312. \text{ Let } I_1 = \int_{-4}^{-5} e^{(x+5)^2} dx$$

$$= (-5+4) \int_0^1 e^{((-5+4)x-4+5)^2} dx$$

$$= - \int_0^1 e^{(-x+1)^2} dx$$

$$= - \int_0^1 e^{(x+1)^2} dx$$

$$\text{Also, let } I_2 = 3 \int_{1/3}^{2/3} e^{9(x-\frac{2}{3})^2} dx$$

$$= 3 \left(\frac{2}{3} - \frac{1}{3} \right) \int_0^1 e^{9 \left(\left(\frac{2}{3} - \frac{1}{3} \right) x + \frac{1}{3} - \frac{2}{3} \right)^2} dx$$

$$= \int_0^1 e^{\left(\left(\frac{1}{3}\right)^{x-\frac{1}{3}}\right)^2} dx$$

$$= \int_0^1 e^{(x-1)^2} dx$$

Thus, $I_1 + I_2 = 0$

313. We have,

$$\int_0^\pi \sin^{2014} x \cos^{2013} x dx$$

$$= \int_{-\pi/2}^{\pi/2} \sin^{2014}\left(\frac{\pi}{2} + x\right) \cos^{2013}\left(\frac{\pi}{2} + x\right) dx$$

$$= - \int_{-\pi/2}^{\pi/2} \sin^{2014}(x) \cos^{2013}(x) dx$$

$$= 0, \quad (\because \text{it is an odd function.})$$

314. Let $I = \int_0^1 |\cos \pi x| dx$

$$= \frac{1}{\pi} \int_0^\pi |\cos x| dx$$

$$= \frac{2}{\pi} \quad \left(\because \int_0^\pi |\cos x| dx = 2 \right)$$

315. We have,

$$\int_0^1 |\sin 2\pi x| dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} |\sin(t)| dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} |\sin(x)| dx$$

$$= \frac{1}{2\pi} \times 4 = 2/\pi$$

316. We have,

$$\int_{-3/4}^{3/2} |x \cos \pi x| dx$$

$$= \frac{1}{\pi} \int_{-3\pi/4}^{3\pi/2} \left| \frac{t}{\pi} \cos(t) \right| dt$$

$$= \frac{1}{\pi^2} \int_{-3\pi/4}^{3\pi/2} |t \cos(t)| dt$$

$$= \frac{1}{\pi^2} \int_{-3\pi/4}^{3\pi/2} |x \cos(x)| dx$$

$$= \frac{1}{\pi^2} \int_{-3\pi/4}^{3\pi/4} |x \cos(x)| dt + \frac{1}{\pi^2} \int_{3\pi/4}^{3\pi/2} |x \cos(x)| dt$$

$$= \frac{2}{\pi^2} \int_0^{3\pi/4} |x \cos(x)| dt + \frac{1}{\pi^2} \int_{-3\pi/4}^{3\pi/2} |x \cos(x)| dt$$

$$= \frac{2}{\pi^2} \int_0^{3\pi/4} |x \cos(x)| dx - \frac{1}{\pi^2} \int_{-3\pi/4}^{3\pi/2} [x \cos(x)] dt$$

$$= \frac{2}{\pi^2} \int_0^{\pi/2} [x \cos(x)] dx - \frac{2}{\pi^2} \int_{-\pi/2}^{3\pi/4} [x \cos(x)] dx$$

$$- \frac{1}{\pi^2} \int_{3\pi/4}^{3\pi/2} (x \cos(x)) dx$$

$$= \frac{2}{\pi^2} (x \sin x + \cos x)_{\pi/2}^0 - \frac{2}{\pi^2} (x \sin x + \cos x)_{\pi/2}^{3\pi/4}$$

$$- \frac{1}{\pi^2} (x \sin x + \cos x)_{\pi/4}^{3\pi/2}$$

$$= \frac{2}{\pi^2} \left(\frac{\pi}{2} - 1 \right) - \frac{2}{\pi^2} \left(\frac{3\pi}{4\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{\pi}{2} \right)$$

$$- \frac{1}{\pi^2} \left(-\frac{3\pi}{2} - \frac{3\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} \right)$$

$$= \frac{1}{\pi^2} \left(\frac{7\pi}{2} - \frac{3\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} - 2 \right)$$

317. Let $I = \int_{-\pi/12}^{-5\pi/12} \left(\frac{\sin^{2014} x}{\sin^{2014} x + \cos^{2014} x} \right) dx$

$$= \int_{\pi/12}^{5\pi/12} \left(\frac{\sin^{2014}(-x)}{\sin^{2014}(-x) + \cos^{2014}(-x)} \right) dx$$

$$= \int_{\pi/12}^{5\pi/12} \left(\frac{\sin^{2014}(x)}{\sin^{2014}(x) + \cos^{2014}(x)} \right) dx \quad \dots(i)$$

$$= \int_{\pi/12}^{5\pi/12} \left(\frac{(\sin^{2014}(\frac{\pi}{2} - x))}{\sin^{2014}(\frac{\pi}{2} - x) + \cos^{2014}(\frac{\pi}{2} - x)} \right) dx$$

$$= \int_{\pi/12}^{5\pi/12} \left(\frac{\cos^{2014}(x)}{\cos^{2014}(x) + \sin^{2014}(x)} \right) dx \quad \dots(ii)$$

Adding Eqs (i) and (ii), we get

$$2I = \int_{\pi/12}^{5\pi/12} \left(\frac{\sin^{2014} x + \cos^{2014} x}{\sin^{2014} x + \cos^{2014} x} \right) dx$$

$$= \int_{\pi/12}^{5\pi/12} dx$$

$$= \left(\frac{5\pi}{12} - \frac{\pi}{12} \right)$$

$$= \frac{\pi}{3}$$

$$\Rightarrow I = \frac{\pi}{6}$$

318. Let $I = \int_{-\pi/3}^{-\pi/6} \frac{\sin^{2n} x}{\cos^{2n} x + \sin^{2n} x} dx \quad \dots(i)$

$$\begin{aligned}
 &= \int_{-\pi/3}^{-\pi/6} \frac{\sin^{2n}x \left(-\frac{\pi}{2} - x\right)}{\cos^{2n}\left(-\frac{\pi}{2} - x\right) + \sin^{2n}\left(-\frac{\pi}{2} - x\right)} dx \\
 &= \int_{-\pi/3}^{-\pi/6} \frac{\cos^{2n}(x)}{\sin^{2n}(x) + \cos^{2n}(x)} dx \quad \dots(ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_{-\pi/3}^{-\pi/6} \left(\frac{\sin^{2n}(x) + \cos^{2n}(x)}{\sin^{2n}(x) + \cos^{2n}(x)} \right) dx \\
 &= \int_{-\pi/3}^{-\pi/6} dx \\
 &= \left(-\frac{\pi}{6} + \frac{\pi}{3} \right) = \frac{\pi}{6}
 \end{aligned}$$

$$\Rightarrow I = \frac{\pi}{12}$$

$$\begin{aligned}
 319. \text{ Let } I &= \int_{-\pi/6}^{-\pi/3} \left(\frac{\sin^3 x + \cos^3 x}{\sin^{10} x + \cos^{10} x} \right) dx \quad \dots(i) \\
 &= \int_{-\pi/6}^{-\pi/3} \left(\frac{\sin^3\left(-\frac{\pi}{2} - x\right) + \cos^3\left(-\frac{\pi}{2} - x\right)}{\sin^{10}\left(-\frac{\pi}{2} - x\right) + \cos^{10}\left(-\frac{\pi}{2} - x\right)} \right) dx \\
 &= \int_{-\pi/6}^{-\pi/3} \left(\frac{-\cos^3(x) - \sin^3(x)}{\cos^{10}(x) + \sin^{10}(x)} \right) dx \quad \dots(ii)
 \end{aligned}$$

Adding Eqs (i) and (ii), we get

$$\begin{aligned}
 2I &= \int_{-\pi/6}^{-\pi/3} \left(\frac{\cos^3(x) + \sin^3(x) - \cos^3(x) - \sin^3(x)}{\cos^{10}(x) + \sin^{10}(x)} \right) dx \\
 &= 0 \\
 \Rightarrow I &= 0
 \end{aligned}$$

$$\begin{aligned}
 320. \text{ Let } I &= \int_{4\pi-2}^{4\pi} \frac{\sin\left(\frac{t}{2}\right)}{4\pi+2-t} dt \\
 &= 2 \int_0^1 \frac{\sin\left(\frac{4\pi-(4\pi-2)t+4\pi-2}{2}\right) dt}{(4\pi+2-(4\pi-(4\pi-2))t+(4\pi-2))} \\
 &= 2 \int_0^1 \frac{\sin(t-1) dt}{(4-2t)} \\
 &= \int_0^1 \frac{\sin(t-1) dt}{(2-t)} \\
 &= \int_0^1 \left(\frac{\sin(1-t-1)}{(2-(1-t))} \right) dt \\
 &= 2 \int_0^1 \left(\frac{\sin(-t)}{(1+t)} \right) dt
 \end{aligned}$$

$$\begin{aligned}
 &= -2 \int_0^1 \left(\frac{\sin t}{(1+t)} \right) dt \\
 &= -\alpha
 \end{aligned}$$

321. We have,

$$\begin{aligned}
 &\int_0^{\pi} \left(\frac{|\sin(2x)|}{|\sin x| + |\cos x|} \right) dx \\
 &= 2 \int_0^{\pi/2} \left(\frac{|\sin(2x)|}{|\sin x| + |\cos x|} \right) dx \\
 &= 2 \int_0^{\pi/2} \left(\frac{\sin 2x}{\sin x + \cos x} \right) dx \\
 &= 2 \int_0^{\pi/2} \left(\frac{(\sin x + \cos x)^2 - 1}{\sin x + \cos x} \right) dx \\
 &= 2 \int_0^{\pi/2} \left((\sin x + \cos x) - \frac{1}{\sin x + \cos x} \right) dx \\
 &= 2 \int_0^{\pi/2} \left((\sin x + \cos x) - \frac{1}{\sqrt{2}} \operatorname{cosec}\left(x + \frac{\pi}{4}\right) \right) dx \\
 &= 2 \left\{ -\cos x + \sin x - \frac{1}{\sqrt{2}} \log \left| \tan\left(\frac{x}{2} + \frac{\pi}{8}\right) \right| \right\}_0^{\pi/2} \\
 &= 2 \left\{ 2 - \frac{1}{\sqrt{2}} \left(\log \left| \tan\left(\frac{3\pi}{8}\right) \right| - \log \left| \tan\left(\frac{\pi}{8}\right) \right| \right) \right\} \\
 &= \left\{ 4 - \frac{1}{\sqrt{2}} \left(\log \left| \frac{\sqrt{2}+1}{\sqrt{2}-1} \right| \right) \right\} \\
 &= \{ 4 - \sqrt{2} \log(\sqrt{2} + 1) \}
 \end{aligned}$$

322. We have,

$$\begin{aligned}
 &\int_a^b \left(\frac{\sin(x-a) - \cos(x-a)}{\sin(b-x) - \cos(b-x)} \right) dx \\
 &= m \int_a^b \left(\frac{\sin(b-x) - \cos(b-x)}{\sin(x-a) - \cos(x-a)} \right) dx \\
 \Rightarrow &\int_a^b \left(\frac{\sin(b-x) - \cos(b-x)}{\sin(x-a) - \cos(x-a)} \right) dx \\
 &= m \int_a^b \left(\frac{\sin(b-x) - \cos(b-x)}{\sin(x-a) - \cos(x-a)} \right) dx \\
 \Rightarrow m &= 1
 \end{aligned}$$

323. Given,

$$\int_0^1 \frac{e^t}{(t+1)} dt = a$$

$$\text{Also, } \int_{b-1}^b \frac{e^{-t}}{(t-b-1)} dt = a$$

$$= \int_0^1 \left(\frac{e^{-((b-(b-1))t+b-1)}}{(b-(b-1)t+(b-1)-b-1)} \right) dt$$

$$\begin{aligned}
 &= \int_0^1 \left(\frac{e^{-t-b+1}}{t-2} \right) dt \\
 &= e^{-b} \int_0^1 \left(\frac{e^{-(t-1)}}{t-2} \right) dt \\
 &= -e^{-b} \int_1^0 \left(\frac{e^y}{-(y+1)} \right) dy, \quad \text{Let } (t-1) = -y \\
 &= -e^{-b} \int_0^1 \left(\frac{e^t}{-(t+1)} \right) dy \\
 &= -ae^{-b}
 \end{aligned}$$

324. Let $I = \int_0^{\pi} \frac{x \sin 2x \sin\left(\frac{\pi}{2} \cos x\right)}{2x - \pi} dx$

$$\begin{aligned}
 &= \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} - x\right) \sin 2\left(\frac{\pi}{2} - x\right) \sin\left(\frac{\pi}{2} \cos\left(\frac{\pi}{2} - x\right)\right)}{2\left(\frac{\pi}{2} - x\right) - \pi} dx \\
 &= \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} - x\right) \sin 2x \sin\left(\frac{\pi}{2} \sin x\right)}{-2x} dx \\
 &= \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} + x\right) (-\sin 2x) \sin\left(-\frac{\pi}{2} \sin x\right)}{-2x} dx \\
 &= \int_0^{\pi/2} \frac{2x (\sin 2x) \sin\left(\frac{\pi}{2} \sin x\right)}{2x} dx \\
 &= \int_0^{\pi/2} \frac{2 \sin x \cos x \sin\left(\frac{\pi}{2} \sin x\right)}{x} dx \\
 &= \frac{2}{\pi} \int_0^{\pi/2} 2\left(\frac{2t}{\pi}\right) \sin t dt, \text{ where } t = \frac{\pi}{2} \sin x \\
 &= \frac{8}{\pi^2} \int_0^{\pi/2} t \sin t dt \\
 &= \frac{8}{\pi^2} (t(-\cos t) + \sin t) \Big|_0^{\pi/2} \\
 &= \frac{8}{\pi^2} (1 - 0) = \frac{8}{\pi^2}
 \end{aligned}$$

325. Given,

$$\begin{aligned}
 &\int_0^1 (1 + \cos^8 x)(ax^2 + bx + c) dx \\
 &= \int_0^1 (1 + \cos^8 x)(ax^2 + bx + c) dx \\
 &\quad + \int_1^2 (1 + \cos^8 x)(ax^2 + bx + c) dx \\
 &\quad - \int_1^2 (1 + \cos^8 x)(ax^2 + bx + c) dx = 0
 \end{aligned}$$

Thus, the equation has at least one root in (1, 2)

326. Since $|\sin x| \leq 1$, for $x \geq 10$

$$\Rightarrow \left| \frac{\sin x}{1+x^8} \right| \leq \frac{1}{1+x^8} \quad \dots(i)$$

For $10 \leq x \leq 19$,

$$\begin{aligned}
 &1 + x^8 > x^8 \geq 10 \\
 &\Rightarrow \frac{1}{1+x^8} < \frac{1}{x^8} \leq \frac{1}{10^8} \\
 &\Rightarrow \frac{1}{1+x^8} \leq \frac{1}{10^8} \quad \dots(ii)
 \end{aligned}$$

From Eqs (i) and (ii), we get

$$\begin{aligned}
 &\left| \frac{\sin x}{1+x^8} \right| \leq 10^{-8} \\
 &\Rightarrow \int_{10}^{19} \left| \frac{\sin x}{1+x^8} \right| dx \leq \int_{10}^{19} 10^{-8} dx \\
 &= (19-10)10^{-8} = 9 \times 10^{-8} \\
 &= 10^{-7} - 10^{-8} < 10^{-7}
 \end{aligned}$$

$$\text{Hence, } \int_{10}^{19} \left| \frac{\sin x}{1+x^8} \right| dx \leq 10^{-7}$$

327. Since $|\cos(x)| \leq 1$, for $x \geq 5$

$$\Rightarrow \left| \frac{\cos(x)}{1+x^4} \right| \leq \frac{1}{1+x^4} \text{ for } x \geq 5 \quad \dots(i)$$

Again, for $5 \leq x \leq 10$,

$$\begin{aligned}
 &1 + x^4 > 5^4 \\
 &\Rightarrow \frac{1}{(1+x^4)} < \frac{1}{5^4} \\
 &\Rightarrow \frac{1}{1+x^4} < 5^{-4} \quad \dots(ii)
 \end{aligned}$$

From Eqs (i) and (ii), we get

$$\begin{aligned}
 &\left| \frac{\cos x}{(1+x^4)} \right| < 5^{-4} \\
 &\Rightarrow \left| \int_5^{10} \frac{\cos x}{(1+x^4)} dx \right| < \int_5^{10} (5^{-4}) dx \\
 &\Rightarrow \left| \int_5^{10} \frac{\cos x}{(1+x^4)} dx \right| < 5^{-3}
 \end{aligned}$$

328. We have,

$$\begin{aligned}
 &1 \leq x \leq 3 \\
 &\Rightarrow 1 \leq x^2 \leq 9 \\
 &\Rightarrow 4 \leq x^2 + 3 \leq 12 \\
 &\Rightarrow 2 \leq \sqrt{x^2 + 3} \leq 2\sqrt{3}
 \end{aligned}$$

$$\Rightarrow 2(3-1) \leq \int_1^3 \sqrt{x^2+3} \, dx \leq 2\sqrt{3}(3-1)$$

$$\Rightarrow 4 \leq \int_1^3 \sqrt{x^2+3} \, dx \leq 4\sqrt{3}$$

Hence, the result.

329. We have,

$$0 \leq x \leq 1$$

$$\Rightarrow 0 \leq x^2 \leq 1$$

$$\Rightarrow e^0 \leq e^{x^2} \leq e^1$$

$$\Rightarrow \int_0^1 (e^0) dx \leq \int_0^1 (e^{x^2}) dx \leq \int_0^1 (e^1) dx$$

$$\Rightarrow (1-0) \leq \int_0^1 (e^{x^2}) dx \leq e(1-0)$$

$$\Rightarrow 1 \leq \int_0^1 (e^{x^2}) dx \leq e$$

330. Let $f(x) = \frac{x}{x^3+16}$.

$$\Rightarrow f'(x) = \frac{16-2x^3}{(x^3+16)^2} > 0, \forall x \in (0, 1)$$

$$\Rightarrow f''(x) =$$

$$\frac{-(x^3+16)^2(6x^2) - (16-2x^3)2(x^3+16) \cdot 3x^2}{(x^3+16)^4} > 0,$$

$$\forall x \in (0, 1)$$

Thus, $f(x)$ is concave upward.

$$\Rightarrow (1-0)f(0) \leq \int_0^1 \left(\frac{x}{x^3+16} \right) dx \leq \left(\frac{1-(0)f(0)+f(1)}{2} \right)$$

$$\Rightarrow 0 \leq \int_0^1 \left(\frac{x}{x^3+16} \right) dx \leq \frac{17}{2}$$

Hence, the result.

331. Let $f(x) = \sqrt{3+x^3}$.

$$\Rightarrow f'(x) = \frac{3x^2}{2\sqrt{3+x^3}}$$

$$\Rightarrow f'(x) = \frac{3x^2}{2\sqrt{3+x^3}} > 0, \forall x \in (1, 3)$$

Thus, $f(x)$ increases in $(1, 3)$.

$$\text{So, } (3-1)f(1) \leq \int_1^3 (\sqrt{3+x^3}) dx \leq (3-1)f(3)$$

$$\Rightarrow 4 \leq \int_1^3 (\sqrt{3+x^3}) dx \leq 2\sqrt{30}$$

332. We have for every x in $[0, 1]$,

$$\sqrt{4-2x^2} \leq \sqrt{4-x^2-x^3} \leq \sqrt{4-x^2}$$

$$\Rightarrow \frac{1}{\sqrt{4-x^2}} \leq \frac{1}{\sqrt{4-x^2-x^3}} \leq \frac{1}{\sqrt{4-2x^2}}$$

$$\Rightarrow \int_0^1 \frac{dx}{\sqrt{4-x^2}} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \int_0^1 \frac{dx}{\sqrt{4-2x^2}}$$

$$\Rightarrow \left(\sin^{-1}\left(\frac{x}{2}\right) \right)_0^1 \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{1}{\sqrt{2}} \left(\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) \right)_0^1$$

$$\Rightarrow \frac{\pi}{6} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{\pi}{4\sqrt{2}}$$

333. We have,

$$1 \leq \sqrt{1+x^3} \leq 1+x^3, x \geq 0$$

$$\Rightarrow \int_0^1 1 dx \leq \int_0^1 \sqrt{1+x^3} dx \leq \int_0^1 (1+x^3) dx$$

$$\Rightarrow 1 \leq \int_0^1 \sqrt{1+x^3} dx \leq \left(x + \frac{x^4}{4} \right)_0^1$$

$$\Rightarrow 1 \leq \int_0^1 \sqrt{1+x^3} dx \leq \frac{5}{4}$$

334. We have,

$$1 \leq x \leq 2$$

$$\Rightarrow 1 \leq x^4 \leq 16$$

$$\Rightarrow 2 \leq (1+x^4) \leq 17$$

$$\Rightarrow \frac{1}{17} \leq \frac{1}{(1+x^4)} \leq \frac{1}{2}$$

$$\Rightarrow \frac{1}{17} \leq \int_0^1 \frac{dx}{1+x^4} \leq \frac{7}{24}$$

335. Let $I = \int_0^\infty \left(\frac{\tan^{-1} ax - \tan^{-1} x}{x} \right) dx$

$$\Rightarrow \frac{dI}{da} = \int_0^\infty \left(\frac{1 \cdot x}{(1+a^2x^2)x} \right) dx$$

$$= \int_0^\infty \left(\frac{dx}{(1+a^2x^2)} \right)$$

$$= \frac{1}{a^2} \int_0^\infty \left(\frac{dx}{(1/a^2)+x^2} \right)$$

$$= \frac{1}{a^2} (a \tan^{-1}(ax)) \Big|_0^\infty$$

$$= \frac{1}{a^2} \left(a \frac{\pi}{2} \right) = \frac{\pi}{2a}$$

$$\Rightarrow I = \frac{\pi}{2} \ln a + c$$

When $a = 1, I = 0 \Rightarrow c = 0$

$$\text{Hence, } I = \frac{\pi}{2} \ln a$$

336. Let $I = \int_0^{\pi/2} \ln \left(\frac{1+a \sin x}{1-a \sin x} \right) \frac{dx}{\sin x}, (|a| < 1).$

$$\begin{aligned}
 \Rightarrow \frac{dI}{da} &= \int_0^{\pi/2} \left(\frac{2 \sin x}{1 - a^2 \sin^2 x} \right) \frac{dx}{\sin x}, (|a| < 1) \\
 &= \int_0^{\pi/2} \left(\frac{2 \sin x}{1 - a^2 \sin^2 x} \right) \frac{dx}{\sin x} \\
 &= \int_0^{\pi/2} \left(\frac{2}{1 - a^2 \sin^2 x} \right) dx \\
 &= \int_0^{\pi/2} \left(\frac{2 \sec^2 x}{\sec^2 x - a^2 \tan^2 x} \right) dx \\
 &= \int_0^{\pi/2} \left(\frac{2 \sec^2 x}{1 + (1 - a^2) \tan^2 x} \right) dx \\
 &= \int_0^{\infty} \left(\frac{2 dt}{1 + (1 - a^2) t^2} \right), \text{ Let } t = \tan x \\
 &= \frac{1}{(1 - a^2)} \int_0^{\infty} \left(\frac{2 dt}{t^2 + \left(\frac{1}{\sqrt{1 - a^2}} \right)^2} \right) \\
 &= \frac{2}{\sqrt{1 - a^2}} \times \left(\tan^{-1} (t \sqrt{1 - a^2}) \right)_0^{\infty} \\
 &= \frac{2}{\sqrt{1 - a^2}} \times \frac{\pi}{2} \\
 &= \frac{\pi}{\sqrt{1 - a^2}}
 \end{aligned}$$

$$\Rightarrow I = \pi \sin^{-1} a + c$$

When $a = 0$, $I = 0$, then $c = 0$

Hence, $I = \pi \sin^{-1} a$

$$\begin{aligned}
 336. \text{ Given } I &= \int_0^{\pi/2} \ln \left(\frac{1 + a \sin x}{1 - a \sin x} \right) \frac{dx}{\sin x}, (|a| < 1) \\
 \Rightarrow \frac{dI}{da} &= \int_0^{\pi/2} \left(\frac{1 - a \sin x}{1 + a \sin x} \right) \times \frac{2 \sin x}{(1 - a \sin x)^2} \times \frac{1}{\sin x} dx \\
 &= \int_0^{\pi/2} \left(\frac{2 dx}{1 - a^2 \sin^2 x} \right) \\
 &= \int_0^{\pi/2} \left(\frac{2 \operatorname{cosec}^2 x}{\operatorname{cosec}^2 x - a^2} \right) dx \\
 &= \int_0^{\pi/2} \left(\frac{2 \operatorname{cosec}^2 x}{\cot^2 x + (1 - a^2)} \right) dx \\
 &= \frac{-2}{\sqrt{1 - a^2}} \tan^{-1} \left(\frac{\cot x}{\sqrt{1 - a^2}} \right) \Big|_0^{\pi/2} \\
 &= \frac{2}{\sqrt{1 - a^2}} \times \frac{\pi}{2} = \frac{\pi}{\sqrt{1 - a^2}}
 \end{aligned}$$

Integrating, we get

$$I = \pi \sin^{-1}(a) + c$$

When $a = 0$, $I(a) = 0$, then $c = 0$

Thus, $I = \pi \sin^{-1}(a)$

$$\begin{aligned}
 337. \text{ Given } I(b) &= \int_0^1 \frac{x^b - 1}{\log x} dx \\
 \Rightarrow \frac{d[I(b)]}{db} &= \int_0^1 \left(\frac{x^b \log x}{\log x} \right) dx \\
 &= \int_0^1 (x^b) dx \\
 &= \frac{x^{b+1}}{b+1}
 \end{aligned}$$

Integrating both sides w.r.t. b , we get

$$I(b) = \log(b+1) + c$$

When $b = 0$, then $c = 0$

Thus, $I(b) = \log(b+1)$

338. We have,

$$\begin{aligned}
 (f(x))^2 &= \int_0^x f(t) \cdot \frac{2 \sec^2 t}{4 + \tan t} dt \\
 \Rightarrow 2f(x)f'(x) &= \frac{f(x) \cdot 2 \sec^2 x}{4 + \tan x} \\
 \Rightarrow f'(x) &= \frac{\sec^2 x}{4 + \tan x}
 \end{aligned}$$

Integrating, we get

$$f(x) = \log|4 + \tan(x)| + c$$

When $x = 0$, then $c = -\log(4)$

$$f(x) = \log|4 + \tan(x)| - \log(4)$$

$$\Rightarrow f\left(\frac{\pi}{4}\right) = \log(4+1) - \log(4) = \log\left(\frac{5}{4}\right)$$

339. Given,

$$\begin{aligned}
 f(x) &= \int_0^{\pi/2} \log(1 + x \sin^2 \theta) / \sin^2 \theta d\theta \\
 \Rightarrow f'(x) &= \int_0^{\pi/2} \left(\frac{\sin^2 \theta}{(1 + x \sin^2 \theta)} \times \frac{1}{\sin^2 \theta} \right) d\theta \\
 &= \int_0^{\pi/2} \left(\frac{d\theta}{(1 + x \sin^2 \theta)} \right) \\
 &= \int_0^{\pi/2} \left(\frac{\operatorname{cosec}^2 \theta d\theta}{(\cot^2 \theta + (1+x))} \right) \\
 &= \left(-\frac{1}{\sqrt{1+x}} \tan^{-1} \left(\frac{\cot \theta}{\sqrt{1+x}} \right) \right) \Big|_0^{\pi/2} \\
 &= \frac{\pi}{2\sqrt{1+x}}
 \end{aligned}$$

Integrating, we get

$$f(x) = \pi \sqrt{1+x} + c$$

When $x = 0$, $f(0) = 0$, then $c = -\pi$

$$f(x) \pi\sqrt{1+x} - \pi = \pi(\sqrt{1+x} - 1)$$

$$340. \text{ Let } I = \int_0^1 \left(\frac{x^{\cos \alpha} - 1}{\log x} \right) dx$$

$$\begin{aligned} \Rightarrow \frac{dI}{d\alpha} &= \int_0^1 \left(\frac{x^{\cos \alpha} \log x}{\log x} \right) dx \\ &= \int_0^1 (x^{\cos \alpha}) dx \\ &= \left(\frac{x^{\cos \alpha + 1}}{\cos \alpha + 1} \right)_0^1 = \frac{1}{1 + \cos \alpha} \end{aligned}$$

Integrating, we get

$$I(\alpha) = \log|1 + \cos \alpha| + c$$

$$\text{When } \alpha = \frac{\pi}{2}, I(\alpha) = 0, \text{ then } c = 0$$

$$\text{Thus, } I(\alpha) = \log|1 + \cos \alpha|$$

341. Do yourself

342. We have,

$$\begin{aligned} &\int_0^2 \sqrt{(1+x)(1+x^4)} dx \\ &\leq \sqrt{\int_0^2 (1+x) dx} \times \sqrt{\int_0^2 (1+x^4) dx} \\ &= \sqrt{\left(x + \frac{x^2}{2}\right)_0^2} \times \sqrt{\left(x + \frac{x^5}{5}\right)_0^2} \\ &= \sqrt{(2+2)} \times \sqrt{\left(2 + \frac{32}{5}\right)} \\ &= 2 \times \sqrt{\frac{42}{5}} \end{aligned}$$

Hence, the maximum value of the given integral is

$$2 \times \sqrt{\frac{42}{5}}.$$

343. We have,

$$\begin{aligned} &\int_0^1 \sqrt{(1+x)(1+x^3)} dx \\ &\leq \sqrt{\int_0^1 (1+x) dx} \sqrt{\int_0^1 (1+x^3) dx} \\ &= \sqrt{\left(x + \frac{x^2}{2}\right)_0^1} \sqrt{\left(x + \frac{x^4}{4}\right)_0^1} \\ &= \sqrt{\frac{3}{2}} \sqrt{\frac{5}{4}} = \sqrt{\frac{15}{8}} \end{aligned}$$

Hence, the maximum value is $\sqrt{\frac{15}{8}}$.

344. We have,

$$\begin{aligned} &\int_a^{a+\frac{\pi}{2}} (\sin^4 x + \cos^4 x) dx \\ &= \int_0^{\frac{\pi}{2}} (\sin^4 x + \cos^4 x) dx \end{aligned}$$

$$\begin{aligned} &= \int_0^{\frac{\pi}{2}} (1 - 2\sin^2 x \cos^2 x) dx \\ &= \int_0^{\frac{\pi}{2}} \left(1 - \frac{\sin^2(2x)}{2} \right) dx \\ &= \int_0^{\frac{\pi}{2}} \left(1 - \frac{2\sin^2(2x)}{4} \right) dx \\ &= \int_0^{\frac{\pi}{2}} \left(1 - \frac{1 - \cos 4x}{4} \right) dx \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{3}{4} + \frac{1}{4} \cos 4x \right) dx \\ &= \left(\frac{3}{4}x + \frac{\sin 4x}{16} \right)_0^{\pi/2} \\ &= \left(\frac{3}{4} \times \frac{\pi}{2} \right) \\ &= \frac{3\pi}{8} \end{aligned}$$

$$\begin{aligned} 345. \text{ Mean Value} &= \frac{1}{(2\pi - 0)} \int_0^{2\pi} \sin^2 x dx \\ &= \frac{1}{4\pi} \int_0^{2\pi} 2\sin^2 x dx \\ &= \frac{1}{4\pi} \int_0^{2\pi} (1 - \cos^2 x) dx \\ &= \frac{1}{4\pi} \left(x - \frac{\sin^2 x}{2} \right)_0^{2\pi} \\ &= \frac{1}{4\pi} (2\pi - 0) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} 346. \text{ Mean value} &= \frac{1}{(2 - 0)} \int_0^2 \frac{dx}{e^x + 1} \\ &= \frac{1}{2} \int_0^2 \frac{e^{-x} dx}{e^{-x} + 1} \\ &= \left(-\frac{1}{2} \log(e^{-x} + 1) \right)_0^2 \\ &= \left(-\frac{1}{2} \log(2(e^{-2} + 1)) \right) \\ &= \left(\frac{1}{2} \log \left| 2 \left(\frac{e^2}{e^2 + 1} \right) \right| \right) \end{aligned}$$

$$\begin{aligned} 347. \text{ Mean value} &= \frac{1}{(1 - 0)} \int_0^1 (\sqrt[3]{x}) dx \\ &= \frac{1}{(1 - 0)} \int_0^1 (\sqrt[3]{x}) dx \end{aligned}$$

$$= \frac{1}{(1-0)} \times \left(\frac{3}{4} x^{4/3} \right)_0^1$$

$$= \frac{3}{4}$$

$$348. \text{ Mean value} = \frac{1}{(2\pi-0)} \int_0^{2\pi} (\sin^3 x) dx$$

$$= \frac{1}{8\pi} \int_0^{2\pi} (3\sin x - \sin 3x) dx$$

$$= \frac{1}{8\pi} \left(-3\cos x + \frac{\cos 3x}{3} \right)_0^{2\pi}$$

$$= \frac{1}{8\pi} \left(-3 + \frac{1}{3} + 3 - \frac{1}{3} \right)$$

$$= 0$$

$$349. \text{ Mean value} = \frac{1}{(1-0)} \int_0^1 \frac{dx}{1+x^2}$$

$$= (\tan^{-1}(x))_0^1$$

$$= \frac{\pi}{4}$$

$$350. \text{ Mean value} = \frac{1}{(1-0)} \int_0^1 \left(\frac{e^x}{e^x+1} \right) dx$$

$$= (\log(e^x+1))_0^1$$

$$= \log\left(\frac{e+1}{2}\right)$$

351. We have,

$$\lim_{\phi \rightarrow 0} \left(\int_{\phi}^1 \log(x) dx \right)$$

$$= \lim_{\phi \rightarrow 0} (x \log(x) - x)_\phi^1$$

$$= \lim_{\phi \rightarrow 0} (\phi - 1 - \phi \log(\phi))$$

$$= \lim_{\phi \rightarrow 0} (\phi - 1) - \lim_{\phi \rightarrow 0} (\phi \log(\phi))$$

$$= \lim_{\phi \rightarrow 0} (\phi - 1) - \lim_{\phi \rightarrow 0} \left(\frac{\log(\phi)}{\frac{1}{\phi}} \right)$$

$$= \lim_{\phi \rightarrow 0} (\phi - 1) - \lim_{\phi \rightarrow 0} \left(\frac{\frac{1}{\phi}}{-\frac{1}{\phi^2}} \right)$$

$$= \lim_{\phi \rightarrow 0} (\phi - 1) + \lim_{\phi \rightarrow 0} (\phi)$$

$$= -1$$

352. We have,

$$\int_0^{\infty} e^{-x} x^6 dx$$

$$= \int_0^{\infty} e^{-x} x^{7-1} dx$$

$$= \Gamma(7)$$

$$= \Gamma(6+1)$$

$$= 6!$$

$$= 720$$

353. We have,

$$\int_0^1 \left(\log\left(\frac{1}{x}\right) \right)^{10} dx$$

$$= - \int_{\infty}^0 e^{-t} t^{10} dt$$

$$= \int_0^{\infty} e^{-x} x^{11-1} dx$$

$$= \Gamma(11)$$

$$= \Gamma(10+1)$$

$$= (10)!$$

354. We have,

$$\int_0^{\infty} e^{-x} x^3 dx$$

$$= \int_0^{\infty} e^{-x} x^{4-1} dx$$

$$= \Gamma(7)$$

$$= (3)!$$

$$= 6$$

355. We have,

$$\int_0^1 \left(\log\left(\frac{1}{x}\right) \right)^{n-1} dx$$

$$= \int_0^1 (-\log x)^{n-1} dx$$

$$= \int_0^{\infty} (e^{-t} t^{n-1}) dt, \quad (\text{Let } t = -\log x)$$

$$= \int_0^{\infty} (e^{-x} x^{n-1}) dx$$

$$= \Gamma(n)$$

$$356. \text{ Let } I = \int_0^1 x^4 \sqrt{1-x^2} dx$$

$$= \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta \quad (\text{Let } \sin \theta = x)$$

$$= \frac{\Gamma\left(\frac{4+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{4+2+2}{2}\right)}$$

$$= \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{3}{2}\right)}{2\Gamma(4)}$$

$$= \frac{\frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} \times \frac{1}{2} \sqrt{\pi}}{2 \times 6}$$

$$= \frac{\pi}{32}$$

357. We have,

$$\begin{aligned}
 I &= \int_0^{\pi/2} \sin^8 x \cos^4 x \, dx \\
 &= \frac{\Gamma\left(\frac{8+1}{2}\right) \Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{8+4+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{9}{2}\right) \Gamma\left(\frac{5}{2}\right)}{2\Gamma\left(\frac{14}{2}\right)} \\
 &= \frac{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} \times \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{2 \cdot (6)!} \\
 &= \frac{7\pi}{2048}
 \end{aligned}$$

358. We have,

$$\begin{aligned}
 \int_0^1 x^6 \sqrt{1-x^2} \, dx &= \int_0^{\pi/2} \sin^6 \theta \cos^2 \theta \, d\theta \quad (\text{Let } \sin \theta = x) \\
 &= \frac{\Gamma\left(\frac{6+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{6+2+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{7}{2}\right) \Gamma\left(\frac{3}{2}\right)}{2\Gamma(5)} \\
 &= \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \times \frac{1}{2} \sqrt{\pi}}{2 \times 24} \\
 &= \frac{5\pi}{256}
 \end{aligned}$$

359. We have,

$$\begin{aligned}
 \int_0^{\pi/2} \sin^4 x \cos^6 x \, dx &= \frac{\Gamma\left(\frac{4+1}{2}\right) \Gamma\left(\frac{6+1}{2}\right)}{2\Gamma\left(\frac{4+6+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{7}{2}\right)}{2\Gamma(6)} \\
 &= \frac{\frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{2 \times 120} \\
 &= \frac{3\pi}{512}
 \end{aligned}$$

360. We have, $\int_0^1 x^{10} \sqrt{1-x^2} \, dx$

$$\begin{aligned}
 &= \int_0^{\pi/2} \sin^{10} \theta \cos^2 \theta \, d\theta \quad (\text{Let } \sin \theta = x) \\
 &= \frac{\Gamma\left(\frac{10+1}{2}\right) \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{10+2+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{11}{2}\right) \Gamma\left(\frac{3}{2}\right)}{2\Gamma(7)} \\
 &= \frac{\frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \cdot \frac{1}{2} \sqrt{\pi}}{2 \times (6)!} \\
 &= \frac{21\pi}{512}
 \end{aligned}$$

361. We have,

$$\begin{aligned}
 \int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}} \times \int_0^{\pi/2} \sqrt{\sin x} \, dx &= \int_0^{\pi/2} \sin^{-1/2} x \cos^0 x \, dx \times \int_0^{\pi/2} \sin^{1/2} x \cos^0 x \, dx \\
 &= \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{3}{4}\right)} \times \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{5}{4}\right)} \\
 &= \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{3}{4}\right)} \times \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{2 \cdot \frac{1}{4} \cdot \Gamma\left(\frac{1}{4}\right)} \\
 &= \left(\Gamma\left(\frac{1}{2}\right)\right)^2 \\
 &= \pi
 \end{aligned}$$

362. We have,

$$\begin{aligned}
 \int_0^{\infty} e^{-a^2 x^2} \, dx &= \frac{1}{2a} \int_0^{\infty} e^{-t} t^{-1/2} \, dt, \quad (\text{Let } (ax)^2 = t) \\
 &= \frac{1}{2a} \int_0^{\infty} e^{-t} t^{(1-\frac{1}{2})} \, dt \\
 &= \frac{\Gamma\left(\frac{1}{2}\right)}{2a} \\
 &= \frac{\sqrt{\pi}}{2a}
 \end{aligned}$$

363. We have,

$$\begin{aligned}
 \int_0^{\pi/2} \sin^7 x \cos^5 x \, dx &= \frac{\Gamma\left(\frac{7+1}{2}\right) \Gamma\left(\frac{5+1}{2}\right)}{2\Gamma\left(\frac{7+5+2}{2}\right)}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{\Gamma(2)\Gamma(3)}{2\Gamma(7)} \\
 &= \frac{3 \cdot 2 \cdot 1 \cdot 2 \cdot 1}{2 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\
 &= \frac{1}{120}
 \end{aligned}$$

364. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \sin^{10} x \, dx \\
 &= \frac{9}{10} \cdot \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\
 &= \frac{63\pi}{512}
 \end{aligned}$$

365. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \cos^7 x \, dx \\
 &= \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \\
 &= \frac{16}{35}
 \end{aligned}$$

366. We have,

$$\begin{aligned}
 &\int_0^{\pi/4} 8 \cos^4 x \sin^4 x \, dx \\
 &= \frac{1}{2} \int_0^{\pi/4} 16 \cos^4 x \sin^4 x \, dx \\
 &= \frac{1}{2} \int_0^{\pi/4} (\sin^2 x)^4 \, dx \\
 &= \frac{1}{2} \int_0^{\pi/2} (\sin^4 t) \, dx \\
 &= \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\
 &= \frac{3\pi}{64}
 \end{aligned}$$

367. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \sin^6 x \cos^5 x \, dx \\
 &= \frac{(5 \cdot 3 \cdot 1)(42)}{(11 \cdot 9 \cdot 7 \cdot 5 \cdot 3 \cdot 1)} \\
 &= \frac{8}{693}
 \end{aligned}$$

368. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \sin^5 x \cos^3 x \, dx \\
 &= \frac{(4 \cdot 2)(2)}{(8 \cdot 6 \cdot 4 \cdot 2)} \\
 &= \frac{1}{24}
 \end{aligned}$$

369. We have,

$$\int_0^{\pi/2} \sin^6 x \cos^4 x \, dx$$

$$\begin{aligned}
 &= \frac{\Gamma\left(\frac{6+1}{2}\right)\Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{6+4+2}{2}\right)} \\
 &= \frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{5}{2}\right)}{2\Gamma(6)} \\
 &= \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}}{2 \times (5!)} \\
 &= \frac{3\pi}{512}
 \end{aligned}$$

370. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \sin^7 x \cos x \, dx \\
 &= \int_0^1 t^7 \, dt, \quad (\text{Let } \sin x = t) \\
 &= \left(\frac{t^8}{8}\right)_0^1 \\
 &= \frac{1}{8}
 \end{aligned}$$

371. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \sin^7 x \cos^5 x \, dx \\
 &= \frac{\Gamma\left(\frac{7+1}{2}\right)\Gamma\left(\frac{5+1}{2}\right)}{2\Gamma\left(\frac{7+5+2}{2}\right)} \\
 &= \frac{\Gamma(4)\Gamma(3)}{2\Gamma(7)} \\
 &= \frac{(3!) \times (2!)}{2 \times (6!)} \\
 &= \frac{6 \times 2}{2 \times 720} = \frac{1}{120}
 \end{aligned}$$

372. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \sin^{11} x \, dx \\
 &= \left(\frac{11-1}{11}\right)\left(\frac{11-3}{11-2}\right)\left(\frac{11-5}{11-4}\right)\left(\frac{11-7}{11-6}\right)\left(\frac{11-9}{11-8}\right) \\
 &= \frac{10}{11} \cdot \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \\
 &= \frac{256}{693}
 \end{aligned}$$

373. We have,

$$\begin{aligned}
 &\int_0^{\pi/2} \cos^8 x \, dx \\
 &= \left(\frac{8-1}{8}\right)\left(\frac{8-3}{8-2}\right)\left(\frac{8-5}{8-4}\right)\left(\frac{8-7}{8-6}\right) \cdot \frac{\pi}{2} \\
 &= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{3} \\
 &= \frac{35\pi}{256}
 \end{aligned}$$